

# Mathematics

## Year 4

### Teacher Guide

This book was developed with the participation of the Department of Basic Education of South Africa (DBE) using the DBE State-owned (Sasol Inzalo)

# **YEAR 4**

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# TERM 1

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## UNIT 1

### WHOLE NUMBERS: COUNTING, ORDERING, COMPARING, REPRESENTING AND PLACE VALUE

#### 1.1 Counting, ordering, comparing and representing

1. Fill in the next number:

- (a) 12 200 000; 15 200 000; 18 200 000; **21 200 000**  
(b) 48 500 750; 48 500 500; 48 500 250; **48 500 000**  
(c) 91 400 000; 88 250 000; 85 100 000; **81 950 000**  
(d) 22 444 111; 44 444 111; 66 444 111; **88 444 111**

2. Copy and complete the table. Count in 55 000s

|                       |                       |                       |
|-----------------------|-----------------------|-----------------------|
| 19 000 000            | (a) <b>19 055 000</b> | 19 110 000            |
| 19 055 000            | 19 110 000            | (b) <b>19 165 000</b> |
| (c) <b>19 110 000</b> | (d) <b>19 165 000</b> | 19 220 000            |

#### Order, compare and represent whole numbers

3. Arrange the following number in ascending order.

- (a) 27 153 999; 27 135 999; 27 315 999; 27 513 999  
**27 135 999; 27 153 999; 27 315 999; 27 513 999**  
(b) 50 101 101; 50 010 101; 50 100 110; 50 001 011  
**50 001 011; 50 010 101; 50 100 110; 50 101 101**

4. Arrange the following number in descending order.

- (a) 33 923 673; 33 926 373; 33 923 763; 33 926 337  
**33 926 373; 33 926 337; 33 923 763; 33 923 673**  
(b) 41 561 152; 41 516 512; 41 615 215; 41 651 125  
**41 651 125; 41 615 215; 41 561 152; 41 516 512**

#### Comparing whole numbers

5. Compare the underlined digits and explain the relationship that you see in your own words.

- (a) 56 101 101 and 65 101 101

1 thousand and 1 Ones

The 1 in the thousands place is nine hundred and ninety-nine more than the 1 in the ones place.

(b) 97 172 851 and 97 271 158

7 Ten thousands and 7 million

The 7 in the ten thousand place is six million nine hundred and thirty thousand thousands less than the 7 in the millions place.

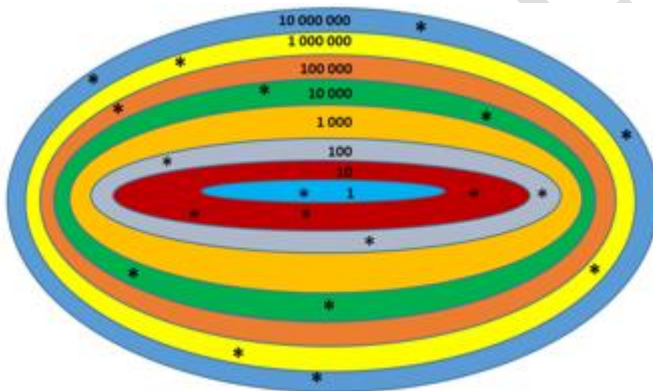
6. Compare these numbers by inserting  $<$ ;  $>$  or  $=$

(a) 90 299 992 ---- 90 292 999  $90\ 299\ 992 > 90\ 292\ 999$

(b) 11 723 723 ---- 11 723 723  $11\ 723\ 723 = 11\ 723\ 723$

### Representing whole numbers

7. Study the following diagram and answer questions.



(a) You have dropped some stars onto a game board. This was the result. If you add the numbers what is the total?

$$40\ 000\ 000 + 3\ 000\ 000 + 100\ 000 + 40\ 000 + 300 + 30 + 1 \\ = 43\ 140\ 331$$

8. You have been given these eight digits:

|   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|
| 1 | 7 | 9 | 3 | 0 | 2 | 4 | 5 |
|---|---|---|---|---|---|---|---|

(a) Use each digit once. Make the smallest 8 - digit whole number

10 234 579

(b) Use each digit once. Make the largest 8- digit whole number

97 543 210

(c) You can use a digit twice. Make the largest 8 - whole number

99 775 544

(d) You can use a digit twice. Make the smallest 8 - digit whole number.

10 023 457

## 1.2 Place value of whole numbers

1. Write the number (digit) in the correct column.

| No. | Number     | TM | M | HTh | Tth | T | H | T | O |
|-----|------------|----|---|-----|-----|---|---|---|---|
| (a) | 27 683 005 | 2  | 7 | 6   | 8   | 3 | 0 | 0 | 5 |
| (b) | 80 765 011 | 8  | 0 | 7   | 6   | 5 | 0 | 1 | 1 |

2. Write the number symbol for each number.

(a) Thirty million and five thousand

30 005 000

(b) Thirty million and five

30 000 005

(c) Thirty million and fifty thousand

30 050 000

## 1.3 Rounding off to the nearest 10, 100, 1 000, 10 000, 100 000 and 1 000 000 to estimate answers.

1. Round off each of the numbers as indicated in the following table.

| Number     | Nearest 10 | Nearest 1 000 | Nearest 10 000 | Nearest 100 000 |
|------------|------------|---------------|----------------|-----------------|
| 23 400 136 | 23 400 140 | 23 400 000    | 23 400 000     | 23 400 000      |
| 45 555 555 | 45 555 560 | 45 556 000    | 45 560 000     | 45 600 000      |

2. The first number below was rounded off to the second number. Was it rounded off to the nearest 10, 100, 1000, 10 000 or 100 000

(a) R1 676 241 was rounded off to R1 676 200

It was rounded off to the nearest 100

(b) R2 575 001 was rounded off to R3 000 000

It was rounded off to the nearest 1 000 000



## 1.4 Solving problems in context involving counting, ordering, comparing, representing, place value and rounding off.

1. My brother remembers seeing an affordable spacious block of flats for sale. He could not remember the exact price, but he knew it was an eight-digit number. He remembered that the last digit was a five and in the hundred thousand place was a three. He remembered that the first digit was one and the ten thousand place was four. In the millions place he remembered the number was two and, in the tens, place the number was three times three. Finally, he said the number nine was in the hundreds place.

What is the price of the block of flats?

$$5 + 300\,000 + 10\,000\,000 + 40\,000 + 2\,000\,000 + 90 + 900$$

| TM | M | HTH | Tth | Th | H | T | O |
|----|---|-----|-----|----|---|---|---|
| 1  | 2 | 3   | 4   |    | 9 | 9 | 5 |

The number is R12 340 995

2. A baker needs to have an estimate of how many biscuits are baked per week in her bakery. She normally rounds off the frequency of the tallies and then adds them. The frequencies are as follows: 7 653 492; 4 892 222; 6 600 154.

$$7\,653\,492 \approx 8\,000\,000$$

$$4\,892\,222 \approx 5\,000\,000$$

$$6\,600\,154 \approx 7\,000\,000$$

The estimated sum will be 20 000 000

3. You live in a street with 6 houses. Each family owns a car. The first family's car costs two hundred thousand rands. The second family's car costs forty-five thousand less than the first family's car. The third family's car costs double the first family's car. The fourth family costs a half million rands for their car. The fifth family paid fifty thousand rand more than the fourth family, and the last family paid twenty thousand rands less than the third family.

Calculate the price of each car and arrange the prices in descending order?

| Family | Cost of cars                    | Descending Order |
|--------|---------------------------------|------------------|
| 1      | R 200 000                       | R 550 000        |
| 2      | R 200 000 - R 45 000 = R155 000 | R 500 000        |

|   |                                          |           |
|---|------------------------------------------|-----------|
| 3 | $R\ 200\ 000 \times 2 = R\ 400\ 000$     | R 400 000 |
| 4 | $R\ 1\ 000\ 000 \div 2 = R\ 500\ 000$    | R 380 000 |
| 5 | $R\ 500\ 000 + R\ 50\ 000 = R\ 550\ 000$ | R 200 000 |
| 6 | $R\ 400\ 000 - R\ 20\ 000 = R\ 380\ 000$ | R 155 000 |

4. I am greater than seventy but less than ninety, the sum of my two digits is thirteen, my one's digit is a multiple of two. What number am I?

The number should be greater than 70 and less than 90 and the sum of its two digits must be equal to 13.

$$76 - (7 + 6 = 13)$$

$$85 - (8 + 5 = 13)$$

Ones digit a multiple of two, I am 76.

## UNIT 2

### WHOLE NUMBERS: ADDITION AND SUBTRACTION

#### 2.1 Use a range of techniques to perform written calculations with whole numbers.

##### Addition and Subtraction in columns

| Addition                                                                                                                | Subtraction                                                                                                          |
|-------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| $21\,768\,421 + 16\,593\,094 =$<br>$\begin{array}{r} 21\,768\,421 \\ + 16\,593\,094 \\ \hline 38\,361\,515 \end{array}$ | $23\,473\,546 - 22\,181\,362$<br>$\begin{array}{r} 23\,473\,546 \\ - 22\,181\,362 \\ \hline 1\,292\,184 \end{array}$ |

1. Calculate the following using the column method. Use your calculator to check your answers.

(a)  $763\,264 + 39\,652 = 802\,916$

(b)  $19\,054\,947 + 11\,593\,204 = 30\,648\,151$

(c)  $798\,456 - 573\,245 = 225\,211$

(d)  $482\,563 - 356\,278 = 126\,285$

(e)  $2\,839\,632 - 1\,573\,245 = 1\,266\,387$

(f)  $31\,695\,481 + 22\,397\,447 - 7\,356\,278 = 46\,736\,650$

$$\begin{array}{r} 31\,695\,481 \\ + 22\,397\,447 \\ \hline 54\,092\,928 \\ - 7\,356\,278 \\ \hline 46\,736\,650 \end{array}$$

##### Adding and subtracting using a calculator

The basics of using a calculator are the same whether one is using a standard or scientific calculator. Once one learns what the buttons do and how to use them for different calculations, one becomes more comfortable using a calculator whenever necessary. Remember to clear the calculator memory before starting a new problem.

##### Example 1

Keystroke sequence gives a guide on the exact keys to press

##### The calculator plan

$$124\,786 + 639\,536$$

##### Calculator keystroke sequence

1 2 4 7 8 6 + 6 3 9 5 3 6 =



$$757\ 869 - 295\ 425$$

|   |   |   |   |   |   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 7 | 5 | 7 | 8 | 6 | 9 | - | 2 | 9 | 5 | 4 | 2 | 5 | = |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

2. Calculate each of the following using the column method and check your answers using a calculator.

(a)  $15\ 987\ 364 + 49\ 790\ 683 = 65\ 778\ 047$

(b)  $76\ 263\ 785 + 57\ 953\ 064 = 134\ 216\ 849$

(c)  $87\ 623\ 735 - 37\ 470\ 184 = 50\ 153\ 551$

(d)  $97\ 654\ 231 - 74\ 933\ 923 = 22\ 720\ 308$

(e)  $31\ 511\ 547 + 14\ 923\ 789 - 21\ 573\ 245 = 24\ 862\ 091$

## 2.2 Problem solving involving addition and subtraction

1. The people in a village get their water from a nearby dam. On a certain day the dam contains 688 000 litres of water. The village people use about 595 000 litres of water per week.

(a) Will the community have enough water for the following week if no rains fall? Support your answer with calculations.

Total number of litres of water available = 688 000

Total consumption of water by village people per week = 595 000

Approximate amount of water left by the end of the week if no rainfalls:  
 $= 688\ 000 - 595\ 000 = 93\ 000$  litres

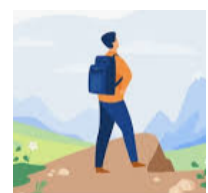
No, the community will not have enough water since only about 93000 litres will be available.

(b) Is it possible to calculate the exact amount of water that will be in the dam after a week?

It is difficult to calculate the exact amount of water that will be in the dam after a week because we do not know the source of the water. Even if the rain does not fall, water might still be coming into the dam from a river.

<https://www.freeepik.com/vectors/hiking>

2. Moses went hiking in the mountains for 6 days. He started on Monday and walked 6 kilometers and his last trip was on Saturday. Each day he walked one kilometer more than the day before. What will be the total distance walked after 6 days?



| Day                                                            | Kilometres walked per day               | Km/Day |
|----------------------------------------------------------------|-----------------------------------------|--------|
| Monday                                                         | 6 km                                    | 6 km   |
| Tuesday                                                        | 6 km + 1 km                             | 7 km   |
| Wednesday                                                      | 6 km + 1 km + 1 km                      | 8 km   |
| Thursday                                                       | 6 km + 1 km + 1 km + 1 km               | 9 km   |
| Friday                                                         | 6 km + 1 km + 1 km + 1 km + 1 km        | 10km   |
| Saturday                                                       | 6 km + 1 km + 1 km + 1 km + 1 km + 1 km | 11 km  |
| Total km walked                                                |                                         | 51 km  |
| The total distance walked by Moses after 6 days will be 51 km. |                                         |        |

3. A bank manager is working on the budget for a year. He has to try to keep the total expenditure on new office equipment below R700 000. He still has to budget for new computers that are badly needed, but this is what he has written so far:

| ITEMS                                     | COST                |
|-------------------------------------------|---------------------|
| 62 × new office chairs                    | R 54 020,00         |
| 38 × new computer screens                 | R 100 800,00        |
| 11 × new printers                         | R 141 600,00        |
| 15 × new tea trolleys                     | R 25 740,00         |
| 10 × new carpets for senior staff offices | R 144 000,00        |
| 120 × small plastic filing cabinets       | R 52 380,00         |
| 1 × new table for the boardroom           | R 48 000,00         |
| 18 × new chairs for the boardroom         | R 41 400,00         |
| TOTAL AMOUNT BUDGETED                     | <b>R 607 940,00</b> |

- (a) How much has the bank manager budgeted for printers and computer screens altogether?

R 100 800,00

+ R 141 600,00

R 242 400,00

[Use any suitable method]

The bank manager budgeted R 242 400.00 for printers and computer screens.

- (b) How much, in total, has the bank manager budgeted for chairs and tables?

R 54 020,00

[Any suitable method]

R 41 400,00

+ R 48 000,00

R 143 420,00

The total amount budgeted for chairs and tables is R 143 420,00

- (a) Work out the total cost of all the items the bank manager has budgeted for?

The total amount budgeted by the bank manager is R 607 940,00

- (b) Can the bank manager add new computers for 3 senior staff members if each computer costs R30 000,00?

Total budget – expenditure = R800 000 – R 607 940 = R 192 060

The cost of 3 computers is R 90 000

Therefore, the bank manager can still add 3 new computers.

4. Benjamin had the following expenses last month:

Movie R80; Bus R280; Donation to welfare R50; Stationery R65; Repairs to bicycle R225; New shirt R135; Ice cream R23. He receives R800 pocket money per month. He has R325 saved thus far.

Complete the budget to find out if he will have enough money or he will need to tap on his savings.

| Item                     | Estimated amount | Actual amount | Difference |
|--------------------------|------------------|---------------|------------|
| <b>Income</b>            | R 800            | R 800         |            |
| <b>Expenses</b>          |                  |               |            |
| Transport                | R200             | R280          | (R80)      |
| Stationery               | R100             | R65           | R35        |
| Clothes                  | R 100            | R135          | (R35)      |
| Food                     | R100             | R23           | R77        |
| Entertainment            | R50              | R30           | R20        |
| Donations                | R50              | R50           | R0         |
| Savings                  | R100             | R0            | R100       |
| Emergencies              | R100             | 225           | (R125)     |
| Estimated total Expenses | R 800            | R808          | (R8)       |

He will not have enough money. He will be short of R8 and he will have to tap into his savings.

## UNIT 3

### WHOLE NUMBERS: MULTIPLICATION AND DIVISION

#### 3.1 Use a range of techniques to perform written calculations with whole numbers

##### Multiples and Factors

1. Complete the following table

| Number | Factors                      | First three multiples |
|--------|------------------------------|-----------------------|
| 101    | 1; 101                       | 101; 202 and 303      |
| 285    | 1; 3; 5; 15; 19; 57; 95; 285 | 285; 570 and 855      |
| 136    | 1; 2; 4; 8; 17; 34; 68; 136  | 136; 272 and 408      |

2. Answer the questions based on the group of numbers given in the diagram.

(a) List the factors of 35

5; 7 and 35

(b) List the multiples of 7

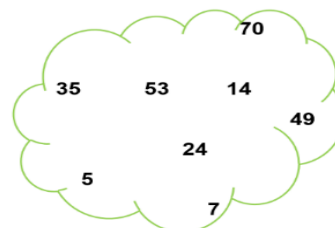
7; 14; 35; 49 and 70

(c) Which numbers are multiples of 14?

14 and 70

(d) Which numbers have 5 as a factor?

5; 35 and 70



3. Write down the answers as quickly as you can.

(a)  $4\,355 \times 1$

$4\,355 \times 1 = 4\,355$

(b)  $7227 \times 0$

$7227 \times 0 = 0$

(c)  $0 + 3\,000$

$0 + 3\,000 = 3\,000$

(d)  $5\,653 + 0$

$5\,653 + 0 = 5\,653$

4. Multiply and check the answers using a calculator.

(a)  $1\,117 \times 522$

$= 583\,074$

(b)  $3\,641 \times 183$

$= 666\,303$

(c)  $2\,954 \times 300$

$= 886\,200$

(d)  $5\,324 \times 219$

$= 1\,165\,956$

5. Divide and check the answers using a calculator.

(a)  $6\,478 \div 111 = 58,36$  or 58 rem 40

(b)  $5\,079 \div 263 = 19,31$  or 19 rem 82

(c)  $9\,864 \div 274 = 36$

(d)  $7\,822 \div 320 = 24,44$  or 24 rem 142

### 3.2 Problem solving involving division and multiplication

1. Mrs. Nkosi is buying apples from the market and sell them to school children. Last weekend she bought three boxes of apples; each box contained fifteen packets of twelve apples each. Each box costs her one hundred and eighty rands. She managed to sell eighty percent of the apples at one rands twenty cents before the rest were eaten by her grandchildren and their friends. Did she make a profit or a loss?

3 boxes of apples  $\times$  R180,00 = R540 (Buying price)

3 boxes  $\times$  15 packets = 45 packets (Each box has 15 packets)

45 packets  $\times$  12 apples = 540 apples (Each packet has 12 apples)

80 % of 540 apples sold will be 432 apples

432 apples  $\times$  R1,20 (price per apple) = R518,40

NB: Answer she made a loss of R21,60 because she bought the apples for R540 and after selling them she got R 518,40.

2. Sihle is buying pens for R5,80 each and he wants to make a 50% profit. How much must he sell them for?

Each pen is R5,80

50% profit will be:  $R5,80 \times 50\% = R\,2,90$

$(R5,80 + R2,90 = R8,70)$

He must sell each per for R8,70

3. The Minister of Mineral Resources and Energy announced that the price of petrol is going to decrease by 20%. If your budget was R4 500 a month for petrol, how much will you budget now?

Monthly budget for petrol before 20% decrease = R4 500

Decrease by 20% will be  $R4\,500 \times 20\% = R900$



New budget =  $R4\ 500 - R900 = R3\ 600$

I will budget R 3 600 a month.

4. My father wanted to purchase a laptop he saw online. When he made the offer to purchase, they told him that the price does not include VAT. The original price was R8 995. How much will the laptop cost including VAT.

Laptop price excluding VAT R8 995

VAT in South Africa is at 15%

VAT for laptop –  $R8\ 995 \times 15\% = R1\ 349,25$

The laptop price including VAT will be  $R8\ 995 + R1\ 349 = R10\ 344,25$

## UNIT 4

### RATIO AND RATE

Ratio is a way of comparing two or more quantities of the same kind. It is a way of comparing one part of a whole with another part of the same whole.

Rate is a way of comparing two different quantities. It is used to indicate how much of a given quantity is associated with one unit of another quantity.

#### Revision Activity

1. A school tap is leaking. 5 drops of water are lost per minute. How many drops will be lost in 15 minutes?

$5 \times 15 = 75$  drops of water.

2. Sihle has R32 and Ntokozo has R4. Write down the ratio of Sihle's money to Ntokozo's money.

$8:1$

3. If 3,5kg of bananas cost R12, what will 1000g of bananas cost?

$R12 \div 3,5\text{kg} = R3,43$  per 1000g

4. Peter and James share 28 sweets in the ratio of 3:1. How many sweets will each one get?

Peter will get 21 sweets. James will get 7 sweets

#### 4.1 Ratio in the context of speed, distance and time

##### Examples

Mpho collects apples in the orchard. She picks about 5 apples each minute. Approximately how many apples will Mpho pick in each of the following periods of time?

- |              |                                                               |
|--------------|---------------------------------------------------------------|
| ▪ 8 minutes  | $8 \text{ min} \times 5 \text{ apples} = 40 \text{ apples}$   |
| ▪ 11 minutes | $11 \text{ min} \times 5 \text{ apples} = 55 \text{ apples}$  |
| ▪ 15 minutes | $15 \text{ min} \times 5 \text{ apples} = 75 \text{ apples}$  |
| ▪ 20 minutes | $20 \text{ min} \times 5 \text{ apples} = 100 \text{ apples}$ |

Musa and Karabo also collect apples in the orchard, but they both work faster than Mpho. Musa collects apples at a rate of about 12 apples per minute and Karabo at a rate of about 15 apples per minute.

- Complete the following table to show approximately how many apples they will each collect in different periods of time.

| Time in minutes    | 1  | 2  | 3  | 8   | 10  | 20  |
|--------------------|----|----|----|-----|-----|-----|
| Mpho               | 5  | 10 | 15 | 40  | 50  | 100 |
| Musa               | 12 | 24 | 36 | 96  | 120 | 240 |
| Karabo             | 15 | 30 | 45 | 120 | 150 | 300 |
| The Three together | 32 | 64 | 96 | 256 | 320 | 640 |

The number of apples picked is directly proportional to the time taken.

If you filled the table correctly, you will notice that during any period of time, Karabo collected 3 times as many apples as Mpho. We can say that during any time interval, the **ratio** between the number of apples collected by Mpho and Karabo is **three is to one**, which can be written as **3:1**. For any period of time, the ratio between the number of apples collected by Musa and Mpho is **12:5**. Ratio should always be given in its simplest form.

- What is the ratio between the number of apples collected by Karabo and Musa in two minutes? **15: 12 or 5:4**
- What is the ratio between the number of apples collected by Karabo and Mpho in three minutes? Write the ratio in simplest form **45:15 or 3: 1**
- How much will each person get, if R14 400 is shared between two people in each of the following ways?
  - In the ratio 1: 3
    - Person 1: R3 600
    - Person 2: R10 800
  - In the ratio 5: 7
    - Person 1: R6 000
    - Person 2: R8 400
- R240 will be divided between Amahle and Mandy in the ratio 3: 5. This means Mandy gets R5 for every R3 Amahle gets. How much will Amahle and Mandy each get in total?
  - Mandy gets R5 from each portion so she gets  $30 \times R5 = R150$ .
  - Amahle gets  $30 \times R3 = R90$ .

5. Norman, Sello and Thabani worked in Mr Manzi's garden. Norman worked for 5 hours, Sello for 4 hours and Thabani for 3 hours. In total Mr Manzi paid the boys R600. If the money was divided in the ratio of 5:4:3, how much did each boy get?

Norman got R250, Sello got R200, Thabani got R150

## 4.2 Rate in the context of speed, distance and time

### Example

Michael drives from Mabeskraal in Rustenburg to Mbombela in Nelspruit at an average speed of 88 km/h. The distance between these two towns is 440 km.

How long did it take Michael to get to Mbombela?

### Solution

Speed = distance  $\div$  time

$\therefore$  Time = distance  $\div$  speed

$$= 440 \div 88$$

$$= 5 \text{ hours}$$

It took Michael 5 hours to get to Mbombela.

1. A motorist covers a distance of 425 km in exactly 5 hours
  - (a) Approximately how far did the motorist drive in 1 hour?  
85 km in one hour
  - (b) Approximately how far will the motorist drive in 9 hours?  
765km
  - (c) Approximately how long will the motorist need to travel 900km?  
 $900 \div 85 = 10,6 \text{ hours}$
2. Mary ran a 20km race. Her average speed was 8km per hour. How long did it take her to complete the race.  
 $20\text{km} \div 8 = 2,5 \text{ hours}$
3. If a car uses 7,5 litres of petrol for every 100 km that it travels, how many litres would it use for a trip of 620 km?  
46,5 litres
4. It takes 20 minutes for 10 teachers to mark 20 scripts. How long will it take 5 teachers to mark 10 scripts?

10 minutes

5. The marathon runner runs 16 km in 35 minutes. How long will it take him to walk the same distance if he walks 4 km in 12 minutes?

48 minutes

6. A car travels 46 km in 16 minutes. At the same average speed, how far will it travel in 1 hour 12 minutes?

207 km

7. If it takes 7 people to make 21 T-shirt per day, how long will it take 14 people?

12 hours

## UNIT 5

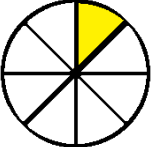
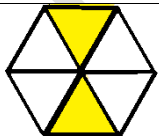
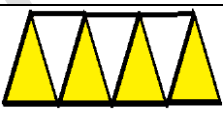
### COMMON FRACTIONS

#### Revision

1. Write the following as common fractions and simplify where possible.

|     |              |                                    |     |            |                                   |
|-----|--------------|------------------------------------|-----|------------|-----------------------------------|
| (a) | $1 \div 12$  | $\frac{1}{12}$                     | (b) | $2 \div 8$ | $\frac{2}{8}$ or $\frac{1}{4}$    |
| (c) | $11 \div 10$ | $\frac{11}{10}$ or $1\frac{1}{10}$ | (d) | $5 \div 9$ | $\frac{5}{9}$                     |
| (e) | 5: 9         | $\frac{5}{9}$                      | (f) | 12: 18     | $\frac{2}{3}$                     |
| (g) | 15:12        | $\frac{15}{12} = 1\frac{1}{4}$     | (h) | 25%        | $\frac{25}{100}$ or $\frac{1}{4}$ |
| (i) | 0,2          | $\frac{1}{5}$                      | (j) | 1,5        | $\frac{3}{2} = 1\frac{1}{2}$      |

2. Copy and complete the table below.

| No  | Diagram                                                                             | Name                    | Symbol          | Equivalent fractions                                 |
|-----|-------------------------------------------------------------------------------------|-------------------------|-----------------|------------------------------------------------------|
| (a) |   | One third or a third    | $\frac{1}{3}$   | $\frac{3}{9}$ or $\frac{2}{6}$ or $\frac{10}{30}$    |
| (b) |  | Three quarter           | $\frac{3}{4}$   | $\frac{6}{8}$ or $\frac{9}{12}$ or $\frac{12}{16}$   |
| (c) |  | One eighth or an eighth | $\frac{1}{8}$   | $\frac{3}{24}$ or $\frac{4}{36}$ or $\frac{2}{16}$   |
| (d) |  | Two sixths              | $\frac{2}{6}$   | $\frac{1}{3}$ or $\frac{4}{12}$ or $\frac{10}{30}$   |
| (e) |  | Four sevenths           | $\frac{4}{7}$   | $\frac{8}{14}$ or $\frac{20}{35}$ or $\frac{12}{21}$ |
| (f) |  | Twelve sixteenths       | $\frac{12}{16}$ | $\frac{3}{4}$ or $\frac{6}{8}$ or $\frac{15}{20}$    |

3. Say whether the following are **true** or **false**. Show how you arrived at an answer.

(a)  $1 = \frac{2}{2} = \frac{5}{5}$  **True**

(b)  $\frac{1}{3} = \frac{1}{6} + \frac{1}{6}$  **True**

(c)  $5,6 = 5\frac{6}{10}$  **True**

(d)  $20\% = \frac{1}{4}$  **False**

(e)  $\frac{3}{8} + \frac{1}{8} = \frac{1}{4}$  **False**

### 5.1 Compare and order common fractions, including tenths and hundredths

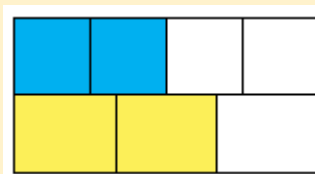
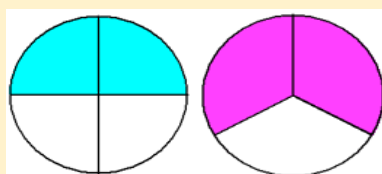
To compare and order common fractions, it is important to express them as fractions that have the same denominator. It then becomes easier to compare numerators of the fractions. If the fraction also includes a whole number, you would compare the whole number before finding the common fraction.

Note: A fraction chart is one of the most appropriate tools to use when comparing fractions.

#### Example:

Replace the \* by <;> or = in order to show the relationship between the two fractions  $\frac{2}{4} * \frac{2}{3}$

Represent both fractions in the diagram and then compare them. The diagrams clearly indicate that two-thirds are greater than two quarters, therefore  $\frac{2}{4} < \frac{2}{3}$



1. Replace the \* with < or > or = in the following:

$$(a) \frac{1}{3} * \frac{1}{6} \qquad \frac{1}{3} > \frac{1}{6}$$

$$(b) \frac{5}{10} * \frac{3}{5} \qquad \frac{5}{10} < \frac{3}{5}$$

$$(c) \frac{9}{10} * \frac{4}{5} \qquad \frac{9}{10} > \frac{4}{5}$$

$$(d) \frac{3}{4} * \frac{10}{12} \qquad \frac{3}{4} < \frac{10}{12}$$

$$(e) \frac{5}{9} * \frac{1}{4} \qquad \frac{5}{9} > \frac{1}{4}$$

2. Rewrite the following fractions so that they all have the same denominator and then arrange the fractions in an ascending order.

|                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| <p>(a) <math>\frac{40}{1000}; \frac{3}{10}; \frac{60}{1000}; \frac{2}{10}</math></p> <p><math>\frac{40}{1000}; \frac{300}{1000}; \frac{60}{1000}; \frac{500}{1000}</math></p> <p><math>\frac{40}{1000}; \frac{60}{1000}; \frac{300}{1000}; \frac{500}{1000}</math></p> <p><math>\therefore \frac{40}{1000}; \frac{60}{1000}; \frac{3}{10}; \frac{5}{10}</math></p> | <p>[Change fractions to thousands]</p> <p>[Order fraction in thousandths]</p> <p>[Final answer to be in original fraction]</p> |
| <p>(b) <math>\frac{5}{6}; \frac{2}{3}; \frac{4}{9}</math></p>                                                                                                                                                                                                                                                                                                      | <p><math>\therefore \frac{4}{9}; \frac{2}{3}; \frac{5}{6}</math></p>                                                           |
| <p>(c) <math>\frac{1}{2}; \frac{1}{4}; \frac{3}{5}</math></p>                                                                                                                                                                                                                                                                                                      | <p><math>\therefore \frac{1}{4}; \frac{1}{2}; \frac{3}{5}</math></p>                                                           |
| <p>(d) <math>\frac{4}{5}; \frac{1}{6}; \frac{1}{3}; \frac{7}{10}; \frac{19}{30}</math></p>                                                                                                                                                                                                                                                                         | <p><math>\therefore \frac{1}{6}; \frac{1}{3}; \frac{19}{30}; \frac{7}{10}; \frac{4}{5}</math></p>                              |
| <p>(e) <math>\frac{8}{9}; \frac{11}{12}; \frac{5}{6}; \frac{2}{3}; \frac{7}{8}</math></p>                                                                                                                                                                                                                                                                          | <p><math>\therefore \frac{2}{3}; \frac{5}{6}; \frac{7}{8}; \frac{8}{9}; \frac{11}{12}</math></p>                               |

## 5.2 Adding and subtracting fractions and calculating fractions of a whole

### Addition and subtraction of fractions

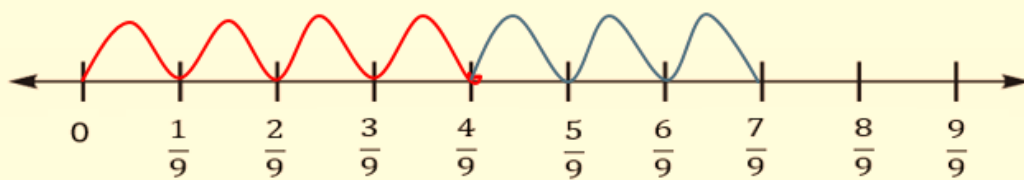
The process for adding and subtracting fractions with the same denominators is quite simple because we just need to work with the numerators. It can be done using a number line.



**Example:**

Calculate  $\frac{4}{9} + \frac{3}{9}$

The denominator shows into how many parts one whole must be divided. Represent this using a number line. In this case, there must be 9 parts.



Four ninths are represented by the red line and 3 ninths are represented by the blue line. The total is seven ninths.

Therefore  $\frac{4}{9} + \frac{3}{9} = \frac{7}{9}$  or 4 ninths + 3 ninths = 7 ninths.

1. Calculate the following. Simplify your answer where possible.

(a)  $\frac{3}{7} + \frac{2}{7} = \frac{5}{7}$

(c)  $\frac{5}{8} - \frac{3}{8} = \frac{2}{8} = \frac{1}{4}$

(b)  $\frac{5}{16} + \frac{6}{16} = \frac{11}{16}$

(d)  $\frac{74}{100} - \frac{23}{100} = \frac{51}{100}$

Sometimes when you add common fractions, the result is more than a whole. The answer becomes a whole number and a common fraction and we call that a **mixed number**.

**Example:**

$$\frac{5}{6} + \frac{3}{6}$$

$$= \frac{5}{6} + \left( \frac{1}{6} + \frac{2}{6} \right)$$

$$\left[ \frac{3}{6} \text{ is broken up into } \frac{1}{6} + \frac{2}{6} \right]$$

$$= \left( \frac{5}{6} + \frac{1}{6} \right) + \frac{2}{6}$$

[Associative property]

$$= \frac{6}{6} + \frac{2}{6}$$

$$= 1 + \frac{2}{6}$$

$$= 1 \frac{2}{6}$$

2. Calculate

(a)  $\frac{9}{10} + \frac{2}{10} = \frac{11}{10} = 1\frac{1}{10}$

(b)  $2\frac{1}{4} + 5\frac{2}{4} = 7\frac{3}{4}$

(c)  $3\frac{1}{3} + 2 = 5\frac{1}{3}$

(d)  $4\frac{8}{10} - 2\frac{7}{10} = 2\frac{1}{10}$

3. A team of road-builders built  $\frac{8}{12}$  km of road in one week, and  $\frac{10}{12}$  km in the next week. What is the total length of road that they built in the two weeks?

The total length of the road built is  $= \frac{18}{12} = 1\frac{1}{2}$  km

4. Margaret has  $5\frac{5}{8}$  bottles of cooking oil. She gives  $3\frac{3}{8}$  bottles to her friend Nelly. How much oil does Margaret have left?

The number of bottles left  $= 2\frac{1}{4}$

5. Sipho cycled  $3\frac{1}{4}$  km on his bike on Monday,  $5\frac{2}{4}$  km on Wednesday and  $4\frac{1}{4}$  km on Friday.

(a) How many kilometers has he covered altogether?

Sipho has travelled 13 km altogether.

(b) If he needs to complete 20 km in a week, how many more kilometers does he need to ride?

He still needs to travel 7km more.

### 5.3 Fraction of a whole

Calculating a fraction of a whole number is the same as multiplying the fraction by the whole number

### Example:

How much is  $\frac{4}{10}$  of R70?

- Instead of writing  $\frac{4}{10}$  of R70, we may write  $\frac{4}{10} \times \text{R70}$
- $\frac{1}{10}$  of 70 = 7, therefore  $\frac{4}{10}$  of 70 is the same as  $(\frac{1}{10} \text{ of } 70) \times 4 = 7 \times 4 = 28$

Calculate the following:

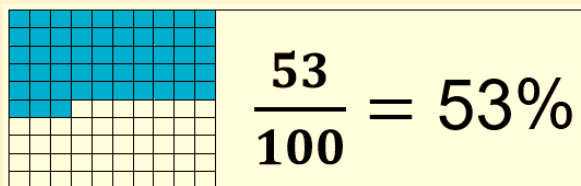
1. What is  $\frac{1}{3}$  of 12 sweets? **4 sweets**
2. How much is  $\frac{1}{100}$  of R400? **R4**
3. How much is  $\frac{7}{100}$  of R250? **R17,50**
4. How much is  $\frac{25}{100}$  of R600? **R150**
5. A friend bakes 12 scones and he gave Bob 4 scones. What fraction of scones did Bob eat? **Bob ate a third ( $\frac{1}{3}$ ) of the scones.**
6. There are 800 learners in a school.  $\frac{3}{5}$  of them are boys.
  - (a) How many learners are boys? **There are 480 boys.**
  - (b) How many learners are girls? **There are 320 girls.**
7.  $\frac{4}{4}$  of \_\_\_\_\_ balls = 20 balls. **20 balls**
8. What is 55% of R800 **R440**

## 5.4 Equivalence, comparing and ordering fractions

It is very easy to understand fractions expressed as **percentages**. Expressing a fraction as a percentage means the same as **converting** a fraction to a percentage. **Percent** means per hundred or **out of 100**. The percentage symbol % is a shorthand representation of this.

**Example:**

What is the meaning of 53 out of 100? It can be expressed in a number of ways as shown:



1. Complete the following table:

| Diagrammatic form | Common fraction                 | Percentage |
|-------------------|---------------------------------|------------|
|                   | $\frac{1}{2} = \frac{50}{100}$  | 50%        |
|                   | $\frac{3}{5} = \frac{60}{100}$  | 60%        |
|                   | $\frac{5}{25} = \frac{20}{100}$ | 20%        |
|                   | $\frac{6}{8} = \frac{75}{100}$  | 75%        |

2. Write each of the following percentages as a fraction. Simplify where possible.

(a) 12%  $= \frac{12}{100} = \frac{3}{25}$

(b) 44%  $= \frac{44}{100} = \frac{11}{25}$

(c) 63%  $= \frac{63}{100}$

$$(d) 75\% = \frac{75}{100} = \frac{3}{4}$$

$$(e) 125\% = \frac{125}{100} = 1\frac{1}{4}$$

3. Express each of the following as a percentage.

$$(a) \frac{12}{100} = 12\%$$

$$(b) \frac{12}{25} = 48\%$$

$$(c) \frac{75}{500} = 15\%$$

$$(d) \frac{600}{1000} = 60\%$$

4. Calculate:

$$(a) 20\% \text{ of } 25 = 5$$

$$(b) 60\% \text{ of R}15 = \text{R } 9,00$$

$$(c) 24\% \text{ of R}72 = 17\frac{7}{25}$$

$$(d) 20\% \text{ of R}300 = \text{R } 60,00$$

$$(e) 130\% \text{ of R}50 = \text{R } 65,00$$

$$(f) \text{ What percentage is 15 out of 25? } = 60\%$$

(g) If the cost price of an item is R900, 00, and the profit was 30%. What is the selling price?  
The profit is = R1 170

(h) A farmer makes 20% profit of R6 250 when she sells 25 of her sheep. If the profit is 20 %, how much is each sheep? R1 500 each sheep

## 5.5 Percentages including percentage increase and percentage decrease

Percentage increase or decrease is the difference between the final value and the initial value, expressed in the form of a percentage.

## Examples

| Percentage increase                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Percentage increase                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>Calculate percentage increase if the price of a train fare increases from R50 to R65.</b></p> <ul style="list-style-type: none"> <li>We first need to calculate the amount of the increase<br/> <math>R65 - R50 = R15</math></li> <li>The amount of the increase is R15 and the original price is R50. Therefore, the fraction of the increase is <math>= \frac{15}{50}</math>.</li> <li>To work out percentage increase, one needs to multiply the fraction by 100</li> </ul> <p>Therefore percentage increase</p> $= \frac{15}{50} \times 100\% = 30\%$ | <p><b>Calculate percentage decrease if the price of a dress decreases from R240 to R180 during a shop sale.</b></p> <ul style="list-style-type: none"> <li>We first need to calculate the amount of the decrease<br/> <math>R240 - R180 = R60</math></li> <li>The amount of the decrease is R60 and the original price is R240. Therefore, the fraction of the decrease is <math>= \frac{60}{240}</math>.</li> <li>To work out percentage decrease, one needs to multiply the fraction by 100</li> </ul> <p>Therefore percentage increase</p> $= \frac{60}{240} \times 100\% = 25\%$ |

Calculate the following:

- Items bought for R150 are sold at a loss for a price of R90. Calculate the percentage loss.

$$\text{Amount of decrease} = R150 - R90 = R 60,00$$

$$\text{Percentage loss/decrease} = \frac{60}{150} \times 100 = 40\%$$

- Joshua buys pencils for R3,30 and sells them for R5,50 each. What percentage profit does he make?

$$\text{Amount of increase} = R5.50 - R 3.30 = R 2,20$$

$$\text{Percentage increase} = \frac{2.20}{3.30} \times 100 = 66,67\%$$

- The production maize increases from 25 tons to 30 tons. The percentage increase in production of maize is \_\_\_\_\_.

$$\text{Amount of increase} = 30 - 25 = 5$$

$$\text{Percentage increase} = \frac{5}{25} \times 100 = 20\%$$

4. Find the percentage change of the first quantity with reference to the second quantity: 72 units, 60 units.

$$\text{Amount of decrease} = 72 \text{ units} - 60 \text{ units} = 12 \text{ units}$$

$$\text{Percentage decrease} = \frac{12}{72} \times 100 = 16,67\%$$

5. Due to an error, Michael misread a number 142 as 124. What is the error in reading the number?

$$\text{Amount of decrease} = 142 - 124 = 18$$

$$\text{Percentage decrease} = \frac{18}{142} \times 100 = 12,68\%$$

6. Tracksuits at a sports store are on sale for R476. If the original price was R560, what discount rate does this represent?

$$\text{Amount of decrease} = \text{R } 560 - \text{R } 476 = 84$$

$$\text{Percentage decrease} = \frac{84}{560} \times 100 = 15 \%$$

## 5.6 Calculating amounts if given percentage increase or decrease

**Example:** Increase R700 by 35%

$$\text{The amount after percentage increase is} = \text{R}700 + \text{R}700 \times \frac{35}{100}$$

$$= \text{R}700 + \text{R}245 = \text{R}945$$

OR

$$\text{The amount after percentage increase is} = \text{R}700 + (100\% + 35\%) \text{ of R}700$$

$$= 135\% \times \text{R}700 = \text{R}945$$

1. Answer the following:

(a) Increase R250 by 25%

$$\text{R}250 + \text{R}250 \times \frac{25}{100} = \text{R}250 + \text{R}62,50 = \text{R}312,50$$

OR

$$\text{The amount after percentage increase is} = (100\% + 25\%) \text{ of R}250$$

$$= 125\% \times R250 = R312,50$$

- (b) Increase R64 by 30%

The amount after percentage increase is = 130% of R64 = R83,20

- (c) Decrease 200 by 50%

The amount after percentage decrease is = (100% – 50%) of 200 = 50% of 200 = 100

- (d) Decrease 900 by 33%

The amount after percentage decrease is = (100 – 33)% of 900 = 67% of 900 = 603

2. Solve the following problems.

- (a) Thabo buys and sells t-shirts. He buys the shirts at a cost of R50 each. He sells them at a profit of 75%. What is his selling price?

The selling price is = 175% of R50 = R87,50

- (b) Colleen estimates that she spend R10 to prepare a hotdog. If she wants to sell her hotdogs at 50% profit, for how much must she sell them?

She must sell hotdogs at 150% of R10 = R15

- (c) A wholesaler decides to give a 30% potatoes discount on all food items. A bag of potatoes originally sold for R20. For how much will they sell it?

A bag of potatoes will sell at 70% of R20 = R14



## UNIT 6

### DECIMAL FRACTIONS

8,235 can be written as  $8,235 = 8 + 0,2 + 0,03 + 0,005$  in expanded notation.

The place values for these parts are as follows:

- the 8 represents the **units**
- the 2 represents the **tenths**
- the 3 represents the **hundredths**
- the 5 represents the **thousandths**

For example, the **value** of the 2 is two tenths and the **place value** of the 2 is **tenths**, because it represents **tenths**. A decimal can be written as a common fraction and or a percentage, for example  $0,69 = \frac{69}{100} = 69\%$

#### 6.1 Counting, ordering and comparing decimal fractions to at least two decimal places

1. Count forwards and backwards

(a) Count forwards in decimals in intervals of 0,4 from 0,4 to 2

0,4 ; 0,8 ; 1,2 ; 1,6 ; 2

(b) Count forwards in decimals in intervals of 0,05 from 0,27 to 0,52

0,27 ; 0,32 ; 0,37 ; 0,42 ; 0,47 ; 0,52

(c) Count forwards in decimals in intervals of 0,2 from 0,37 to 1,37

0,37 ; 0,57 ; 0,77 ; 0,97 ; 1,17 ; 1,37

(d) Count backwards in decimals in intervals of 0,07 from 0,82 to 0,47

0,82 ; 0,75 ; 0,68 ; 0,61 ; 0,54 ; 0,47

(e) Count backwards in decimals in intervals of 0,55 from 0,79 to -1,96

0,79 ; 0,24 ; - 0,31 ; - 0,86 ; -1,41 ; -1,96

2. Fill in < ; > or =

(a) 22,10 = 22,1

(b) 12,34 > 2,34

(c) 0,10 > 0,01

(d) 5.70 > 5.07

(e) 3,6 < 3,62

3. Arrange the following temperatures from coolest to hottest.

| Day            | M    | T    | W    | T  | F     | S     |
|----------------|------|------|------|----|-------|-------|
| Temperature °C | 14,8 | 23,6 | 13,2 | -2 | 18,32 | -4,33 |

| Day            | S     | T  | W    | M    | F     | T    |
|----------------|-------|----|------|------|-------|------|
| Temperature °C | -4,33 | -2 | 13,2 | 14,8 | 18,32 | 23,6 |

4. Arrange the following decimal fractions in descending order.

(a) 2,43; 1,76; 1,61; 1,04; 1,53; 1,21; 1,4

2,43 ; 1,76 ; 1,61 ; 1,53 ; 1,4 ; 1,21 ; 1,04

(b) 0,21; 0,06; 0,6; 0,16; 0,08; 0,8; 0,41

0,8 ; 0,6 ; 0,41 ; 0,21 ; 0,16 ; 0,08 ; 0,06

(c) 1,18; 1,06; 0,7; 2,057; 1,02; 1,2; 1,61

2,057 ; 1,61 ; 1,2 ; 1,18 ; 1,06 ; 1,02 ; 0,7

(d) 23,7; 23,07; 23,5; 23,90; 23,06

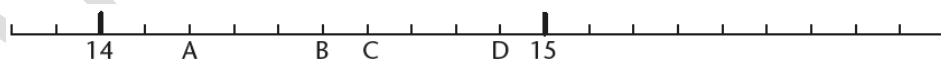
23,90 ; 23,7 ; 23,5 ; 23,07 ; 23,06

(e) 6,49; 6,93; 6,09; 6,6; 5,88; 5,8

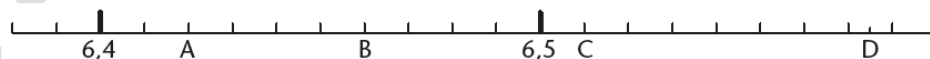
6,93 ; 6,6 ; 6,49 ; 6,09 ; 5,88 ; 5,8

5. Write the value of the letters A, B, C, D using a number line

(a) A = 14,2 ; B = 14,5 ; C = 14,6 ; D = 14,9



(b) A = 6,42 ; B = 6,46 ; C = 6,51 ; D = 6,575



## 6.2 Calculations with decimal fractions:

### Addition and subtraction of decimals

1. Add or subtract decimals

(a)  $23,45 + 12,5 + 1,677 = 37,627$

(b)  $17,12 + 0,23 + 2,01 = 19,36$

$$\begin{aligned} \text{(c)} \quad & -31,59 - 64,21 & = -95,8 \\ \text{(d)} \quad & 214,06 - 10,01 & = 204,05 \\ \text{(e)} \quad & 498,34 - 762,02 & = -263,68 \end{aligned}$$

2. Calculate.

$$\begin{aligned} \text{(a)} \quad & 2,15 + 8,21 - 7,21 & = 3,15 \\ \text{(b)} \quad & 9,43 - (3,61 + 1,14) & = 4,68 \\ \text{(c)} \quad & 1,75 \text{ m} - 1,2 \text{ m} + 0,256 \text{ m} & = 0,806 \text{ m} \\ \text{(d)} \quad & 359,88 - 12,27 + 221,6 & = 569,21 \\ \text{(e)} \quad & 28 \text{ cm} + 305,08 \text{ cm} - 111,052 \text{ cm} & = 222,028 \text{ cm} \end{aligned}$$

3. Multiply

$$\begin{aligned} \text{(a)} \quad & 4,8 \times 10 & = 48 \\ \text{(b)} \quad & 12,21 \times 10 & = 122,1 \\ \text{(c)} \quad & 144,36 \times 100 & = 14\,436 \\ \text{(d)} \quad & 9,49 \times 100 & = 949 \\ \text{(e)} \quad & 1,89 \times 10 & = 18,9 \end{aligned}$$

### 6.3 Recognize equivalence between common fraction, decimal fraction and percentage forms of the same number

1. Copy and complete the table

| Common fraction    | Decimal fraction | Percentages |
|--------------------|------------------|-------------|
| (a) $\frac{1}{2}$  | 0,5              | 50%         |
| (b) $\frac{8}{10}$ | 0,8              | 80%         |
| (c) $\frac{1}{10}$ | 0,1              | 10%         |
| (d) $\frac{3}{4}$  | 0,75             | 75%         |
| (e) $\frac{1}{3}$  | 0,33             | 33%         |
| (f) $\frac{3}{8}$  | 0,375            | 37,5%       |

## 6.4 Consolidation activity

Solve the following problems

1. A roll of material is 25 metres long. 1,35 metres of material is needed to make one dress.
  - (a) How many dresses can be made from the roll of material?  
**18 dresses can be made**
  - (b) How much material will be left over?  
**0,7metres will be left over.**
2. A farmer owns a land with an area of 25,05 hectares and is given a further 30,07 hectares of land. He then inherits 3 more hectares from his father. What is the total area of his land?  
**58,12 hectares**
3. Zamokuhle is an athlete. His most recent records for 100 m sprints are: 10,54 seconds; 9,33 seconds and 11,15 seconds respectively.
  - (a) What is the difference between Zamokuhle,s shortest record and longest record?  
**1,82 seconds**
  - (b) What is the sum of his most recent records?  
**31.02 seconds.**

## UNIT 7

### WHOLE NUMBERS: FINANCE

#### 7.1 Solving problems in financial context that involve profit, loss and discount

1. Copy and complete the table.

|     |                             |                |          |               |
|-----|-----------------------------|----------------|----------|---------------|
| (a) | Item                        | Original price | % profit | Selling price |
|     | Cakes                       | R80,00         | 50%      | R120,00       |
|     | Packet of sweets            | R35,00         | 25 %     | R43,75        |
|     | Packet of birthday balloons | R40,00         | 50 %     | R60,00        |

|     |                                      |                |        |               |
|-----|--------------------------------------|----------------|--------|---------------|
| (b) | Item                                 | Original price | % loss | Selling price |
|     | A bunch of bananas                   | R15,00         | 20 %   | R12,00        |
|     | A plate of samp, meat and vegetables | R 60,00        | 25%    | R45,00        |
|     | A box of sausage rolls               | R150,00        | 40 %   | R90,00        |

|     |           |            |            |            |
|-----|-----------|------------|------------|------------|
| (c) | Item      | Cost price | % discount | Sale price |
|     | Laptop    | R6 990     | 35 %       | R4 543,50  |
|     | Cellphone | R1 300     | 30,77 %    | R 900      |
|     | Printer   | R 2000     | 25 %       | R1 500     |

2. Solve the word problems

(a) Nelly bought a packet of ginger juice powder for R18.

The powder makes 30 litres of ginger juice. She sells the juice at a fundraising event at church for five rands a litre and hired a stall for R50. What is her percentage profit?

Price of ginger powder = R18

The packet makes 30 litres which are sold for R5 each

= 30 litres  $\times$  5 = R150

Profit = R150 - ( R18+ R50) = R82

Percentage profit =  $\frac{R82}{R150} \times 100 \% = 54,67\%$



- (b) Pinky wants a space to rent for her laundry business. She got a house for R4 000 per month. The landlord offered a 30 % discount, if she can pay cash for the first three months. How much must she pay in order to get the 30 % discount?

Rent for the house is R4 000 per month

The 30 % discount of R4 000 = R1 200

$R4\ 000 \times 3\ \text{months} = R12\ 000$

3 months discount =  $R1\ 200 \times 3 = R3\ 600$

Total discount =  $R12\ 000 - R3\ 600 = R8\ 400$

She must pay R8 400 the discount

- (c) Vuyo bought a delivery bakkie for R125 000. After driving it for three months, it gave him problems. He decided to take it back to the dealership and ask for the refund as it was still under warranty. After negotiating with the dealership, they refunded him R80 000. Calculate the percentage loss.

Original price for the bakkie = R125 000

Refund from the dealership = R80 000

Loss =  $R125\ 000 - R80\ 000 = R45\ 000$

Percentage Loss =  $\frac{R45\ 000}{R125\ 000} \times 100 = 36\%$

The percentage loss is 36%

## 7.2 Solving problems in financial context that involve budgets

Mandla wants to develop his barbershop business. He asked the District Youth Development Agency to fund him. They requested him to provide his budget so that they can assist him. He does not have a budget but he knows the money he receives per month and how he spent it.

1. Complete Mandla's budget using the following information:
  - (a) Taxi fare to the barbershop per day is thirty rands per day.
  - (b) Refreshments is eight hundred rands per month
  - (c) He receives three hundred and fifty rand social grant.
  - (d) He offers fifty rands each Sunday to his church.
  - (e) He receives a four hundred rands allowance from his parents.
  - (f) He spends one thousand two hundred rands a month on toiletries.

- (g) The monthly rent for his barber shop is one thousand five hundred rands.
- (h) The monthly maintenance for his machines is seven hundred rands.
- (i) His weekly income is one thousand eight hundred and fifty rands.

| INCOME              |               | EXPENDITURE              |               |
|---------------------|---------------|--------------------------|---------------|
| Item                | Amount        | Item                     | Amount        |
| Social grant        | R350          | Transport                | R900          |
| Allowance           | R400          | Refreshments             | R800          |
| Income              | R7 400        | Church offering          | R200          |
|                     |               | Toiletries               | R1 200        |
|                     |               | Rent                     | R1 500        |
|                     |               | Machine maintenance      | R700          |
| <b>Total income</b> | <b>R8 150</b> | <b>Total expenditure</b> | <b>R5 300</b> |

2. Mr Vanda's grass cutting business is making **nineteen thousand two hundred rands** a month. Weekly he spends five hundred rands on petrol, eighty rands on two-stroke oil, and five hundred rands for his helper. Monthly he spends seven hundred rands for the string and eight thousand for his family's needs. Compile his monthly budget and decide if the **nineteen thousand two hundred rands** will be sufficient for his monthly expenses.

| INCOME                 |                | EXPENDITURE    |                |
|------------------------|----------------|----------------|----------------|
| Item                   | Amount         | Item           | Amount         |
| Grass cutting business | R19 200        | Petrol         | R2 000         |
|                        |                | Two-stroke oil | R320           |
|                        |                | Helper         | R2 000         |
|                        |                | String         | R700           |
|                        |                | Family needs   | R8 000         |
| <b>Total</b>           | <b>R19 200</b> |                | <b>R13 020</b> |

The R19 200 is sufficient for his monthly expenses as the add up to R13 020

## 7.3 Solving problems in financial context that involve accounts and loans

Solve the following problems.

1. Elise borrowed R25 000 from a cash loan for a period of eighteen months, at a simple interest rate of 12 % . How much must she repay at the end of the term.

Amount borrowed R25 000, interest 12%, period 18 months

The formula for calculating the simple interest is:

$$SI = P \times \frac{R}{100} \times T$$

$$R25\,000 \times 12\% \times 18 = R54\,000$$

At the end of the term, she must repay R54 000.

2. Study Mr Fowler's cellphone account statement and answer questions

**Statement**



Mr D Fowler  
P O Box 13426  
Cape Town  
8000

**Inquiries:**  
Tel: 021 344 7453  
Email: cellphone@acc.net

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Email address      dfowler@gmail.com  
Statement date      15 - 07 - 2022  
Billing account      BA 222769  
Previous account      PA 222768  
Primary contact      021-3432121

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**Statement**

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**Statement includes all transactions from**      14 - 06 - 2022 to 14 - 07 - 2022

|                  |   |                            |   |                    |   |                               |   |                        |
|------------------|---|----------------------------|---|--------------------|---|-------------------------------|---|------------------------|
| Previous balance | + | Adjustments (credit/debit) | + | Charges this month | - | Payments received (thank you) | = | Amount due (VAT incl.) |
| R2,068.31        |   | R0.00                      |   | R1,031.18          |   | -R1,039.01                    |   | R2,060.48              |

Account aging      Total overdue : R 0.00

|                  |                 |                 |                 |             |                           |
|------------------|-----------------|-----------------|-----------------|-------------|---------------------------|
| 120 days overdue | 90 days overdue | 60 days overdue | 30 days overdue | Current due | Total due (incl. overdue) |
| R0.00            | R0.00           | R0.00           | R0.00           | R1,031.18   | R2,060.48                 |

**Statement details below are VAT inclusive**

| Date                     | Transaction  | Reference         | Amount    | Due Date   | Open Amount       |
|--------------------------|--------------|-------------------|-----------|------------|-------------------|
| 14-06-2022               | Invoice      | I500000162876413  | 1,031.18  | 01-02-2022 | 1,031.18          |
| 14-07-2022               | Past Invoice | I500000149977609  | 1,029.30  | 01-01-2022 | 1,029.30          |
| 01-08-2022               | PAYMENT      | P5000000180732934 | -1,039.01 |            |                   |
| <b>Total amount due:</b> |              |                   |           |            | <b>R2 060, 48</b> |



(a) What is Mr Fowler's cellphone number?

Mr Fowler cellphone number is 021-3432121

(b) What is his previous balance?

His previous balance is R2 068, 31

(c) What is the billing period for this statement?

The billing period for this statement is 14-06-2022 to 14-07-2022

(d) How much is due on the first of August?

Total amount due is R2 060,48

(e) Does the amount due include or excludes VAT?

The amount includes VAT

FINAL DRAFT COPY

# TERM 2

## UNIT 1

### WHOLE NUMBERS: COUNTING, ORDERING, COMPARING, REPRESENTING AND PLACE VALUE

#### 1.1 Counting, Ordering, Comparing and Representing

##### Counting

1. Complete the following by counting forwards and backwards

(a) 22 132 570; 22 133 815; 22 135 060; 22 136 305

(b) 8 345 271; 8 345 304; 8 345 337; 8 345 370

(c) 32 746 188; 32 746 165; 32 746 142; 32 742 119

##### Ordering

2. Write the numbers in ascending order.

(a) 331 136; 329 245; 21 645; 360 001; 331 111; 392 245

21 645; 331 111; 331 136; 329 245; 360 001; 392 245

(b) 5 561 344; 5 156 344; 5 561 434; 5 516 344; 5 561 443

5 156 344; 5 516 344; 5 561 344; 5 561 434; 5 561 443

##### Comparing

3. Replace the \* with  $>$  ;  $<$  or  $=$  to make the mathematical sentence true.

(a) 91 457 132 \* 91 457 213

91 457 132  $<$  91 457 213

(b) 9 111 161 \* 1 101 615

9 111 161  $>$  1 101 615

(c) 132 570 \* 132 570

132 570  $=$  132 570

#### 1.2 Rounding-off

1. Study the picture and indicate if the boy in the middle should move backwards or forwards if he rounds off his position to the nearest 5.



He will move backwards to 8 345

2. Copy and complete the table.

| Round off to the nearest | 5           | 10 000      | 100 000     | 1 000 000   |
|--------------------------|-------------|-------------|-------------|-------------|
| 47 526                   | 47 525      | 50 000      | 0           | 0           |
| 235 079                  | 235 080     | 240 000     | 200 000     | 0           |
| 378 900 352              | 378 900 350 | 378 900 000 | 378 900 000 | 379 000 000 |
| 976 315 463              | 976 315 465 | 976 320 000 | 976 300 000 | 976 000 000 |
| 13 415 978               | 13 415 980  | 13 426 000  | 13 400 000  | 13 000 000  |

### 1.3 Place value

1. Write the place values for each of the underlined digits:

- (a) 97 832 435      Millions
- (b) 490 841      Hundred thousands
- (c) 6 487 032      Ten thousands
- (d) 11 673 880      Tens
- (e) 88 091 023      Hundreds

2. Write the value of each underlined digit:

- (a) 44 567      40 000
- (b) 65 567 567      5 000 000
- (c) 11 757 458      10 000 000
- (d) 89 561 337      500 000
- (e) 70 702 670      2 000

### 1.4 Problem Solving

Solve the following problems.

1. Peter has four digits and they are 5, 2, 7, 6. Make the highest and smallest four digit numbers he can make without repeating the numbers.

Highest is 7652 and smallest is 2567

2. Bontle bought shoes for R249 and a dress for R399. Round off each amount to the nearest ten rand.

Shoes cost R250 and dress cost R400

3. I am a six-digit number, the sum of my ten thousand, hundred thousand and thousands digits is an odd number which is a square root of one hundred and twenty-one and a factor of fifty-five. The difference between my tens and my unit digits is two. Who am I?

Possible response 344 697

4. Given the number 335 436.

(a) Find the product of the tens digit and the hundreds digit

$$4 \times 3 = 12$$

(b) Add the difference between ten thousand digit and hundred thousand digit to the product in (a).

$$12 + 0 = 12$$

5. You have a five-digit number, after rounding it off to the nearest 1000, the answer is a six-digit number. What was the possible original number?

99 543 (accept any reasonable correct answer)

6. The cyclists travelled 2 745 m on the first day, another 679 456cm on the second day and 345 445cm on the third day.

(a) On which day did they cycle the furthest?

On the second day

(b) On which day did they cycle the least?

The first day

(c) Round off the total kilometers cycled to the nearest thousandths

$$12,994 \text{ km} \approx 13 \text{ km}$$

## UNIT 2

### WHOLE NUMBERS: FINANCE

#### 2.1. Solve problems that involve whole numbers, percentages and decimal fractions in financial contexts

##### Value Added Tax (VAT)

The acronym VAT stands for Value Added Tax. It is a tax levied on goods and services at each stage of production and distribution of sale from the primary to the secondary consumer. A value added tax, known in some countries as goods and services tax, is a type of tax that is assessed incrementally. It differs from country to country, in South Africa, the VAT standard rate is currently 15%. In South Africa some goods are zero rated.

##### What is zero-rated VAT South Africa?

Zero rated goods are products that are exempt from value added taxation (VAT).

##### Examples of VAT – exempt goods (zero rated goods)

Dried fruit, samp, maize meal, rice, brown bread, rated goods, edible legumes, mealie rice, pilchards in tins, eggs, milk, dried mealies, dairy powder milk, dried and powered milk, brown wheaten meat, bread flour and sanitary pads

##### Who must pay VAT in South Africa?

All consumers are legible to pay VAT directly or indirectly. When a company sells goods or services, they generally charge VAT to their customers as part of their selling price. It is mandatory for **any business** to register for VAT if the income earned in any consecutive twelve-month period exceeded or is likely to exceed R1 million. Any business may choose to register voluntarily if the income earned, in the past twelve-month period, exceeded R50 000

1. How much will these items be including VAT? Some of them are zero rated.

| No. | Items                     | Price excluding VAT | Price including VAT |
|-----|---------------------------|---------------------|---------------------|
| (a) | 5 <i>ℓ</i> cooking oil    | R 256,90            | R 256,90            |
| (b) | 500 <i>g</i> margarine    | R 26,50             | R 30,48             |
| (c) | 500 <i>mℓ</i> liquid soap | R 24,99             | R28,74              |

| No. | Items             | Price excluding VAT | Price including VAT |
|-----|-------------------|---------------------|---------------------|
| (d) | 75 ml tooth paste | R35,95              | R41,34              |
| (e) | 5 kg rice         | R89,99              | R89,99              |
| (f) | 1 kg sausage      | R 99,95             | R103,49             |
| (g) | 2 l milk          | R37, 49             | R37, 49             |
| (h) | 500 g yoghurt     | R18,99              | R21,83              |

The prices that stay the same are zero rated.

- (a) Monique buys a hairdryer for R395 excluding VAT. How much will she pay in total including VAT?

Price excluding VAT is R 395

$$\begin{aligned}
 \text{VAT} &= \text{R } 395 \times 15\% \\
 &= \text{R } 59,25 \\
 \text{Price including VAT} &= \text{R } 395 + \text{R } 59,25 \\
 &= \text{R } 454,25
 \end{aligned}$$

She will pay R 454,25 in total.

- (b) Sizwe bought an iPhone 11 for R13 000 excluding VAT. He pays cash and gets a 5% cash discount. How much will he pay in total including VAT?

Price of the phone excluding VAT is R13 000

$$\begin{aligned}
 5\% \text{ discount} &= \text{R } 13\,000 \times 5\% \\
 &= \text{R } 650 \\
 \text{Price minus } 5\% &= \text{R } 13\,000 - \text{R } 650 \\
 &= \text{R } 12\,350 \\
 \text{VAT} &= \text{R } 12\,350 \times 15\% \\
 &= \text{R } 1\,852,50
 \end{aligned}$$

$$\begin{aligned}
 \text{Price including VAT} &= \text{R } 12\,350 + \text{R } 1\,852,50 \\
 &= \text{R } 14\,202,50
 \end{aligned}$$

He will pay R 14 202,50 including VAT?



## 2.2. Simple interest, hire purchase and exchange rates

### Interest

In finance, **interest** is the money charged for borrowing money. It is expressed as a percentage of the borrowed amount. There are different types of interest e.g. simple interest and compound interest.

If you have some money, you could either keep it in your piggy bank account with no interests. If you deposit it into a bank account, you are effectively lending money to the bank. Because you are lending the bank money, you expect some extra money back. This is known as interest. Similarly, if you borrow money from a bank, then you can expect to pay interest on loan. Interest is charge at a percentage of the money owed over the period of time it takes to pay back the loan. This means that the longer the loan exists; the more interest will have to be paid on.

### What is simple interest?

**Simple interest** is the interest calculated only on the initial amount that you invested over a certain period. When someone lends money to someone else the borrower usually pays a fee to the lender. This fee is called 'interest'. Simple or flat rate interest is payed each year as a fixed percentage of the amount borrowed or lent at the start.

**The simple interest formula** is as follows:

Simple Interest = Principal amount (P)  $\times$  Rate (R)  $\times$  Time (T), where:

- Simple Interest: is the total amount of interest to be paid
- Principal (P): is the amount lent or borrowed
- Rate (R): is the percentage of the principal charged as interest each year
- Time (T): is the period in years to pay back the loan.

### What is Hire Purchase?

**Higher purchase** is a system buyer where a buyer pays for an asset in regular instalments while enjoying the use of it. The buyer has an option to make an initial down payment and pays the balance plus interests in instalments. During the repayment period ownership of the item does not pass to the buyer (it is on 'hire').

Upon the full payment of the loan plus interests the title passes to the buyer (the purchase is now complete).

(Gr 8 DBE workbook 1 Pa 20)

The repayment period is normally the same as the production life of the item bought on hire purchase. For example, a farmer buys a tractor and pays it off over five years. After five years he typically has to replace the tractor.



Solve the following problems.

1. Ali borrowed R 5 000 from a bank over a period of 3 years at a simple annual interest rate of 8,5%. How much must he repay at the end of the term period

$$\begin{aligned} SI &= PRT \\ &= R5\,000 \times 8,5\% \times 3 \\ &= R1\,275 \\ &= R5\,000 + R1\,275 \\ &= R6\,275 \end{aligned}$$

He must pay R6 275 at the end of the term.

2. Bulela wants to buy a racing bicycle for R 2 400 cash. He is having R1 800. What annual rate return will he need to invest to accomplish his goal in two years?

$$\begin{aligned} SI &= PRT \\ P &= R1\,800 \\ SI &= R2\,400 \\ R &= \frac{SI}{PT} \times 100 \\ R &= R2400 \times 100 \div (1\,800 \times 2) \\ &= 66,6\% \end{aligned}$$

He will need to invest 66,6 % in order to accomplish his goal.

3. Karin, a farmer wants to buy a new tractor. The tractor costs R340 000 including VAT. She can pay a deposit of R50 000. She decides to buy the tractor on hire purchase over 60 months at a simple interest rate of 10%.

Cost of the tractor minus the deposit will give us the principal amount for the hire purchase.

$$= R340\,000 - R50\,000 = R290\,000$$

(a) How much interest will she pay?

$$\begin{aligned} SI &= PRT \\ &= R290\,000 \times 10\% \times 5 \\ &= R145\,000 \end{aligned}$$

She will pay R145 000 interest.

(b) How much will she pay in total for the tractor over 60 months?

Principal amount + interest

$$R290\,000 + R145\,000$$

$$= R\,435\,000$$

She will pay R435 000 in a period of 60 months

(c) What will her instalment be?

**Total instalment**  $\div$  period

$$R435\,000 \div 60 \text{ months}$$

$$= R\,7\,250$$

Her monthly instalment will be R 7 250

### Exchange Rate

An **exchange rate** is the current market price for which one currency can be changed for another. Currency is money in the form of paper or coins usually issued by a government and generally accepted at its face value and as a method of payment. Different countries have their own currencies.

A currency gets stronger when the money is invested in the country. When we buy products that are made in South Africa, we are investing in South African business and keeping the money in the country. When we buy products imported from other countries, we are investing money in those countries and as a result,

the rand will weaken. The more South African products we buy, the greater the demand for them will be and more jobs will become available for South Africans. The price of oil increases when the South African rand weakens. This is because when the rand is weaker, we can buy less of other currencies with the same amount of money.

Exchange rates are determined by factors such as, economic growth, interest rates, confidence, inflation rates, etc. The Central Bank is mandated by Government to determine exchange rates. The rate is set against another major world currency.

| Country                  | Currency             | Sign | Code       |
|--------------------------|----------------------|------|------------|
| South Africa             | Rand                 | R    | ZAR        |
| United States of America | United States Dollar | \$   | USD / US\$ |
| European Union           | Euro                 | €    | EUR        |
| United Kingdom           | Pound                | £    | GBP        |
| Canada                   | Canadian Dollar      | \$   | CAD        |
| Australia                | Australian Dollar    | \$   | AUD / A\$  |

Solve the following problems.

4. Rita lives in Umlazi in South Africa, her brother lives in London in England. His brother deposited her £70 for her birthday. How many South African Rands (ZAR) can she buy with her birthday money? (Rate £1 = R20,26).

$$£1 = R20,26$$

$$\text{therefore } £70 = 70 \times R20,26$$

$$= R1\,418,20$$

She will buy R1 418,20 (ZAR)

5. Collen earns R800 from his part – time job. He wants to buy a t- shirt which costs \$30 USD. After his purchase how much money will he have left in ZAR? (Rate \$1 = R17,09).

$$R\,800 \div R17,09 = \$46,81$$

$$\$46,81 - \$30 = \$16,81$$

$$= \$16,81 \times R17,09$$

$$= R287,28$$

He will be left with R287,28

6. Lulamile wants to finish her engineering course in Germany, the total amount needed for the course is R150 000. BMW gave her a sponsor of €8 000. How much in rands does Lulamile needs to raise now? (Rate €1 = R17,22).

Money needed for the course: R150 000

Sponsor from BMW: €8 000

Rate €1 = R17,22

€8000 changed to rands =  $R17,22 \times €8000 = R137\,760$

She still needs to raise:  $R150,000 - R137\,760 = R12\,240$

## UNIT 3

### ALGEBRAIC LANGUAGE

An **algebraic expression** is made up of numbers, variables and operation signs. Algebraic expressions can also be described in words, flow diagrams and tables.

**Equivalent expressions** are algebraic expressions that have different sequences of operations, but have the same numerical value for any given value of  $x$ .

A **convention** is something that people have agreed to do in the same way. We write  $3 \times x$  rather than  $x \times 3$ , and we write  $3x$  rather than  $x3$ . In algebraic expressions, when multiplying with variables the multiplication sign is often not shown: We normally write  $4x$  instead of  $4 \times x$  and  $4(x - 5)$  instead of  $4 \times (x - 5)$ .

#### Possible misconceptions:

- $x + x = 2x$  and **NOT**  $x^2$  (note the convention is to write  $2x$  rather than  $x2$ )
- $x^2 + x^2 = 2x^2$  and **NOT**  $2x^4$

1. Translate the following sentences into algebraic language

- (a) The sum of a number and eight is multiplied by six.  $6(m + 8)$
- (b) The product of a number and 3 is increased by 22.  $(t \times 3) + 22$
- (c) Add the sum of  $a$  and  $b$  to the product of  $c$  and  $d$ .  $(a + b) + (c \times d)$
- (d) 7 multiplied by  $h$   $7h$
- (e) The product of a number and 34 added to 12.  $12 + 34b$
- (f) The difference between two hundred and 137.  $200 - 137$
- (g) Divide a number by 2 and subtract 9 from the answer.  $(n \div 2) - 9$
- (h) One third of  $b$ .  $\frac{1}{3}b$
- (i) Add 2 to the product of double a number and seven.  $(2r \times 7) + 2$
- (j) The sum of a number squared and its square root.  $w^2 + \sqrt{w}$
- (k) Seven less than the cube root of a number.  $\sqrt[3]{m} - 7$
- (l) The difference between the square of a number and three times the cube root of the number.  $c^2 - 3\sqrt[3]{c}$
- (m) What is different between  $3(x + 5)$  and  $3x + 5$ .
- $3(x + 5)$       The sum of a number and five multiplied by three
- $3x + 5$       Add 5 to a product of a number and 3

2. Interpret and write the following rules (algebraic expression) in words

- (a)  $\frac{1}{2}n + 2$       Half a number plus 2.  
 (b)  $10(n + 3)$       Multiply the sum of the number and 3 by 10.  
 (c)  $4n - 5$       Multiply a number by 4 and subtract 5 from the answer.  
 (d)  $(l \times 3) + 22$       The product of a number and 3 is increased by 22.  
 (e)  $w^2 + \sqrt{w}$       The sum of a number squared and its square root.

3. Complete the table

| No. | Expression         | Variable | Constant      | Coefficient of         |            |
|-----|--------------------|----------|---------------|------------------------|------------|
| (a) | $4x^2 + 6x - 10$   | $x$      | -10           | $x^2$ is 4             | $4x$ is 6  |
| (b) | $6s^3 + s^2 + 5$   | $s$      | 5             | $s^3$ is 6             | $s^2$ is 1 |
| (c) | $4p^7$             | $p$      | 0             | $p^7$ is 4             |            |
| (d) | $\frac{y^2+1}{2}$  | $y$      | $\frac{1}{2}$ | $y^2$ is $\frac{1}{2}$ |            |
| (e) | $\frac{k}{3} - 15$ | $k$      | -15           | $k$ is $\frac{1}{3}$   |            |

4. Consider the expression  $3x^2 + 4y - 4xy + 7y^2 + 22$

- (a) What is the coefficient of the fourth term?      7  
 (b) What is the constant in this expression?      22  
 (c) What is the variable in the second term?       $y$

5. Rewrite the following expressions, applying algebraic conventions:

- (a)  $3 \times (11 - x) + (2 \times x + 3) \times 13$        $33 - 3x + 26x + 39$   
 (b)  $-2 \times a + b$        $-2a + b$   
 (c)  $a2$        $2a$

6. Solve by inspection and check each answer by substitution

| No. | Expression    | Inspection | Substitution   |  |
|-----|---------------|------------|----------------|--|
| (a) | $v + 10 = 44$ | $v = 34$   | $34 + 10 = 44$ |  |
| (b) | $n - 4 = 6$   | $n = 10$   | $10 - 4 = 6$   |  |

|     |                        |                   |                                |              |
|-----|------------------------|-------------------|--------------------------------|--------------|
| (c) | $7t + 8 = 9$           | $t = \frac{1}{7}$ | $7 \times \frac{1}{7} + 8 = 9$ | $1 + 8 = 9$  |
| (d) | $\frac{x}{5} + 9 = 12$ | $x = 15$          | $\frac{15}{5} + 9 = 12$        | $3 + 9 = 12$ |
| (e) | $x^2 = 16$             | $x = 4$           | $4^2 = 16$                     |              |
| (f) | $a^3 = 125$            | $a = 5$           | $5^3 = 125$                    |              |

7. Solve the following equations using trial and improvement

(a)  $x - 6 = 10$

| $x$      | $x - 6$       | Comment         |
|----------|---------------|-----------------|
| 7        | $7 - 6 = 1$   | Smaller than 10 |
| 10       | $10 - 6 = 4$  | Smaller than 10 |
| 14       | $14 - 6 = 8$  | Smaller than 10 |
| $x$      | $x - 6$       | Comment         |
| 18       | $18 - 6 = 12$ | Bigger than 10  |
| 15       | $15 - 6 = 9$  | Closer to 10    |
| 16       | $16 - 6 = 10$ | Correct answer  |
| $x = 16$ |               |                 |

| No. | Equation           | Solution |
|-----|--------------------|----------|
| (b) | $x^3 - x = 120$    | $x = 5$  |
| (d) | $4a - 3a + 5 = 12$ | $a = 7$  |

8. Substitute to determine the numerical value of the following expressions.

| No. | Expression                        | Substitute by: |    |    |    |   |
|-----|-----------------------------------|----------------|----|----|----|---|
|     | $x$                               | -2             | -1 | 0  | 1  | 2 |
| (a) | $3x - 2$                          | -8             | -5 | -2 | 1  | 4 |
| (b) | $3x + 2 + 2x - 3$                 | -11            | -5 | -1 | 4  | 9 |
| (c) | $(3x + 2)(2x - 3)$                | 28             | 5  | -6 | -5 | 8 |
| (d) | $\frac{(3x + 2)(2x - 3)}{3x + 2}$ | -7             | -5 | -3 | -5 | 1 |
| (e) | $\frac{6x^2 - 5x - 6}{3x + 2}$    | -7             | -5 | -3 | -1 | 1 |



9. Solve the following problems

- (a) Lebo is a florist. She makes a certain number of bouquets, and sells them at R40 each. The money received from the sale of bouquets is R2 000. How many bouquets did Lebo make?

The unknown that we want to find is the number of bouquets Lebo had. So let the total number of bouquets be  $x$

The money received from the sale is:

$$x \times R40 = R2\ 000 \text{ divide both sides by } R40$$

$$x = 50$$

Lebo made 50 bouquets.

- (b) A builder delivers a number of tons of timber to a customer. He has a large truck that holds three tons of timber and a small truck that holds two tons of timber. The driver makes six journeys with the large truck and one journey with the small truck. Each time the truck is fully loaded. How many tons of timber did the builder deliver to his customer?

Let the total number of tons of timber delivered be  $x$

The large truck holds three tons and makes six journeys:  $(3 \times 6)$

The small truck holds two tons and makes one journey:  $(2 \times 1)$

$$(3 \times 6) + (2 \times 1) = x$$

$$20 = x$$

- (c) Naledi sells six cups of hot chocolate for R30. She writes the record of her sales in the equation  $6x = 30$ . If  $x$  is the cost of each cup of hot chocolate, at what price will Naledi sell each cup?

$6x = 30$ . Divide both sides by 6 to get the cost of each cup

$$\frac{6x}{6} = \frac{30}{6}$$

$$x = 5$$

$\therefore$  each cup will sell at R5

- (d) Mazwi and Themba pick pears off the tree. For every 17 pears that Mazwi picks, Themba picks 20. Together they picked 148 pears.

- (i) Represent that in an algebraic equation

$$17x + 20x = 148$$

- (ii) Calculate how many pears did Themba pick

$$17x + 20x = 148$$

$$\frac{37x}{37} = \frac{148}{37}$$

$$x = 4$$

Themba picked  $20x = 20 \times 4 = 80$  pears

## UNIT 4

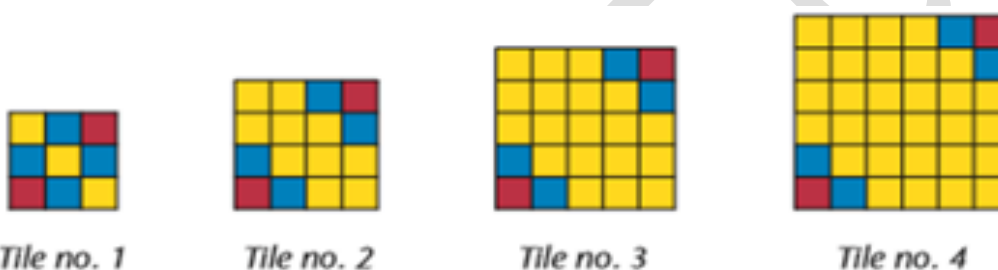
### NUMERIC AND GEOMETRIC PATTERNS

#### 4.1 Geometric Patterns

The “**structure**” in geometric patterns refers to the physical arrangement of geometric shapes at every stage of the pattern.

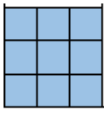
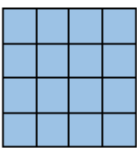
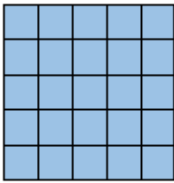
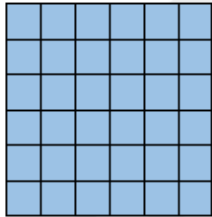
The **structural analysis** is very useful to “see” the **recursive relationship** to help formulate the general rule for the pattern even before the table or flow diagram can be drawn. A calculation plan can be derived from the structure and subsequently the rule.

1. Study the figures of tiles below, draw tile number 4 and describe tile no.5



43 – yellow squares, 4 – blue squares and 2 – red squares

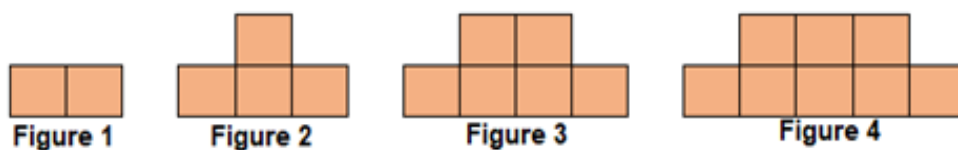
2. Write the calculation plan and the general rule for each tile. The first one has been done.

|                  | Tile 1                                                                              | Tile 2                                                                              | Tile 3                                                                               | Tile 4                                                                                |
|------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| Structure        |  |  |  |  |
| Calculation plan | $3 \times 3 = 9$<br>$(1 + 2)^2 = 9$                                                 | $4 \times 4 = 16$<br>$(2 + 2)^2 = 16$                                               | $5 \times 5 = 25$<br>$(3 + 2)^2 = 25$                                                | $6 \times 6 = 36$<br>$(4 + 2)^2 = 36$                                                 |
| Rule             | $(n + 2)^2$                                                                         |                                                                                     |                                                                                      |                                                                                       |

**NOTE:** The calculation plan has been matched to the position of the tile in the pattern.

3. Study the diagrams and answer the questions that follow:

(a) Extend the pattern by drawing figure 4



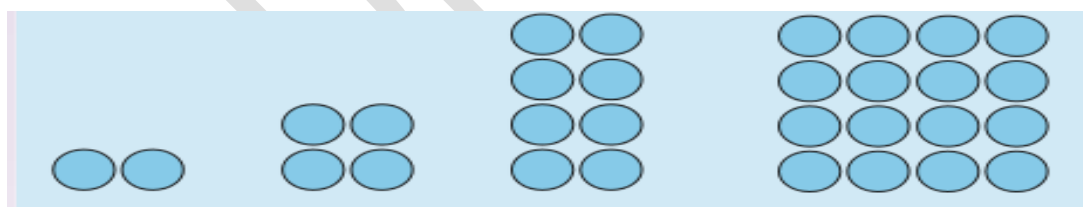
(b) Write the calculation plan

|                  | Figure 1                 | Figure 2                         | Figure 3                             | Figure 4                                 |
|------------------|--------------------------|----------------------------------|--------------------------------------|------------------------------------------|
| Structure        |                          |                                  |                                      |                                          |
| Calculation plan | $1 + 1$<br>or<br>$2 + 0$ | $1 + 2 + 1$<br>or<br>$2 + 1 + 1$ | $1 + 2 + 2 + 1$<br>or<br>$2 + 2 + 2$ | $1 + 2 + 2 + 2 + 1$<br>or<br>$2 + 3 + 3$ |
| Rule             | $2n$                     |                                  |                                      |                                          |

(c) Describe the pattern in words.

Double the figure number or double the position number.

4. Study the pattern below



(a) Represent the pattern in table

| Position        | 1              | 2                       | 3                                | 4                                         |
|-----------------|----------------|-------------------------|----------------------------------|-------------------------------------------|
| Number of balls | $2 =$<br>$2^1$ | $2 \times 2 =$<br>$2^2$ | $2 \times 2 \times 2 =$<br>$2^3$ | $2 \times 2 \times 2 \times 2 =$<br>$2^4$ |

(b) Write the general formula or rule.

$2^n$

(c) Describe the pattern in words.

2 raised to the position number.

5. Investigate how the pattern grows and answer the questions below:

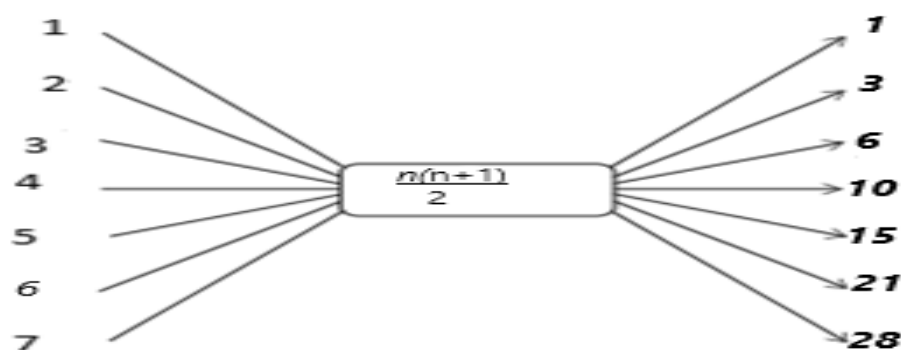
(a) Extend the pattern by drawing **figure 5**.

| Picture 1                                                                         | Picture 2                                                                         | Picture 3                                                                         | Picture 4                                                                          | Picture 5                                                                           |
|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
|  |  |  |  |  |

(b) Describe the pattern in words

(Position squared plus position) divided by 2

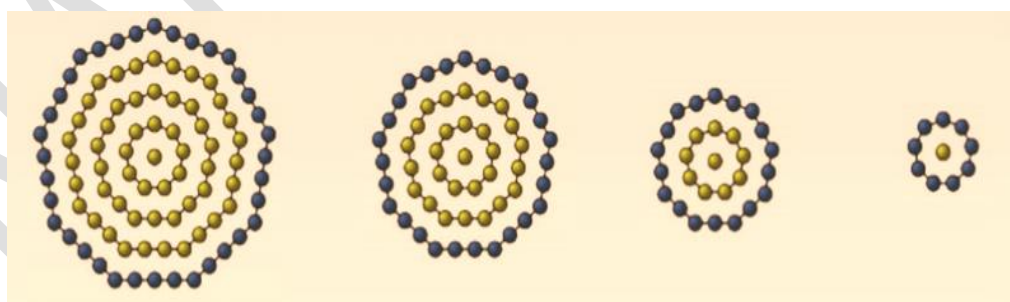
(c) Represent the pattern in a flow diagram



(d) Write the general rule

$$T_n = n(n + 1) \div 2 \text{ or } T_n = (n^2 + n) \div 2$$

6. Study the pattern below and answer the questions that follows:



(a) Represent the pattern in a table

| Position     | 1  | 2  | 3  | 4  |
|--------------|----|----|----|----|
| No. of beads | 91 | 55 | 28 | 10 |

(b) How is the pattern in question 6 different from the pattern in question 5?

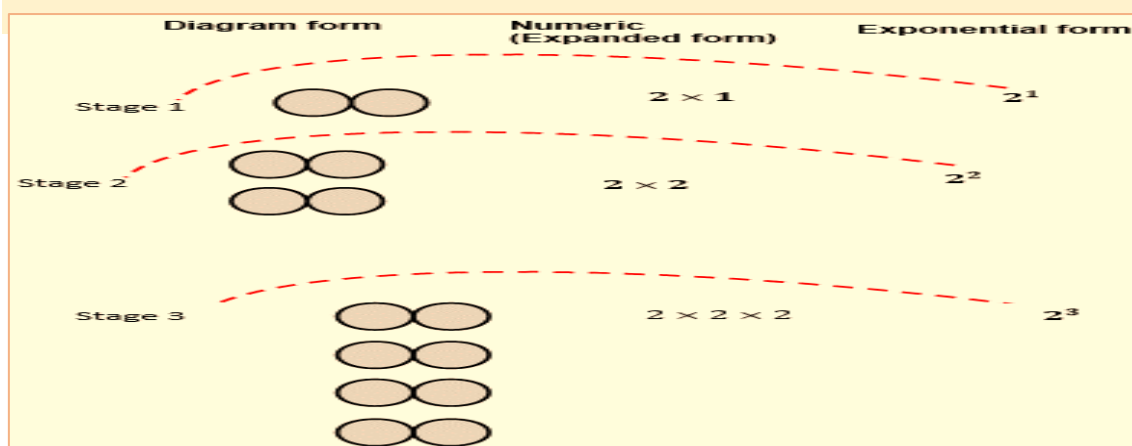
The pattern in question 5 is an increasing pattern, whereas the pattern in question 6 is a decreasing pattern.

## 4.2 Numeric Patterns

There is a connection between the **physical structure** (construction) and the **numeric representation** at each stage of the pattern. Studying and analyzing the structure assists in developing a numeric pattern and subsequently the general rule.

### Example

#### Physical representation



#### Numeric representation in a table.

| Stage number           | 1            | 2            | 3                     | 4                              | 5     | n     |
|------------------------|--------------|--------------|-----------------------|--------------------------------|-------|-------|
| Number of circles      | 2            | 4            | 8                     | 16                             |       |       |
| Equivalent description | $2 \times 1$ | $2 \times 2$ | $2 \times 2 \times 2$ | $2 \times 2 \times 2 \times 2$ |       |       |
| Equivalent description | $2^1$        | $2^2$        | $2^3$                 | $2^4$                          | $2^5$ | $2^n$ |

1. What are the next three numbers in each of these patterns?

(a) 15; 20; 25; 30; 35; 40; 45

(b) 8; 16; 24; 32; 40; 48; 56

(c) 3; 15; 27; 39; 51; 63; 75

(d) 0,9; 0,7; 0,5; 0,3; 0,1; -0,1; -0,3

(e)  $\frac{1}{4}$ ;  $\frac{1}{2}$ ;  $\frac{3}{4}$ ; 1;  $1\frac{1}{4}$ ;  $1\frac{1}{2}$

(f) 42;35 ;28;21; 14; 7; 0

(g) 128;116;104; 92; 80; 68; 56

Apply the knowledge learnt in number sentences and algebraic language to interpret the following:

2. Extend the following patterns and indicate the rule

| No. | Pattern                | Rule             |
|-----|------------------------|------------------|
| (a) | 3; 8; 13; 18; 23; 28   | $T_n = 5n - 2$   |
| (b) | 10; 14;18; 22; 26; 30  | $T_n = 4(n) + 6$ |
| (c) | 10; 25; 40; 55; 70; 85 | $T_n = 15n - 5$  |
| (d) | 5; 10; 15; 20; 25; 30  | $T_n = 5n$       |
| (e) | 2; 4; 8; 16 32; 64     | $T_n = 2^n$      |
| (f) | 15; 25; 35; 45; 55; 65 | $T_n = 10n + 5$  |
| (g) | 1; 4; 9;16; 25; 36     | $T_n = n^2$      |

3. In what way are patterns B and E different?

Pattern B has **constant difference** of 4; 4 is added to get to the next number, while pattern B have a **constant ratio** of 2

4. Given that any term in a number pattern is :

$$T_n = 2 \times n + 1$$

(a) Determine the value of the 1<sup>st</sup> term

$$2 \times 1 + 1 = 3$$

(b) Determine the value of the 4<sup>th</sup> term

$$2 \times 4 + 1 = 9$$

(c) What will be the value of the 100<sup>th</sup> term

$$2 \times 100 + 1 = 201$$

(d) Write the rule for this pattern in words.

**Multiply the term number by 2 and add 1**

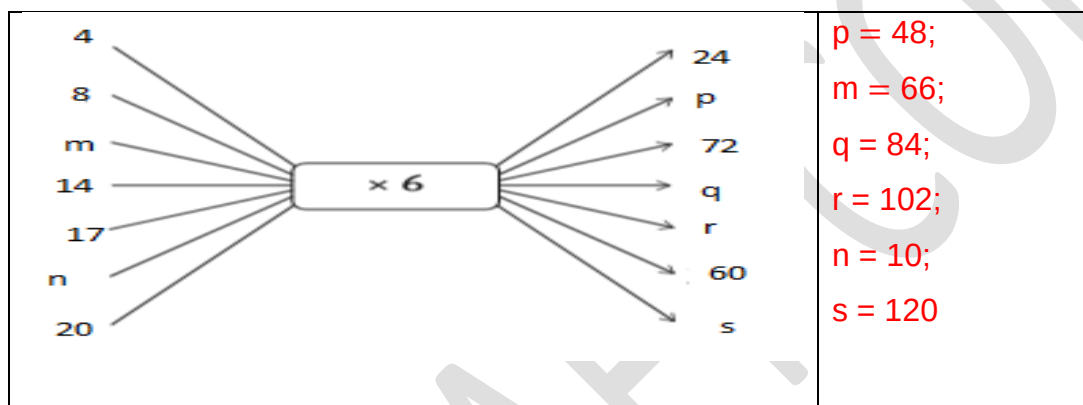
5. Which of the following rules will produce the number pattern given in the second row of the table?

|             |   |    |    |    |    |    |
|-------------|---|----|----|----|----|----|
| Term number | 1 | 2  | 3  | 4  | 5  | 6  |
| Term value  | 8 | 13 | 18 | 23 | 28 | 33 |

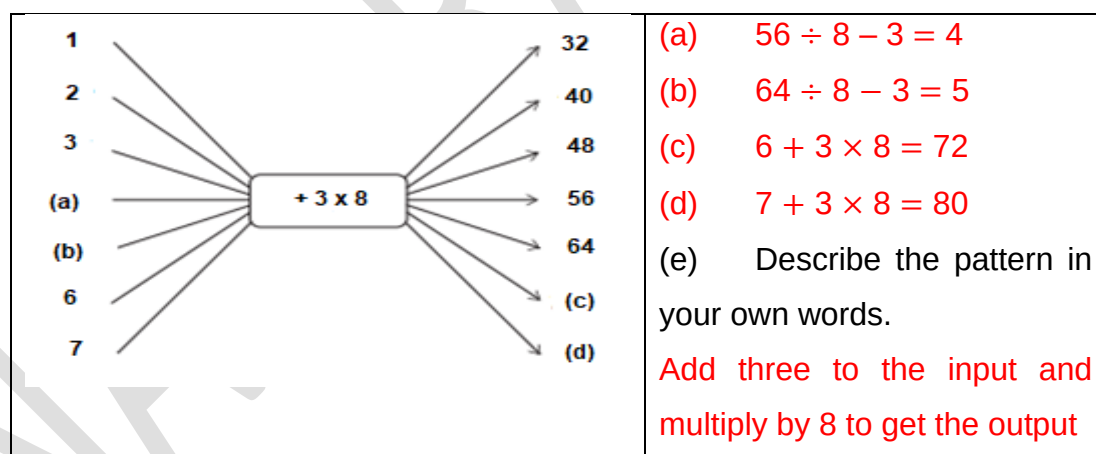
- (a) Term value =  $8n$  where  $n$  is the term number
- (b) Term value =  $6n - 1$  where  $n$  is the term number
- (c) Term value =  $6n + 2$  where  $n$  is the term number
- (d) Term value =  $10n - 2$  where  $n$  is the term number
- (e) Term value =  $5n + 3$  where  $n$  is the term number

### Input and output

6. Copy and complete the following flow diagram:



7. Fill in the missing output and input numbers:



### Problem solving

8. Mr Twala pays a fee to park his car in a parking lot every day. He has to pay R3 to enter the parking lot and then a further R2 for every hour that he leaves his car there.
- (a) Copy and complete the following table to show how much his parking costs him per day for various numbers of hours



|                      |    |    |    |     |     |     |     |
|----------------------|----|----|----|-----|-----|-----|-----|
| Number of hours      | 1  | 2  | 3  | 4   | 5   | 6   | 7   |
| Cost of parking in R | R5 | R7 | R9 | R11 | R13 | R15 | R17 |

(b) How did you complete this table? Describe your method in words.

Number of hours multiplied by the cost of parking per hour (R2), added to the parking fee (R3)

(c) Is there another way that you could complete the table? Describe the rule.

$$Tn = n \times 2 + 3$$

9.1. Thapelo is building a house model using cubes. If he uses 2 cubes in the first model, 5 in the second and 10 in the third model.

(a) How many cubes will he need to complete the fourth model, if the pattern continues?

He will need 17 cubes to complete the fourth model.

(b) Write down a mathematical rule to determine the value of any term in the sequence.

$$Tn = n^2 + 1$$

(c) Use your rule to determine the 10th term of the pattern.

$$Tn = n^2 + 1$$

$$100^2 + 1 = 100$$

9. Write the symbolic formula for the following statements:

(a) Output number is equal to input number multiplied by four, added to twelve

$$y = 4x + 12$$

(b) Output number is equal to 2, multiplied by the input number added to four

$$y = 2(x + 4)$$

(c) Create your own numeric pattern in a table and a flow diagram

Any mathematically correct pattern

(d) Write the formula for your pattern.

Any mathematically correct formula

(e) Write your pattern in a sequence.

Any mathematically correct sequence

(f) Describe the pattern that you created in words.

Any mathematically sound description

## UNIT 5:

### MEASUREMENT

#### 5.1 Length

- Convert the following lengths to the given units.

| No. | Length                 | Conversions |
|-----|------------------------|-------------|
| (a) | 1,2 m to cm            | 120 cm      |
| (b) | 13 478 mm to cm        | 13,478 cm   |
| (c) | $3\frac{1}{2}$ m to mm | 3 500 mm    |
| (d) | 639,2 cm to mm         | 6 392 mm    |
| (e) | 4 593 cm to m          | 45,93 m     |
| (f) | 4 071 mm to cm         | 407,10 cm   |

- My car has to go for a service in 2 871 km. I drove 1 264 km during the month. How many kilometres left before I have to take my car for the service?

ODOMETER READING



(DBE WORBOOK 2 Grade 6,  
Page 97 - 98 of 11<sup>th</sup> Edition)

$$2\,871\text{ km} - 1\,264\text{ km} = 1\,607\text{ km}$$

I am left with 1 607 km before I take my car for the service.

- Bongi travelled 3,5 km to the event and Sipho travelled 3 250 m to the same event. Who travelled the furthest between the two? Write your answer in metres.

$$3,5\text{ km} = 3\,500\text{ m}$$

$$3\,500\text{ m} - 3\,250\text{ m} = 250\text{ m}$$

Bongi travelled 250 metres furthest.

- David measured the distance between two trees with a stick. He found that the distance was  $12\frac{1}{3}$  sticks. When he measured the stick with his ruler, it was

78 cm long. What was the distance between the two trees? Give your answer in centimetres and metres.

$$12\frac{1}{3} \times 78 \text{ cm}$$

$$= (12 \times 78 \text{ cm}) + (\frac{1}{3} \text{ of } 78 \text{ cm})$$

$$= 936 \text{ cm} + 26 \text{ cm}$$

$$= 962 \text{ cm}$$

The distance between the two trees was 962 cm or 9,62 m.

5. Calculate:

(a) Calculate in centimetres and round off your answer to the nearest 100 mm:

$$6 \text{ m} + 157 \text{ cm} - 1 \text{ 145 mm.}$$

$$6 \text{ m} = 6 \text{ 000 mm}$$

$$157 \text{ cm} = 1 \text{ 570 mm}$$

$$6 \text{ 000 mm} + 1 \text{ 570 mm} - 1 \text{ 145 mm} = 6 \text{ 425 mm}$$

6 425 mm to the nearest 100 mm is 6 400 mm.

(b) Calculate in millimetres and round off your answer to the nearest 1 000 mm:

$$23 \times (1 \text{ 380 mm} - 78 \text{ cm})$$

$$78 \text{ cm} = 780 \text{ mm}$$

$$23 \times (1 \text{ 380 mm} - 780 \text{ mm})$$

$$23 \times 600 = 13 \text{ 800 mm}$$

$\therefore$  13 800 mm to the nearest 1000 mm is 14 000 mm.

(c) Calculate in metres and round off your answer to the nearest kilometre:

$$5,4 \text{ km} - 204 \text{ m} \times 14.$$

$$5,4 \text{ km} = 5 \text{ 400 m}$$

$$5 \text{ 400 m} - (204 \text{ m} \times 14)$$

$$5 \text{ 400 m} - 2 \text{ 856 m} = 2 \text{ 544 m}$$

$$\therefore 2 \text{ 544 m} = 2,54 \text{ km}$$

2,54 km and to the nearest km it will be 3 km.

## 5.2 Mass

Calculate the following problems in your writing books.

1. A 500 g bag of flour costs R3,50. How much will a 1,5 kg cost?

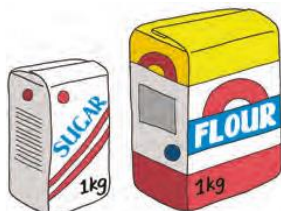


R3,50

(DBE WORBOOK 2  
Grade 6 LB. Page 10)

1,5 kg bag of flour will cost R10,50

2. Compare a 1 kg packet of sugar and a 1 kg packet of flour.



DBE Textbook  
Gr 6 LB, Page 216

- (a) Does the mass of the two packets differ?

No

- (a) Does the size of the two packets differ?

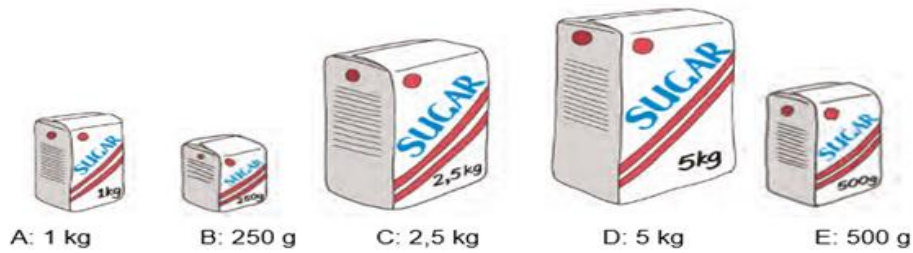
Yes, the 1 kg packet of flour is bigger than the 1 kg packet of sugar.

- (b) Explain why it is that a 1 kg packet of sugar and a 1 kg packet of flour have different sizes.

Consider learners' explanations. A reasonable answer would be, for example: Flour is a different substance than sugar; you need more flour to make up a kilogram mass compared to the amount of sugar needed to make up a kilogram mass.

Learners might simply say that "sugar is heavier than flour". If they say this, ask them if the small packet of sugar in question 3 is heavier than the big packet of flour in question 2. (The answer to this is no, because the packet of flour is heavier than the small packet of sugar.) So, it is incorrect to say "sugar is heavier than flour". We always have to compare the heaviness of equal volumes of a substance.

3. Compare the mass of the packets of sugar below and answer questions that follow.



- (a) How many of Packet B have the same mass as Packet A?

There are 4 packets of Packet B.

- (b) What fraction of the sugar in Packet A has the same mass as Packet B?

It is a quarter.

- (c) Which packet holds half as much sugar as Packet A?

It is packet E

- (d) Which packet holds  $2\frac{1}{2}$  times as much sugar as Packet A?

It is packet C

### 5.3 Capacity/Volume

1. Diesoline is used to generate electricity at a small power station. The power station uses 684 ℓ of diesoline each day. For how many days can the power station operate if there is a stock of 9 765 ℓ of diesoline available?

$$9\,765\text{ ℓ} \div 684\text{ ℓ} = 14,276\text{ ℓ}$$

The power station can operate for fourteen days.

2. Thuli adds 250 mℓ of concentrated fruit juice to 2 ℓ of water, to make drinks for the athletes in a long-distance race.

- (a) How much concentrated juice should she add to 5 ℓ of water?

250 mℓ is added to 2 litres of water, therefore 125 mℓ will be added to 1 litre.

She should add 625 litres of concentrated juice to 5 litres of water.

- (b) How many athletes can she provide with 400 ml of juice each, with the juice she made by adding concentrate to 5 ℓ of water?

She can provide 2 812 athletes.

3. A bucket holds 3,2 litres, a second holds 575 millilitres and a third holds 4 litres 920 millilitres. What is the three containers total capacity? Write the answer in millilitres.

The total capacity of the three containers is 8695 millilitres

4. 30 teachers from the same school organised 8 times 2,25 litres of soft drinks. If they share it equally, how much each is going to drink in millilitres?

Each teacher will drink  $8 \times 2,25 \text{ l} = 18 \text{ l}$

$18 \text{ l} \div 30 \text{ teachers} = 0,6 \text{ l}$

$0,6 \text{ l} = 600 \text{ millilitres}$

Therefore, each teacher will drink 600 millilitres

5. Mila and Bongi are selling water in bottles, Mila sold 750 millilitres of water and Bongi sold 4 250 millilitres.

- (a) How much more did Bongi sell than Mila?

Bongi sold 3 500 millilitres more than Mila.

- (b) Write the answer in litres.

3 500 millilitres = 3,5 l

- (c) Write the answer as a common fraction.

3 500 millilitres =  $3\frac{1}{2} \text{ l}$

6. Match the common fractions below with the mass of the illustrated items.

$\frac{1}{4} \text{ kg}; \frac{1}{2} \text{ kg}; \frac{3}{4} \text{ kg}; 2\frac{1}{2} \text{ kg}; 1\frac{1}{2} \text{ kg}$



|                                                                                     |                                                                                     |                                                                                       |                                                                                       |                                                                                       |
|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
|  |  |  |  |  |
| A = $\frac{3}{4} \text{ kg}$                                                        | B = $1\frac{1}{2} \text{ kg}$                                                       | C = $\frac{1}{4} \text{ kg}$                                                          | D = $2\frac{1}{2} \text{ kg}$                                                         | E = $\frac{1}{2} \text{ kg}$                                                          |

7. Find the equivalent mass of the following objects (in grams):

- (a) a baby with a mass of 2,8 kg

2,8 kg is equivalent to 2 800 g

- (b) a book with a mass of 0,5 kg

0,5 kg is equivalent to 500 g

(c) a brick with a mass of 1,5 kg

1,5 kg is equivalent to 1 500 g

8. Find the equivalent mass of the following objects (in kilograms):

(a) a 250 g bag of flour

250 g is equivalent to 0,25 kg

(b) an apple of 100 g

100 g is equivalent to 0,10 kg

(c) a 750 g stone

750 g is equivalent to 0,75 kg

9. A bag of 30 oranges, all of the same mass, weighs 5 kg. Calculate the mass of:



(a) 15 oranges

15 oranges have a mass of  $5\ 000\text{ g} \div 2$   
 $= 2\ 500\text{ g}$  or 2,5 kg

(b) 5 oranges

5 oranges have a mass of  $5\ 000\text{ g} \div 6$   
 $= 833\text{ remainder }2\text{ g}$ , or about 833 g

(c) 1 orange.

1 orange has a mass of  $5\ 000\text{ g} \div 30$   
 $= 166\text{ remainder }20\text{ g}$ , or about 166 g

# TERM 3



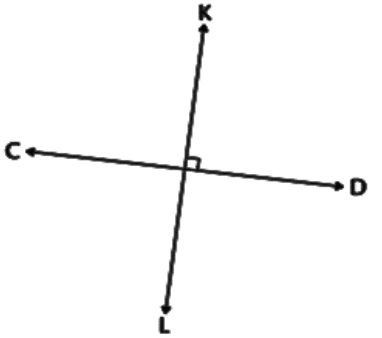
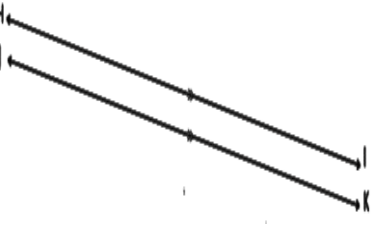
## UNIT 1

### GEOMETRY OF STRAIGHT LINES

1. Match the term in column A with the definition in Column B

| Column A               | Column B                                                                     |
|------------------------|------------------------------------------------------------------------------|
| (a) Ray                | (iv) A set of points with definite starting-point and no definite end-point. |
| (b) Straight line      | (i) A set of points with no definite starting-point and end-point.           |
| (c) Line segment       | (v) A set of points with a definite starting-point and an end-point.         |
| (d) Parallel line      | (ii) When two lines on the same plane are a constant / same distance apart   |
| (e) Perpendicular line | (iii) When two lines intersect at right angle                                |

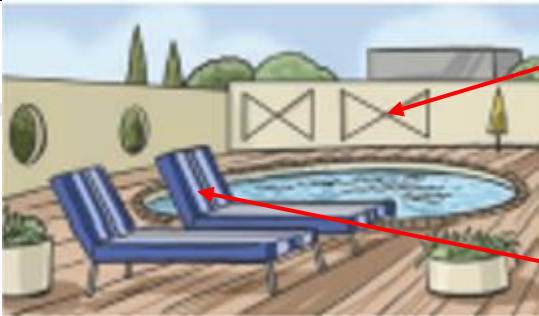
2. Identify the types of lines:

|     |                                                                                     |                                            |
|-----|-------------------------------------------------------------------------------------|--------------------------------------------|
| (a) |  | AB is a <b>straight line</b>               |
| (b) |  | MN is a <b>ray</b>                         |
| (c) |  | GH is a <b>line segment</b>                |
| (d) |  | Line CD is <b>perpendicular</b> to Line KL |
| (e) |  | Line HI is <b>parallel</b> to line JK      |

3. Study the following pictures and mark the parallel and perpendicular lines

| No.                                                                            | Picture                                                                            | Hints                                                                                                             |
|--------------------------------------------------------------------------------|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| (a)                                                                            |   | The opposite faces of the diagram are parallel to each other.<br>The 90° sign can be shown in any of the vertices |
| (b)                                                                            |   | The opposite faces of the diagram are parallel to each other.<br>The 90° sign can be shown in any of the vertices |
| (c)                                                                            |  | The 90° sign can be shown in any of the vertices<br>The opposite faces of the diagram are parallel to each other  |
| The parallel lines markings can be shown the pillars, stairs, wall and windows |                                                                                    |                                                                                                                   |

4. Identify types of lines formed in the following pictures. [Any suitable answer]

|     |                                                                                      |
|-----|--------------------------------------------------------------------------------------|
| (a) |  |
| (b) |  |

## 1.2 ANGLE RELATIONSHIPS

### Remember:

The sum of the angles on a straight line is  $180^\circ$

If lines are perpendicular, then adjacent supplementary angles are each equal to  $90^\circ$ .

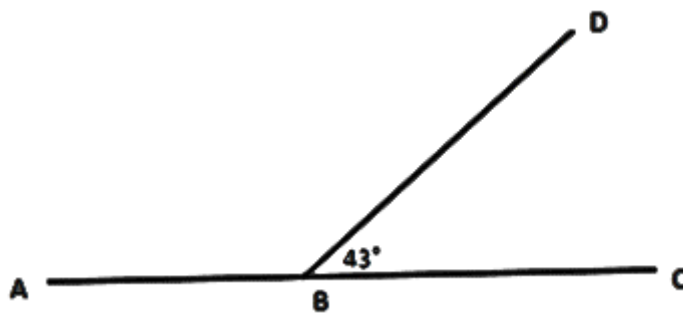
If lines intersect, then vertically opposite angles are equal.

If parallel lines are cut by a transversal, then

- corresponding angles are equal
- alternate angles are equal
- co-interior angles are supplementary

1. Given the diagrams below

(a) Calculate the size of angle  $ABD$  if angle  $DBC$  is equal to  $43^\circ$



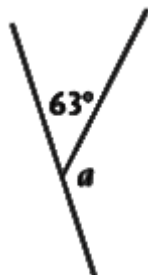
$$\angle ABD + \angle DBC = 180^\circ$$

sum of angles in a straight line

$$\angle ABD = 180^\circ - 43^\circ$$

$$\angle ABD = 137^\circ$$

(b) Calculate Angle  $a$



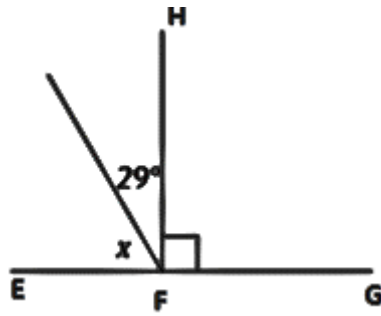
$$63^\circ + a = 180^\circ$$

sum of angles in a straight line

$$a = 180^\circ - 63^\circ$$

$$a = 117^\circ$$

(c) If line HF is perpendicular to line EG calculate angle  $x$



$$29^\circ + x = 90^\circ$$

adjacent angles are supplementary

$$x = 90^\circ - 29^\circ$$

$$x = 61^\circ$$

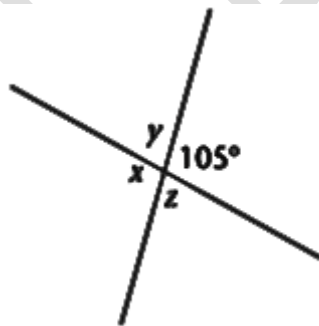
2. Given the lines that intersect below. What is the size of angle  $j$ ? Give reason for your answer.



$$\text{Angle } j = 64^\circ$$

vertically opposite angles are equal

3. Calculate angle  $x$ ,  $y$  and  $z$  by completing the steps below:



$$x = 105^\circ$$

$$y + 105^\circ = 180^\circ$$

$$y = 180^\circ - 105^\circ$$

$$= 75^\circ$$

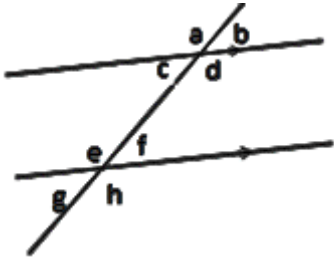
$$y = z$$

$$z = 75^\circ$$

Vertically opposite angles  
angles on a straight line

Vertically opposite angle

4. Which set of pairs are:



(a) Corresponding angles

Angles b and f ; d and h ; a and e ; c and g

(b) Alternating angles

Angles c and f ; d and e

(c) Co interior angles

Angles d and f ; c and e

5. Study the picture of the house. Recognize, mark and describe pairs of angles formed by:



a) perpendicular lines

The angle formed is  $90^\circ$  (right angle), it can be marked in the windows, wall, door and floor

b) parallel lines cut by a transversal

Pairs of angles formed are corresponding angles marked on the roof

## UNIT 2

### GEOMETRY OF 2-D SHAPES

1. Match the properties in Column A with the name of the 2D shape in column B

| Column A                                                                                                        | Column B              |
|-----------------------------------------------------------------------------------------------------------------|-----------------------|
| (a) A triangle with three equal sides and angles                                                                | Equilateral triangle  |
| (b) A triangle with the hypotenuse side                                                                         | Right angled triangle |
| (c) A quadrilateral with all four sides equal                                                                   | Rhombus               |
| (d) A quadrilateral with only one pair of opposite sides parallel                                               | Trapezium             |
| (e) A quadrilateral with two pairs of adjacent sides equal and opposite angles between the unequal sides equal. | Kite                  |
| (f) A triangle with two equal sides and two equal angles                                                        | Isosceles triangle    |

2. Look at the figures and name each triangle

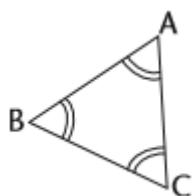


Figure A

Equilateral triangle

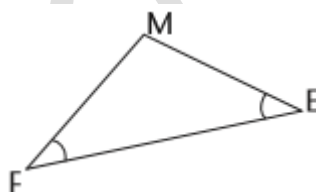


Figure B

Isosceles triangle

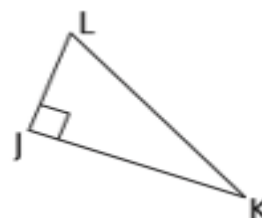


Figure C

Right-angled triangle

3. Identify types of triangles in the pictures

(a)



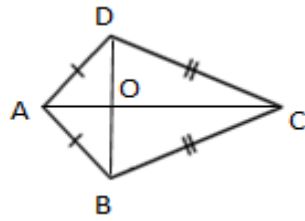
Equilateral triangles; Isosceles triangles and right-angled triangles

(b)



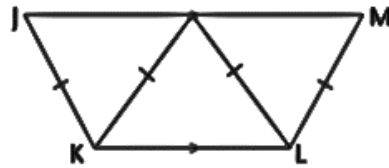
Isosceles triangle, equilateral triangle and right-angled triangle

4. Divide a kite into triangles and name the triangles.'



Right-angled triangle, Isosceles triangle

5. Which 2D shapes are found in the diagram below?



Isosceles triangles, parallelogram and trapezium

6. Sort and group the 2D shapes according to their properties.

|   |  |   |  |   |  |
|---|--|---|--|---|--|
| A |  | B |  | C |  |
| D |  | E |  | F |  |

- (a) All four sides of equal length
- (b) Opposite sides are equal
- (c) At least one pair of adjacent sides are equal
- (d) Perpendicular sides
- (e) All four angles are right angles
- (f) Two pairs of opposite sides parallel
- (g) Only one pair of opposite sides parallel

D and C  
A; C; D and F  
C, D and E  
C and A  
A and C  
A; C and F  
B

7. Name quadrilaterals in number 6 above.

A: Rectangle

B: Trapezium

C: Square

D: Rhombus

E: Kite and

F: Parallelogram

8. Mention 2 properties that are common in all the quadrilaterals.

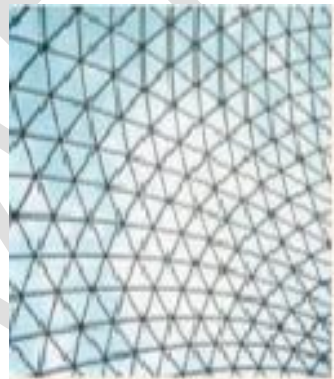
All have four sides and four angles

Sum of their interior angles equals to  $360^\circ$

9. Determine if the following statements are **true** or **false**.

- (a) A rhombus is a quadrilateral with all four angles of equal size **False**
- (b) A square has four right angles **True**
- (c) A kite has each pair of opposite angles equal **False**
- (d) A rectangle has 2 right angles and 2 acute angles **False**
- (e) A parallelogram is a quadrilateral with four angles of different sizes **False**
- (f) Total sum of angles in a trapezium is  $360^\circ$  **True**

10. The quadrilaterals are evident in common structures around the world. Study the following tower and mesh wire and name the quadrilaterals noticeable.



**Tower:** Trapezium and Parallelogram

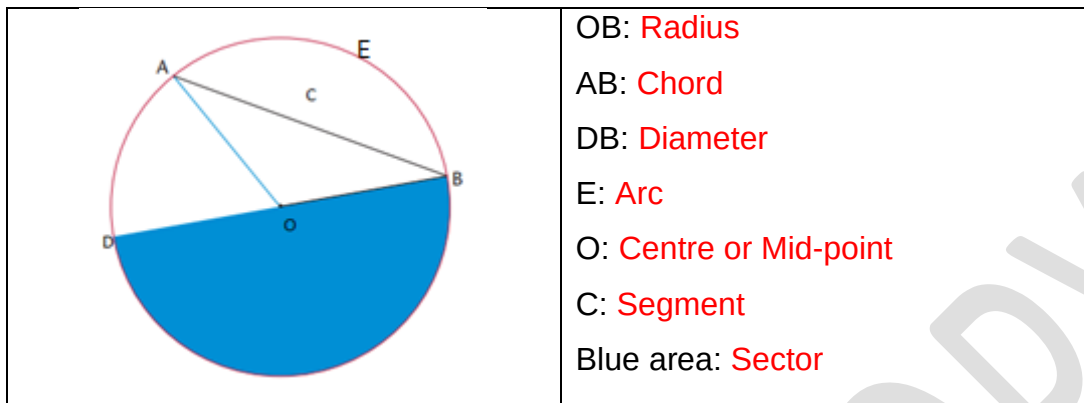
**Mesh Wire:** Rhombus; square; parallelogram; trapezium

11. Define parts of the circles

- (a) Centre: A point exactly in the middle of the circle
- (b) Chord: A line that joins any two points on a circumference
- (c) Arc: The curved outside edge of the sector is an arc.
- (d) Sector: A section of the circle created by two radii. Segment
- (e) Diameter: A distance from one side of the circumference, through the centre of the circle to the other side that cuts the circle in half.
- (f) Radius: The distance from the centre of the circle to any point on the circumference

12. Identify parts of the circle as indicated by letters in the diagram:



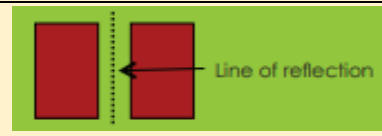
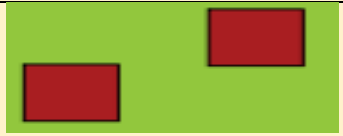
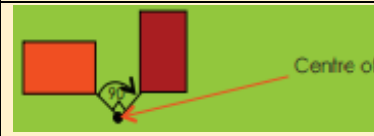


## UNIT 3

### TRANSFORMATIONS

Transformation is changing the form of something according to specific rules. A transformation is what brings about the change in position, shape or size of an object. Common transformations are **translations**, **rotations**, **reflections**, and **enlargements**.

#### Reflection, Translation and Rotation

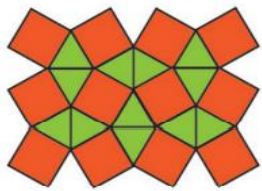
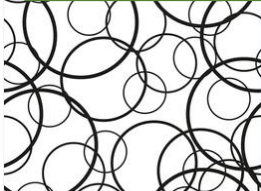
| Reflection                                                                          | Translation                                                                                                   | Rotation                                                                                                                                  |
|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| A reflection is a transformation that has the same effect as a mirror.              | A translation is the movement of an object to a new position without changing its shape, size or orientation. | Rotation: a rotation is a transformation that moves points so that they stay the same distance from a fixed point, the centre of rotation |
|  |                            |                                                       |

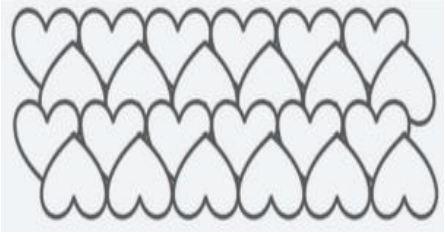
The cited transformation only changes the position of the object. The size does not change, therefore, the image is always congruent to the original figure.

### 3.1 Use transformations to make composite shapes and tessellations

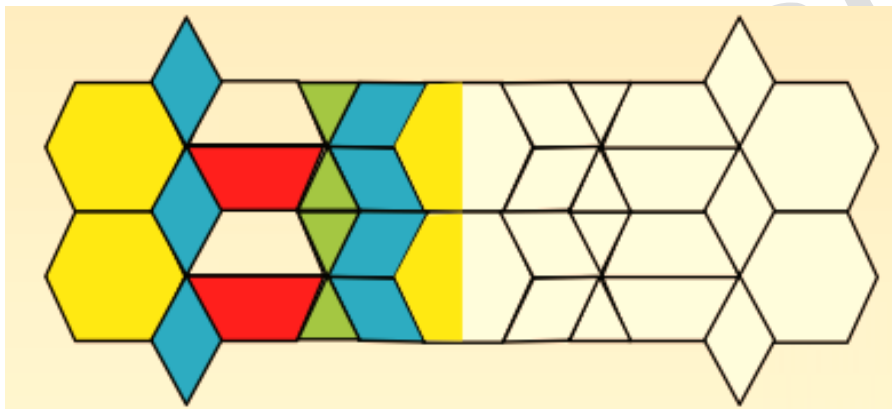
- What shapes have been used in the patterns below? Are these patterns tessellations?

**Tessellation** is the patterns formed by shapes, which fit together without any gaps and do not overlap

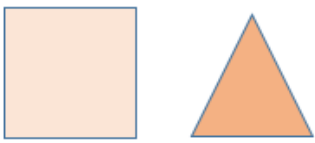
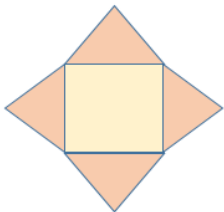
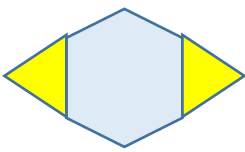
|                                                                                                                                                   |                                                                                                                                            |                                                                                                                                        |
|---------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| <p>(a)</p>  <p>Squares and triangles<br/>Shapes tessellate</p> | <p>(b)</p>  <p>Circles<br/>Shapes do not tessellate</p> | <p>(c)</p>  <p>Octagons<br/>Shapes tessellate</p> |
|---------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|

|                                                                                                                                  |                                                                                                                                                                    |
|----------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>(d)</p>  <p>Hearts<br/>Do not tessellate</p> | <p>(e)</p>  <p>Triangle, trapezia, hexagons and undecagon. Shapes tessellate</p> |
|----------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|

2. Complete this reflective tessellation.

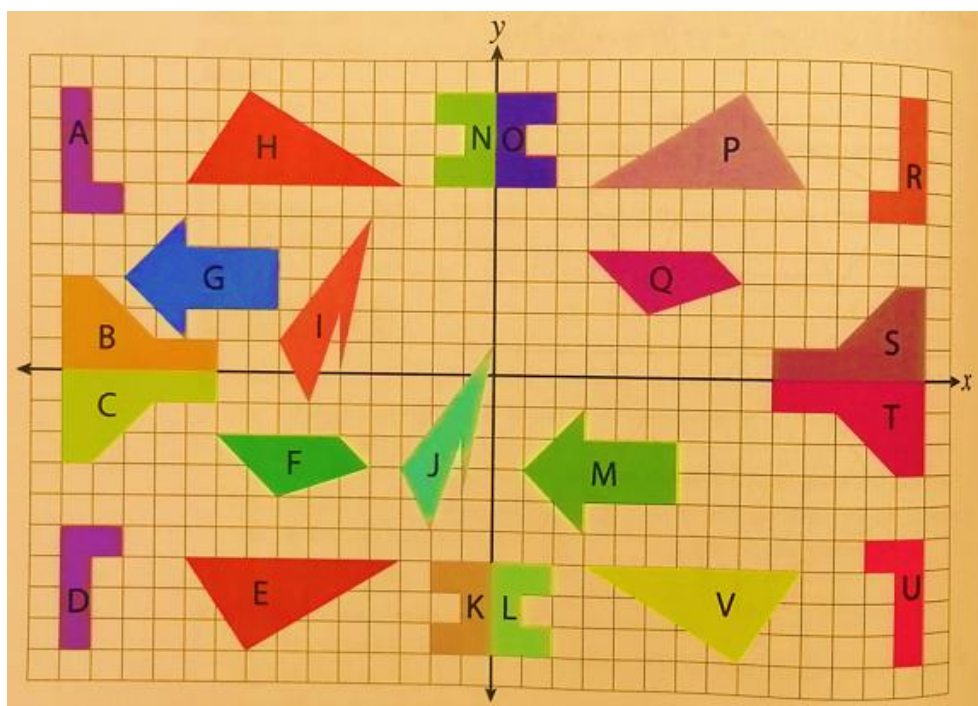


3. Why is this a reflective tessellation?  
 The second half is the mirror image of the first half.
4. Show a tessellation pattern using the following shapes.

|                                                                                                |                                                                                                 |
|------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| <p>(a)</p>  | <p>(b)</p>  |
|             |             |

### 3.2 Recognize, describe and perform translations, reflections and rotations with geometric figures and shapes.

1. Study the following diagram and answer questions.



(a) List the pairs of shapes that are a reflection of each other.

A and R; B and S; C and T; B and C; S and T; D and U; E and V; N and O; K and L; H and P; H and E; P and V; R and U; A and D; N and K; O and L

(b) List the pairs of shapes that are a translation of each other.

N and K; O and L; G and M; I and J; F and Q;

(c) List the pairs of shapes that are a rotation of each other.

A and U; R and D; E and P; H and V; S and C; B and T

(d) Are there any pairs of shapes that are an enlargement of each other? **No**

(e) Consider shape V and describe the transformations that could be carried out on it.

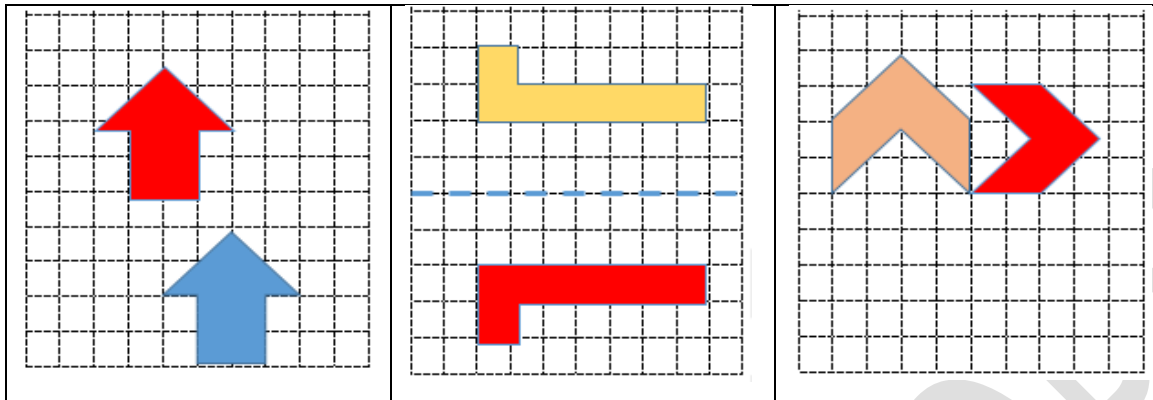
Image E is the reflection of shape V on y-axis

Image P is the reflection of shape V on x-axis

Image H is the 180° rotation of shape V

2. Show the following transformations on a grid.

|                                                             |                                                            |                                     |
|-------------------------------------------------------------|------------------------------------------------------------|-------------------------------------|
| (a) Translate the shape 2 units to the left and 5 units up. | (b) Reflect the figure below about the line of reflection. | (c) Rotate the shape 90° clockwise. |
|-------------------------------------------------------------|------------------------------------------------------------|-------------------------------------|



### 3.3 Enlargement

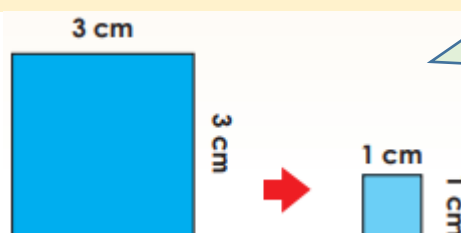
Another important type of transformation is resizing. Resizing can result in a larger shape and that is called enlargement or it can result in a smaller shape and is thus called reduction. If the shape is enlarged proportionally, all their angles are equal and all the sides are increased or decreased in the same ratio. This type of transformation produces similar figures as illustrated.



#### The scale factor

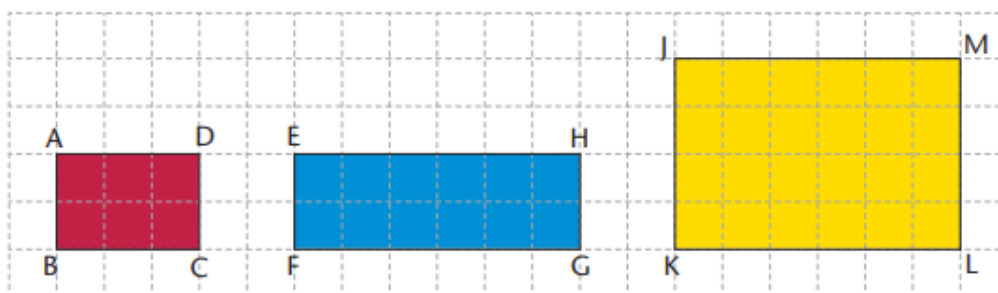
A **scale factor** is a number by which the size of any geometric figure or shape can be changed with respect to its original size.

- When the scale factor is 1, the image is the same size as the original shape.
- When the scale factor is less than 1, the image is a reduced. For example, if the scale factor is  $\frac{1}{2}$  or 0,5, each side of the image is half the length of its corresponding side in the original figure.
- When the scale factor is greater than 1, the image is enlarged. For example, if the scale factor is 2, each side of the image is double the length of its corresponding side in the original figure.



Each dimension of the original shape has been multiplied by  $\frac{1}{3}$  in order to get the image, therefore, the scale factor is  $\frac{1}{3}$ .

3. Look at the following rectangles and answer the questions below.



**Rectangle EFGH:**

- (a) How many times is FG longer than BC? **2 times**  
 (b) How many times is EF longer than AB? **One times /  $EF = AB$**   
 (c) Is EFGH an enlargement of ABCD? Give a reason for your answer.

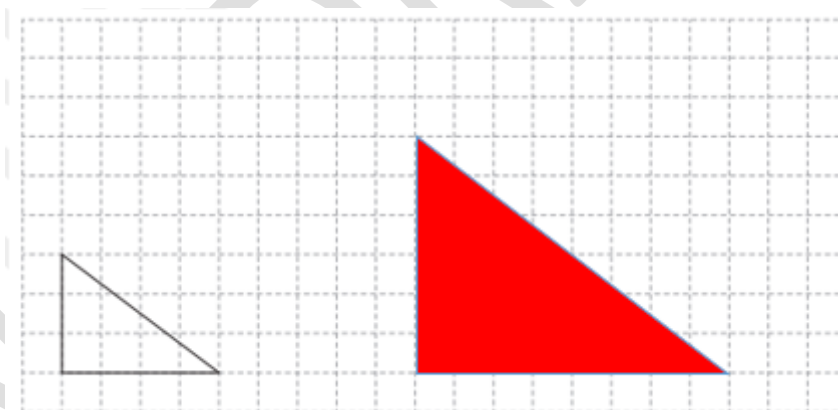
**No, the two shapes are not similar**

**Rectangle JKLM:**

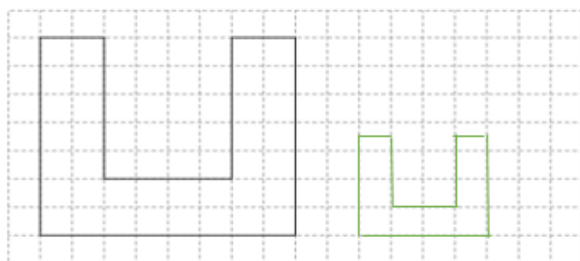
- (d) How many times is KL longer than BC? **2 times**  
 (e) How many times is JK longer than AB? **2 times**  
 (f) Is JKLM an enlargement of ABCD? Give a reason for your answer.

**Yes, the two shapes are similar. All the sides of rectangle JKLM are twice the sides of rectangle ABCD.**

4. On grid paper, enlarge the triangle below with a scale factor of 2.

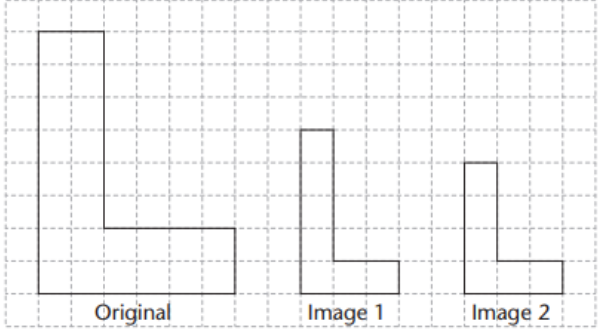
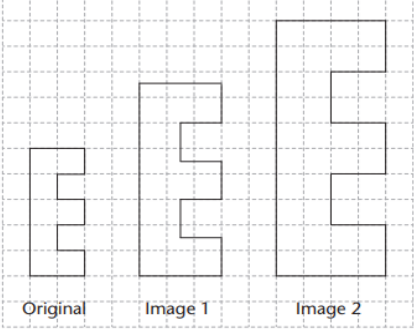


5. On grid paper, resize the following figure. Use a scale factor of 0,5





6. Which image is similar to the original shape? State the scale factor by which it has been resized.

|                                                                                                                                        |                                                                                                                                         |
|----------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| <p>(a)</p>  <p>Original      Image 1      Image 2</p> | <p>(b)</p>  <p>Original      Image 1      Image 2</p> |
| <p>Answer: Image 2 Scale factor = <math>\frac{1}{2}</math></p>                                                                         | <p>Answer: Image 2 Scale factor = 2</p>                                                                                                 |

### 3.4 Describing pattern in nature

1. The patterns have been developed using transformation. Describe how each of the patterns were developed.

|                                                                                     |                                                                                     |                                                                                       |
|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
|  |  |  |
| <p>(a) Enlargement and translation</p>                                              | <p>(b) Reflection; translation and rotation.</p>                                    | <p>(c) Translation and Reflection</p>                                                 |
|  |  |  |
| <p>(d) Reflection</p>                                                               | <p>(e) Rotation and translation</p>                                                 | <p>(f) Translation and enlargement</p>                                                |

## UNIT 4

### GEOMETRY OF 3-D OBJECTS

Most objects we see around us, like fruit, animals, trees, rocks, people and motor cars, have curved or round surfaces. Some objects, like a saucepan or other cooking vessel, have both round and flat surfaces. Other objects only have flat surfaces.



1. Study the following objects.



A brick



A potato



A pyramid



A battery



A clay pot



A cooking pot

(a) Which objects have curved surfaces only?

A potato

(b) Which objects have flat surfaces only?

A brick and a pyramid

(c) Which objects have both curved and flat surfaces?

A battery, a clay pot and a cooking pot

(d) Which of the objects has the shape of a sphere?

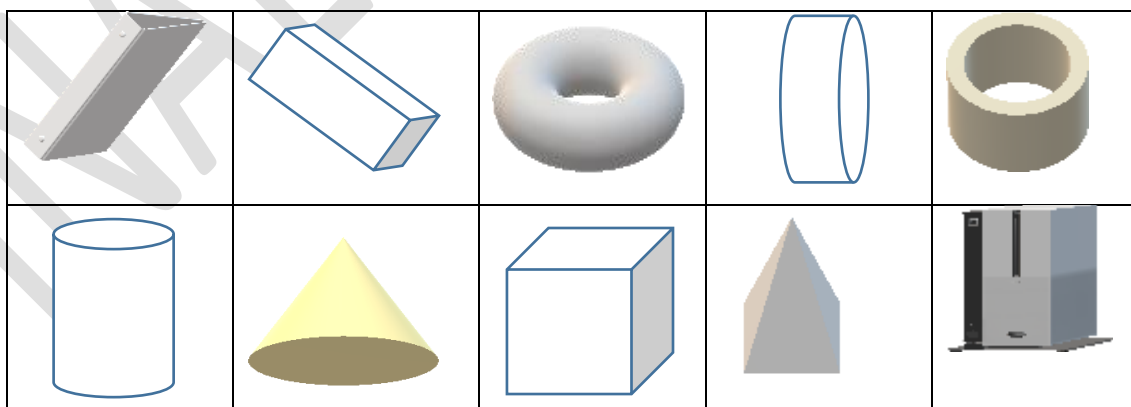
A potato

(e) Which of the objects have the shape of a cylinder?

A battery

(f) Which of the objects have the shape of a rectangular prism? A brick

2. Study the following diagram and answer questions.

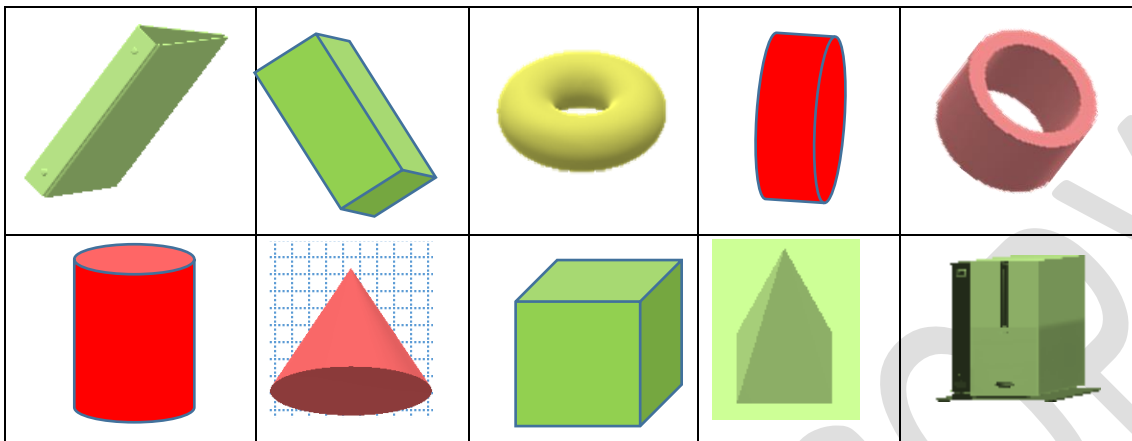


(a) Shade objects with curved surfaces only yellow.

(b) Shade objects with flat surfaces only, green.



(c) Shade the objects with both curved and flat surfaces red.



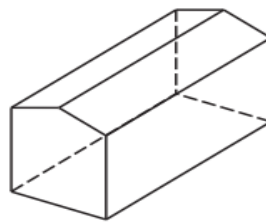
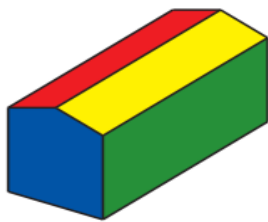
(d) Which of the objects are polyhedra?

Triangular prism, rectangular prism, cube, square-based pyramid and the box.

#### 4.1 Describe, sort and compare polyhedra

A 3D object with flat surfaces (faces) only is called a polyhedron. A straight edge is formed where two flat surfaces meet. The point where two or more edges meet is called a vertex.

1. In the drawing of the same box on the right, dotted lines are used to indicate edges and surfaces that are hidden in the coloured drawing.



(a) How many faces (different flat surfaces) does this object have altogether?

Seven (7)

(b) How many faces cannot be seen in the coloured drawing on the left?

Three (3)

(c) How many of the faces are rectangles?

Five (5)

(d) How many of the faces are pentagons?

Two (2)

(e) How many edges does the coloured polyhedron have?

Fifteen (15)

(f) How many vertices does it have?

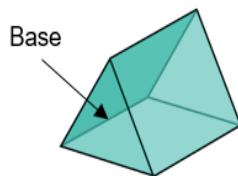
Ten (10)

There are two special groups of **polyhedra** i.e. **prisms** and **pyramids**.

### Prisms

Prism are polyhedra with two faces that are congruent and parallel polygons. These two faces are called bases and are connected by lateral faces that are parallelograms. Prisms are named using their base. The two opposite congruent faces are the base.

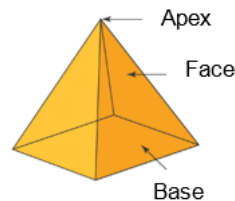
**Example:** Triangular prism



### Pyramids

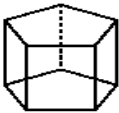
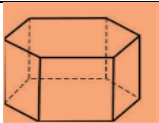
Pyramid have one base. The lateral faces of a pyramid are always triangles. These triangles meet at the same vertex. This vertex is called the **apex** of the pyramid. Pyramids are named using their base. The base is not always the face it is sitting on.

**Example:** Square-based pyramid

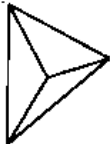
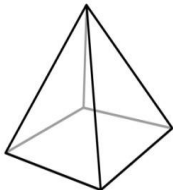
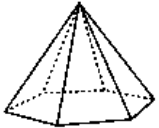


1. Copy and complete the following table of **prisms**.

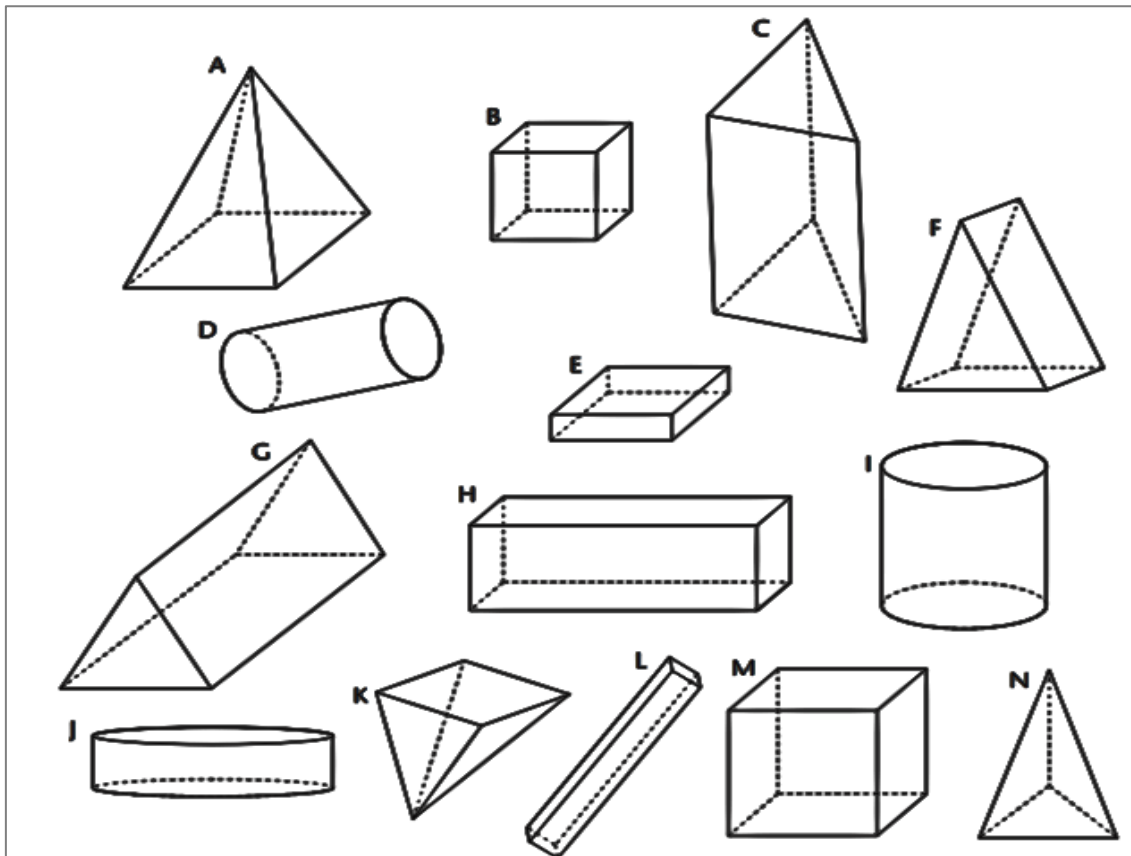
| No. of sides of a base | Sketch | Name of the object             | No. of faces | No. of vertices | No. of edges |
|------------------------|--------|--------------------------------|--------------|-----------------|--------------|
| 3                      |        | Triangular-based prism         | 5            | 6               | 9            |
| 4                      |        | Rectangular-based prism / cube | 6            | 8               | 12           |

|   |                                                                                   |                  |   |    |    |
|---|-----------------------------------------------------------------------------------|------------------|---|----|----|
| 5 |  | Pentagonal prism | 7 | 10 | 15 |
| 6 |  | Hexagonal prism  | 6 | 12 | 18 |

2. Copy and complete the following table of **pyramids**

| Number of sides of a base | Sketch                                                                              | Name of the object       | Number of faces | No. of vertices | No. of edges |
|---------------------------|-------------------------------------------------------------------------------------|--------------------------|-----------------|-----------------|--------------|
| 3                         |    | Triangular-based pyramid | 4               | 4               | 6            |
| 4                         |   | Square-based pyramid     | 5               | 5               | 8            |
| 5                         |  | Pentagonal pyramid       | 6               | 6               | 10           |
| 6                         |  | Hexagonal pyramid        | 7               | 7               | 12           |

3. Study the following 3D objects.



(a) Sort the 3D objects below into the three groups, namely prisms, pyramids and cylinders. Write down the correct letter under each group.

Prisms – B, C, E, F, G, H, L and M

Pyramids – A, K and N

Cylinders – D, I and J

(b) Sort the prisms into three groups (cubes, rectangular prisms and triangular prisms). Write the letter under each group.

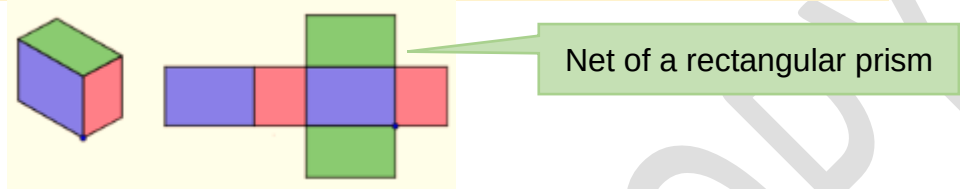
Cubes – B and M

Rectangular prisms – E, H and L

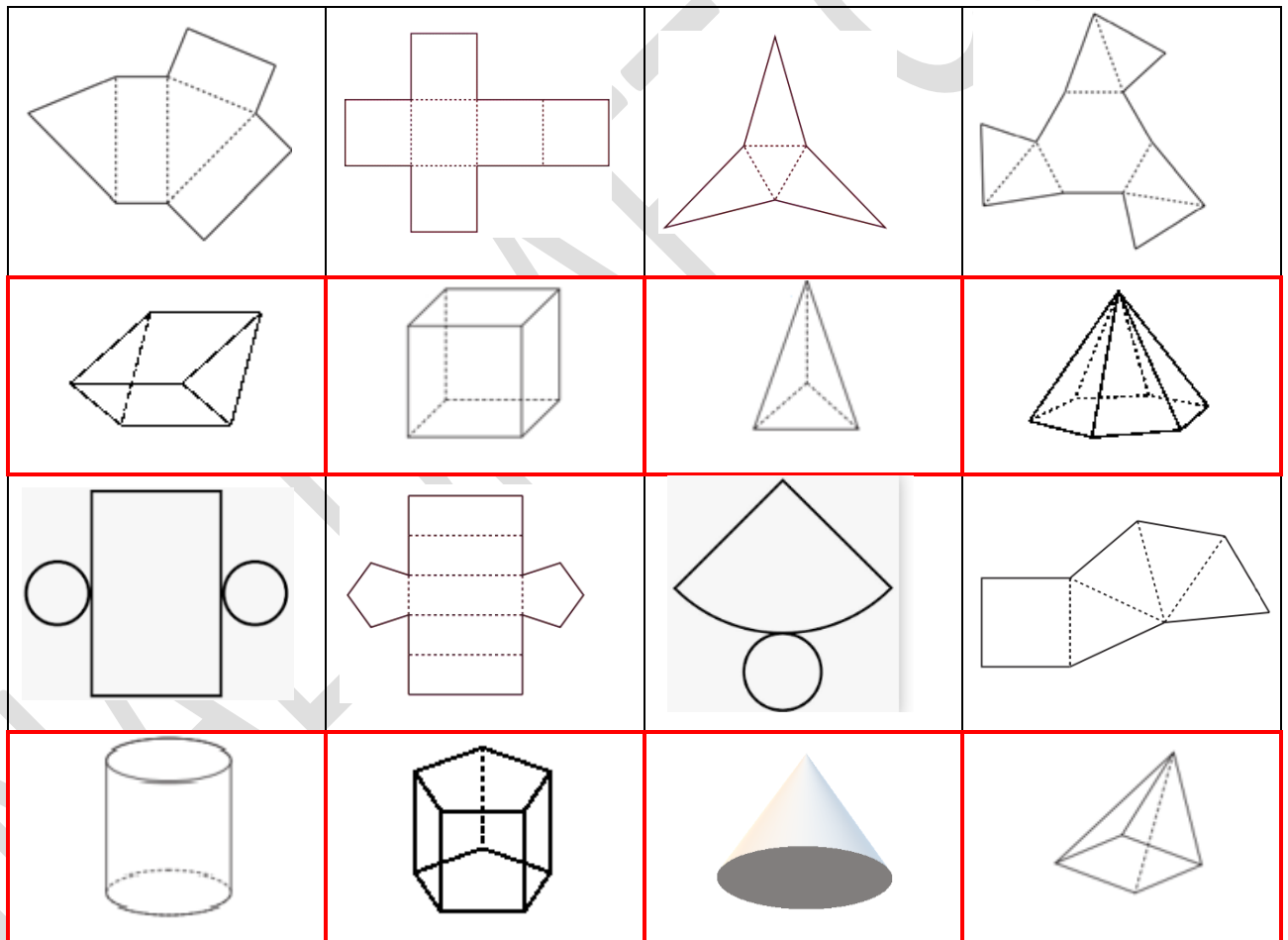
Triangular prisms – C, F and G

## 4.2 Building 3-D models using nets

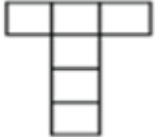
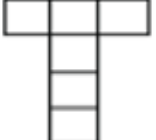
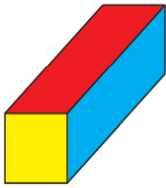
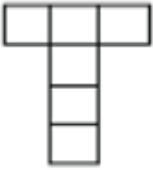
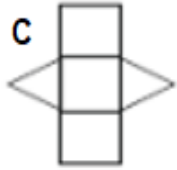
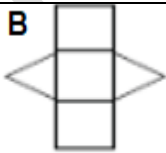
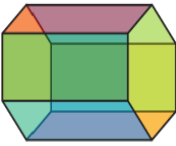
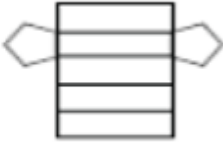
In mathematics, a **net** is a flat pattern that can be folded to form a 3D object. Different 3D objects have different nets. Sometimes the same 3D object can have different nets.



1. Make geometric objects using the nets provided. You will need some sticky tape and a pair of scissors. Examples of nets have been provided but more nets can be added.



2. Choose the matching net for each geometric solid. Write the number and corresponding letter of alphabet.

|      |                                                                                     |   |                                                                                       |                                                                                        |                                                                                         |
|------|-------------------------------------------------------------------------------------|---|---------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| I.   |    | A | A    | B    | C    |
| II.  |    | B | A    | B    | C    |
| III. |    | A | A    | B    | C    |
| IV.  |   | B | A   | B   | C   |
| V.   |  | C | A  | B  | C  |

## UNIT 5

### AREA AND PERIMETER

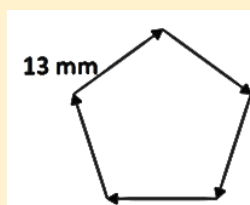
#### 5.1 Calculate the perimeter of regular and irregular polygons

The perimeter of a polygon is the total distance around the polygon.

##### Regular polygons

Regular polygons have all sides and angles equal. The perimeter of regular polygons is found by adding all the sides of a regular polygon or multiplying one side of the regular polygon by the number of its sides.

The pentagon below is an example of a regular polygon. Its perimeter is found by multiplying the 13 mm by 5.



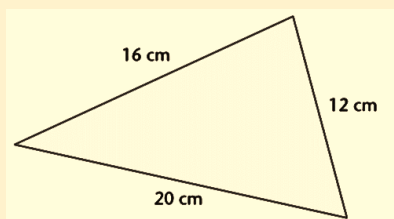
$$\begin{aligned}\text{Perimeter}(P) &= 5 \times 13 \text{ mm} \\ &= 65 \text{ mm}\end{aligned}$$

##### Irregular polygons

A polygon is said to be irregular when its sides are not equal to each other, and the measure of angles may also be unequal. The perimeter of irregular polygons is found by adding all the sides of the irregular polygon.

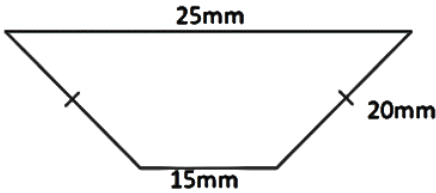
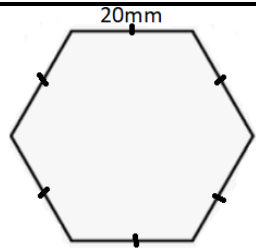
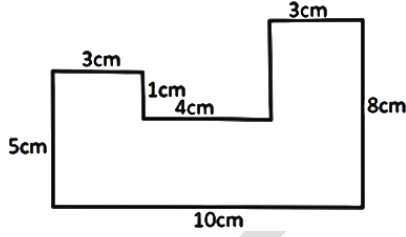
##### Example

Calculate the perimeter of the following shape



$$\begin{aligned}P &= 16 \text{ cm} + 12 \text{ cm} + 20 \text{ cm} \\ &= 48 \text{ cm}\end{aligned}$$

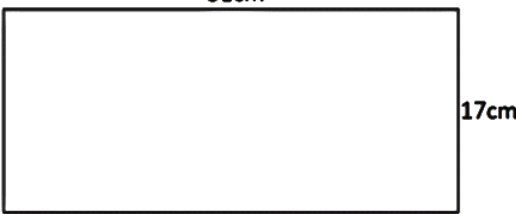
1. Calculate the Perimeter of the polygons below using appropriate formulae. Leave your answer in centimetres.

|                                                                                                                                                                                                                                                                                |                                                                                                                                                                                 |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>(a)</p>  <p><math>P = 25 \text{ mm} + 20 \text{ mm} + 20 \text{ mm} + 15 \text{ mm}</math><br/> <math>= 80 \text{ mm}</math></p>                                                           | <p>(b)</p>  <p><math>P = 6 \times 20 \text{ mm}</math><br/> <math>= 120 \text{ mm}</math></p> |
| <p>(c)</p>  <p><math>P = 3 \text{ cm} + 1 \text{ cm} + 4 \text{ cm} + 4 \text{ cm} + 3 \text{ cm} + 8 \text{ cm} + 10 \text{ cm} + 5 \text{ cm}</math><br/> <math>= 38 \text{ cm}</math></p> |                                                                                                                                                                                 |

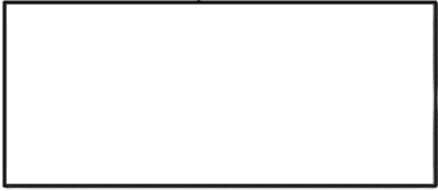
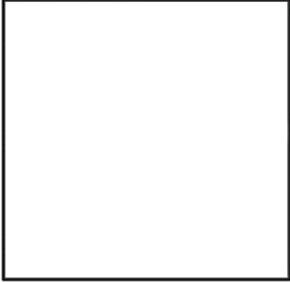
## 5.2 Use appropriate formulae to calculate the perimeter of squares and rectangles

- To calculate the perimeter of a square, we add all the sides of a square. The formula to calculate the perimeter of a square is  $P = 4s$  (where  $s$  represents side).
- The formula to calculate the perimeter of a rectangle is  $P = 2l + 2b$  or  $P = 2(l + b)$ .

- Calculate the perimeter of the following quadrilaterals with the given lengths and breadths. Leave your answer in centimetres.

|                                                                                                |                                                                                             |
|------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| <p>(a)</p>  | <p><math>P = 2 (31 \text{ cm} + 17 \text{ cm})</math><br/> <math>= 96 \text{ cm}</math></p> |
|------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|



|     |                                                                                   |                                                                                        |
|-----|-----------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| (b) |  | $P = 2 (11,3 \text{ cm} + 5,9 \text{ cm})$ $= 2 (17,2 \text{ cm})$ $= 34,4 \text{ cm}$ |
| (c) |  | $P = 4 (9,3 \text{ cm})$ $= 37,2 \text{ cm}$                                           |

(d) If the perimeter of the square is 480mm. What is the length of each side?

$$4s = 480 \text{ mm}$$

$$s = \frac{480}{4} \text{ mm}$$

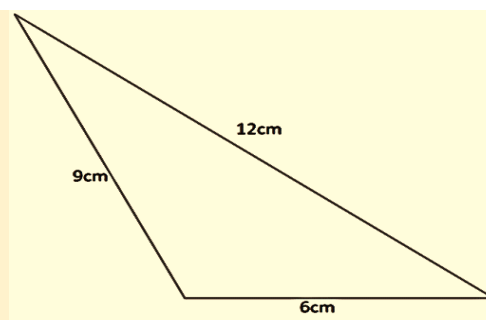
$$= 120 \text{ mm}$$

The length of each side is 120 mm

### 5.3 Use appropriate formulae to calculate the perimeter of triangles and circles

To calculate the perimeter of a triangle, we add all the sides of a triangle. The formula to calculate perimeter of a triangle is  $P = s_1 + s_2 + s_3$

#### Example



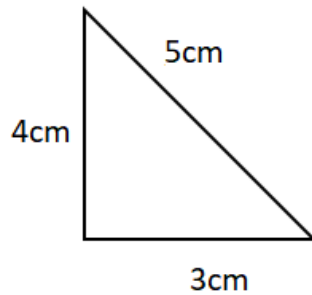
$$P = s_1 + s_2 + s_3$$

$$= 12 \text{ cm} + 9 \text{ cm} + 6 \text{ cm}$$

$$= 27 \text{ cm}$$

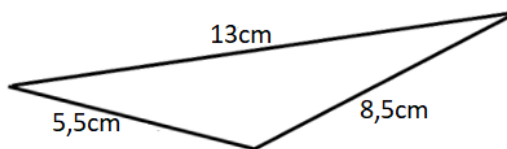
1. Calculate the perimeter of the following triangles given the lengths of the sides.

(a)



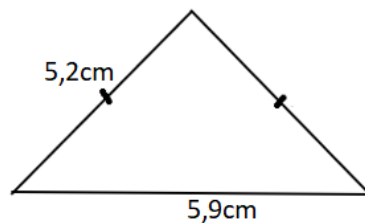
$$\begin{aligned} P &= 5 \text{ cm} + 4 \text{ cm} + 3 \text{ cm} \\ &= 12 \text{ cm} \end{aligned}$$

(b)



$$\begin{aligned} P &= 13 \text{ cm} + 8,5 \text{ cm} + 5,5 \text{ cm} \\ &= 27 \text{ cm} \end{aligned}$$

(c)



$$\begin{aligned} P &= 5,2 \text{ cm} + 5,2 \text{ cm} + 5,9 \text{ cm} \\ &= 16,3 \text{ cm} \end{aligned}$$

To calculate the perimeter of a circle, we take the whole distance around the circle. The circumference of a circle is the same as its perimeter. The formula to calculate the perimeter of a circle is  $P = 2\pi r$

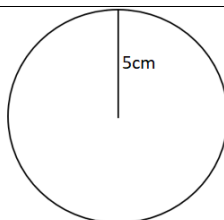
### Example

Calculate the perimeter of the given circle.

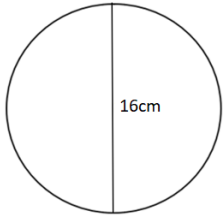
|  |                                                                                                                                         |
|--|-----------------------------------------------------------------------------------------------------------------------------------------|
|  | $\begin{aligned} P &= 2\pi r \\ &= 2\pi(4,5 \text{ cm}) \\ &= 2 \times 3,14 \times 4,5 \text{ cm} \\ &= 28,26 \text{ cm} \end{aligned}$ |
|--|-----------------------------------------------------------------------------------------------------------------------------------------|

2. Calculate the perimeter of a circles below:

(a)



$$\begin{aligned} P &= 2\pi r \\ &= 2 \times 3,14 \times 5 \\ &= 31,4 \text{ cm} \end{aligned}$$

|     |                                                                                                                                                                   |                                                                                                   |
|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| (b) |                                                                                  | $P = 2 \times 3,14 \times \frac{d}{2}$ $= 2 \times 3,14 \times \frac{16}{2}$ $= 50,24 \text{ cm}$ |
| (c) | What is the perimeter of the circle with a diameter of 19cm?<br>$P = 2 \times 3,14 \times \frac{d}{2}$ $= 2 \times 3,14 \times \frac{19}{2}$ $= 59,66 \text{ cm}$ |                                                                                                   |
| (d) | What is the circumference of a pizza if one straight side of each slice equals 3,1cm?<br>$C = 2\pi r$ $= 2 \times 3,14 \times 3,1$ $= 19,47 \text{ cm}$           |                                                                                                   |

## 5.4 Use appropriate formulae to calculate the area of squares and rectangles

### Area of a square

Area of the square is calculated by the formula:  $A = l \times l = l^2$



12mm

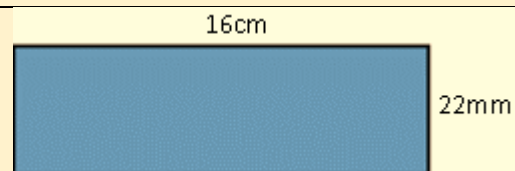
$$A = l \times l$$

$$= 12 \text{ mm} \times 12 \text{ mm}$$

$$= 144 \text{ mm}^2$$

### Area of a rectangle

Formula to calculate Area of a rectangle is  $A = l \times b$



22mm

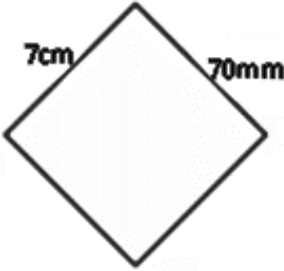
$$A = l \times b$$

$$A = 16 \text{ cm} \times 22 \text{ mm}$$

$$= 16 \text{ cm} \times 2,2 \text{ cm}$$

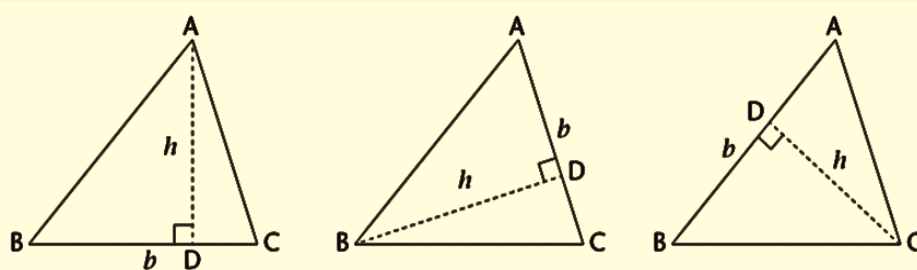
$$= 35,2 \text{ cm}^2$$

1. Calculate the area of the following rectangles and squares

|     |                                                                                                                                                                                                                                                                                                           |                                                                |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|
| (a) |                                                                                                                                                                                                                          | $A = 13 \text{ cm} \times 3 \text{ cm}$<br>$= 39 \text{ cm}^2$ |
| (b) |                                                                                                                                                                                                                          | $A = 7 \text{ cm} \times 7 \text{ cm}$<br>$= 49 \text{ cm}^2$  |
| (c) | <p>Given the area of a square <math>121 \text{ m}^2</math> calculate its length.</p> $121 \text{ m}^2 = s^2$ $\therefore s = \sqrt{121 \text{ m}^2}$ $= 11 \text{ m}$                                                                                                                                     |                                                                |
| (d) | <p>If the area of a rectangle is <math>200 \text{ cm}^2</math>, and its length is <math>50 \text{ cm}</math>, what is its breadth.</p> $200 \text{ cm} = 50 \text{ cm} \times b$ $\therefore b = \frac{200 \text{ cm}^2}{50 \text{ cm}}$ $= 4 \text{ cm}$ <p>The breadth is <math>4 \text{ cm}</math></p> |                                                                |

## 5.5 Use appropriate formulae to calculate the area of triangles and circles

The formula to calculate area of a triangle is  $A = \frac{1}{2} \times b \times h$  (where  $b$  represents the base and  $h$  the height). The height is a line segment drawn from the vertex perpendicular to the base.



AD = height

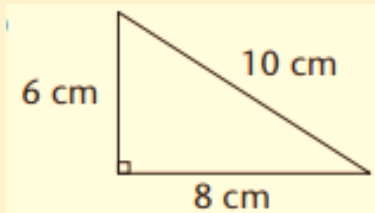
BC = base

BD = height

AC = base

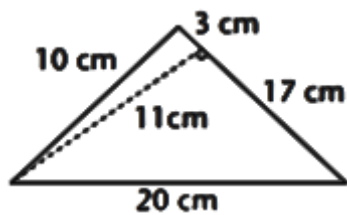
DC = height

AB = base



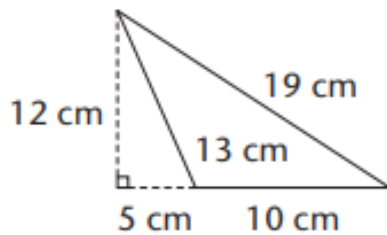
$$\begin{aligned} A &= \frac{1}{2} \times b \times h \\ A &= \frac{1}{2} \times 8 \text{ cm} \times 6 \text{ cm} \\ &= \frac{1}{2} \times 48 \text{ cm}^2 \\ &= 24 \text{ cm}^2 \end{aligned}$$

1. Calculate area of the given triangle.



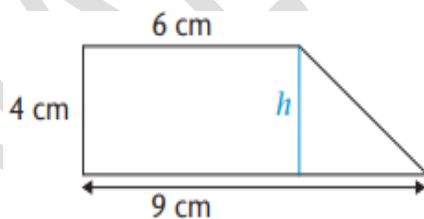
$$\begin{aligned} A &= \frac{1}{2} \times 20 \text{ cm} \times 11 \text{ cm} \\ &= 110 \text{ cm}^2 \end{aligned}$$

2. Calculate the area of the triangle below.



$$\begin{aligned} A &= \frac{1}{2} \times 10 \text{ cm} \times 12 \text{ cm} \\ &= 60 \text{ cm}^2 \end{aligned}$$

3. Work out the length of the sides you need and then calculate the area of the shape.



$$\begin{aligned} \text{Area of rectangle} &= 6 \text{ cm} \times 4 \text{ cm} \\ &= 24 \text{ cm}^2 \end{aligned}$$

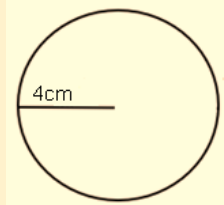
$$\begin{aligned} \text{Height of triangle} &= 9 \text{ cm} - 6 \text{ cm} \\ &= 3 \text{ cm} \end{aligned}$$

$$\begin{aligned} \text{Area of triangle} &= \frac{1}{2} \times 4 \text{ cm} \times 3 \text{ cm} \\ &= 6 \text{ cm}^2 \end{aligned}$$

$$\therefore \text{Area of a trapezium} = 24 \text{ cm}^2 + 6 \text{ cm}^2 = 30 \text{ cm}^2$$

## Area of a circle

The formula to calculate the area of circles is  $A = \pi r^2$  (where  $r$  is the radius of the circle and  $\pi = 3.14$ ). The radius of the circle below is 3cm. The area of the circle is:



$$\begin{aligned} A &= \pi r^2 \\ &= 3,14 \times (4 \text{ cm})^2 \\ &= 3,14 \times 16 \text{ cm}^2 \\ &= 50,24 \text{ cm}^2 \end{aligned}$$

The formula to calculate area of a circle with a given diameter is  $A = \pi \left(\frac{d}{2}\right)^2$

**NB: Diameter of a circle is double its radius.**

1. Calculate the area of circles with the following:

(a) Radius of 83 cm

$$\begin{aligned} A &= 3,14 \times (83 \text{ cm})^2 \\ &= 21\,631,46 \text{ cm}^2 \end{aligned}$$

(b) Diameter of 257 mm

$$\begin{aligned} A &= 3,14 \times \left(\frac{257}{2} \text{ mm}\right)^2 \\ &= 51\,848,47 \text{ mm}^2 \end{aligned}$$

2. Calculate the circumference of a circle with

(a) Radius of 2 cm

$$\begin{aligned} \text{Circumference} &= 2\pi r \\ &= 2 \times 3,14 \times 2 \text{ cm} \\ &= 12,56 \text{ cm} \end{aligned}$$

(b) Diameter of 25 mm.

$$\begin{aligned} \text{Circumference} &= 2\pi r \\ &= 2 \times 3,14 \times \frac{25}{2} \text{ mm} \\ &= 78,5 \text{ mm} \end{aligned}$$

## 5.6 Problem solving involving perimeter and area of polygons

Solve the following problems.

1. A man wants to put a fence around a piece of land to keep his sheep away from his cows. One side fence 6m, the second 7m, the third 8m and the fourth 10m. What is the total perimeter of the fence?

$$\begin{aligned}\text{Perimeter of the fence} &= 6 \text{ m} + 7 \text{ m} + 8 \text{ m} + 10 \text{ m} \\ &= 31 \text{ m}\end{aligned}$$

2. You need to tile the floor of your room with an area of 4,2m x 3,5m. The tiles you want to buy are 45cm x 45cm. How many tiles do you need?

$$\text{Area of the room} = 147\,000 \text{ cm}^2$$

$$\text{Area of each tile} = 2\,025 \text{ cm}^2$$

$$\begin{aligned}\text{Number of tiles needed} &= \frac{147\,000 \text{ cm}^2}{2\,025 \text{ cm}^2} \\ &= 72,6\end{aligned}$$

∴ I will need 73 tiles

3. Mandla draws a circle with a diameter of 16cm. He colours one half of the circle. What is the area of the shaded part?

$$\begin{aligned}\text{Area shaded by Mandla} &= \frac{1}{2} \times 3,14 \times (8 \text{ cm})^2 \\ &= 100,48 \text{ cm}^2\end{aligned}$$

## UNIT 6:

### SURFACE AREA, VOLUME AND CAPACITY

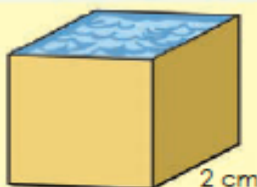
#### 6.1 Using appropriate formulae to calculate the surface area, volume and capacity of cubes

(DBE Workbook 1 - Grade 7, Page 131)

**Example:**

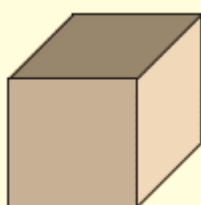
The formula for the

volume of a cube is  $\ell^3$



$$\begin{aligned} 2 \text{ cm} \times 2 \text{ cm} \times 2 \text{ cm} \\ = 8 \text{ cm}^3 \\ = 8 \text{ ml} \\ = 0,008 \ell \end{aligned}$$

1. Calculate the total surface area of the given cube and give the answer in  $\text{cm}^2$

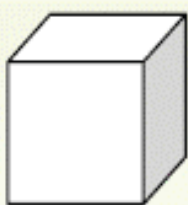


8 mm

**Total Surface Area of a cube**

$$\begin{aligned} &= 6s^2 \\ &= 6 \times (8\text{mm} \times 8\text{mm}) \\ &= 384 \text{ mm}^2 \\ 384 \text{ mm}^2 \div 100 &= 3,84 \text{ cm}^2 \end{aligned}$$

2. Use the formulae to calculate the volume and capacity of water that will fill the following cube.



4 cm

**Volume of a cube =  $\ell^3$  and  $\ell = 4 \text{ cm}$**

$$\begin{aligned} \text{Volume} &= 4 \text{ cm} \times 4 \text{ cm} \times 4 \text{ cm} \\ &= 64 \text{ cm}^3 \end{aligned}$$

**For capacity**

$$1 \text{ cm}^3 \text{ volume} = 1\text{ml capacity}$$

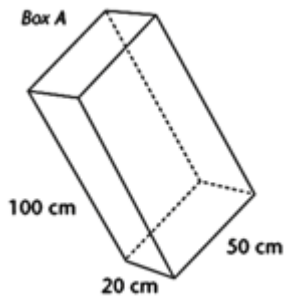
$$\therefore \text{capacity is } 64 \text{ ml}$$



## 6.2 Using appropriate formulae to calculate the surface area, volume and capacity of rectangular prisms.

1. The two boxes on the following page are rectangular prisms. The boxes must be painted.

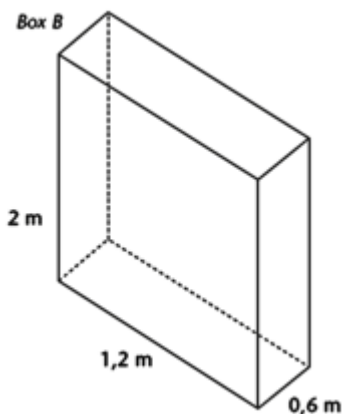
(a) Calculate the total surface area of Box A and of Box B.



Box A

The surface area of a rectangular prism

$$\begin{aligned} &= 2lb + 2lh + 2bh \\ &= 2(100 \times 20 + 20 \times 50 + 100 \times 50) \\ &= 2(2\,000 + 1\,000 + 5\,000) \\ &= 2 \times 8\,000 \\ &= 16\,000 \text{ cm}^2 \end{aligned}$$



Box B

The surface area of a rectangular prism

$$\begin{aligned} &= 2lb + 2lh + 2bh \\ &= 2(2 \times 1,2 + 1,2 \times 0,6 + 0,6 \times 2) \\ &= 2(2,4 + 0,72 + 1,2) \\ &= 2 \times 4,32 \\ &= 8,64 \text{ m}^2 \end{aligned}$$

- (b) What will it cost to paint both boxes if the paint costs R1,34 per square metre?

$$\text{Box A} = 16\,000 \text{ cm}^2 = 1,6 \text{ m}^2$$

$$\text{Total surface area of box A + box B: } 1,6 + 8,64 = 10,24 \text{ m}^2$$

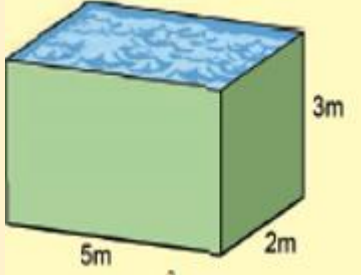
$$\text{Cost of paint: } 10,24 \times 1,34 = \text{R}13,72$$

### The relationship between capacity and volume

The volume of a solid is the amount of space it occupies. Capacity is the amount of space a liquid or other substance can occupy. (DBE Workbook Grade 7, Page 129)

### Example

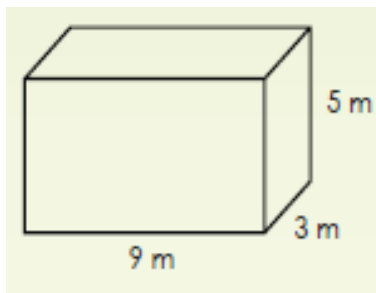
Use the formula to calculate the volume of water that will fill each cube.



This container will hold 30 000 000 ml or 30 000 ℓ of water.

|                                                                                                                   |                                                                                                                                        |                                                                                                                                                      |
|-------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\frac{\text{m}^3}{l \times b \times h}$ $= 5 \text{ m} \times 2 \text{ m} \times 3 \text{ m}$ $= 30 \text{ m}^3$ | $\frac{\text{cm}^3}{l \times b \times h}$ $= 500 \text{ cm} \times 200 \text{ cm} \times 300 \text{ cm}$ $= 30\,000\,000 \text{ cm}^3$ | $\frac{\text{mm}^3}{l \times b \times h}$ $= 5\,000 \text{ mm} \times 2\,000 \text{ mm} \times 3\,000 \text{ mm}$ $= 30\,000\,000\,000 \text{ mm}^3$ |
|-------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|

2. Calculate the volume of the following and give your answer in  $\text{m}^3$ ,  $\text{cm}^3$  and  $\text{mm}^3$ . Also say what the capacity of each container is when filled with water.



$$\begin{aligned} \text{Volume} &= (l \times b) \times h \\ &= (9 \text{ m} \times 3 \text{ m}) \times 5 \text{ m} \\ &= 135 \text{ m}^3 \end{aligned}$$

$$\begin{aligned} \text{Volume} &= (l \times b) \times h \\ &= (900 \text{ cm} \times 300 \text{ cm}) \times 500 \text{ cm} \\ &= 135\,000\,000 \text{ cm}^3 \end{aligned}$$

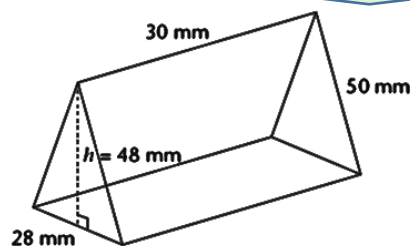
$$\begin{aligned} \text{Capacity} &= 135 \text{ m}^3 \\ &= 135\,000\,000 \text{ cm}^3 \\ &= 135\,000\,000\,000 \text{ mm}^3 \end{aligned}$$

$$\begin{aligned} \text{Volume} &= (l \times b) \times h \\ &= (9\,000 \text{ m} \times 3\,000 \text{ m}) \times 5\,000 \text{ m} \\ &= 135\,000\,000\,000 \text{ mm}^3 \end{aligned}$$

### 6.3 Using appropriate formulae to calculate the surface area, volume and capacity of triangular prisms.

1. Work out the surface areas of the following object. Give all the answers in  $\text{cm}^2$ .

(DBE Textbook- Grade 8 LB, Pg 78)



Surface area of a triangular Prism =  $2 \times \frac{1}{2}bh + lw + 2(lw)$

$$= 2 \times \left( \frac{1}{2} \times 28 \text{ mm} \times 48 \text{ mm} \right) + (28 \text{ mm} \times 30 \text{ mm}) + 2 \times (30 \text{ mm} \times 50 \text{ mm})$$

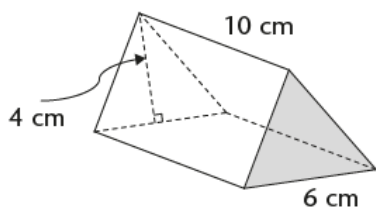
$$= 2 \times 672 \text{ mm}^2 + 840 \text{ mm}^2 + 2 \times 1\,500 \text{ mm}^2$$

$$= 1\,344 \text{ mm}^2 + 840 \text{ mm}^2 + 3\,000 \text{ mm}^2$$

$$= 5\,184 \text{ mm}^2$$

$$= 51,84 \text{ cm}^2$$

2. Calculate the volume and capacity of the following objects using the formulae



$$\text{Volume} = \left( \frac{1}{2}bh \right) \times H$$

$$\text{Volume} = \frac{1}{2} \times (6 \text{ cm} \times 4 \text{ cm}) \times 10 \text{ cm} = 120 \text{ cm}^3$$

$$1 \text{ cm}^3 = 1 \text{ ml}$$

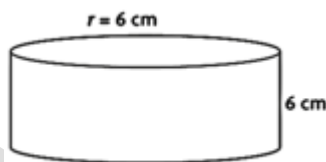
$$\therefore \text{Capacity} = 120 \text{ ml}$$

#### 6.4 Using appropriate formulae to calculate the surface area, volume and capacity of cylinders.

DBE Textbook LB Gr 9 page 80)

1. Calculate the surface areas of the following objects. Use  $\pi = 3,14$  and round off all your answers to two decimal places.

(a)

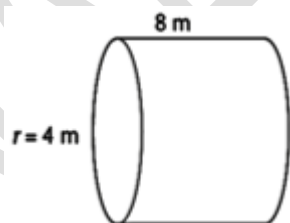


$$SA = 2\pi r (r + h)$$

$$= 2 \times 3,14 \times 6 \times (6 + 6)$$

$$= 452,16 \text{ cm}^2$$

(b)



$$SA = 2\pi r (r + h)$$

$$= 2(3,14) \times 4 \times (4 + 8)$$

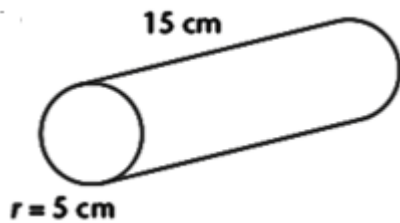
$$= 301,44 \text{ m}^2$$

2. Calculate the volume and capacity of the following cylinders. Use  $\pi = 3,14$  and round off all answers to two decimal places.

DBE Textbook LB Gr 9 page 83)

$$\text{Volume of a cylinder} = \text{Area of base} \times h = \pi r^2 \times h$$

(a)



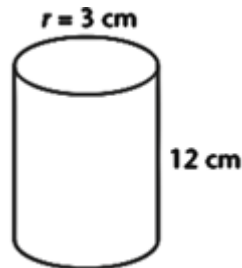
$$V = \pi r^2 \times h$$

$$= 3,14 \times 5 \times 5 \times 15$$

$$= 1\,177,5 \text{ cm}^3$$

The capacity is = 1 177,5 cm<sup>3</sup>/ml

(b)



$$V = \pi r^2 \times h$$

$$= 3,14 \times 9 \times 12$$

$$= 339,12 \text{ cm}^3$$

The capacity is 339,12cm<sup>3</sup> / ml

## 6.5 Solving problems in context involving.

1. A rectangular – shaped petrol tank is used to store a reserved petrol. The tank has a side length of 8m, width of 3m and a height of 4m.

(a) What is the capacity of the tank? Write your answer in litres?

$$\text{Volume} = (l \times b) \times h$$

$$= (8 \text{ m} \times 3 \text{ m}) \times 4 \text{ m}$$

$$= 96 \text{ m}^3$$

(b) If the tank is quarter full of petrol, what does it hold in kilolitres?

$$1 \text{ m}^3 \text{ volume} = 1\text{k}\ell \text{ capacity}$$

$$\therefore \text{capacity is } 96 \text{ k}\ell = 96\,000 \text{ litres}$$

## UNIT 7 DATA HANDLING

Data handling refers to the process of collecting the set of information, organize and represent in different forms. Data can be about many different things, for example, people's opinions on politics or the success rates of treating people with a certain kind of medicine. We use data to help us make decisions and solve problems about the world around us.

### 7.1 Collect and organize data

#### Collecting Data

Data collection is a systematic process of gathering and measuring information. A questionnaire with suitable multiple choice questions can be a tool to collect the desired information.

The method of collecting data will help us to identify the most appropriate tool to be used when collecting the data. There are various sources of data collection.

#### Multiple-choice Questionnaire

Year 4 learners are involved in a recycling project. You want to determine which type of material they are mostly bringing to school. You want to investigate this by means of a survey. A multiple-choice questionnaire can be used to record your data.

1. Design a simple questionnaire for your survey, using multiple-choice questions. Your questionnaire must also include:
  - (a) Gender
  - (b) Type of material

| SURVEY                                            |              |       |      |         |       |
|---------------------------------------------------|--------------|-------|------|---------|-------|
| TYPE OF RECYCLING MATERIAL YOU BROUGHT TO SCHOOL  |              |       |      |         |       |
| Please assist by answering a few questions        |              |       |      |         |       |
| Gender (tick the correct box)                     | Boy          |       | Girl |         |       |
| Type of recycling material (tick the correct box) | Plastic bags | Paper | Tins | Bottles | Boxes |
|                                                   |              |       |      |         |       |

2. Use the multiple-choice question that you developed in Question (1) to collect the data from 60 learners (30 boys and 30 girls). Keep the results for the next section.

**NB: Any suitable questionnaire will be acceptable.**

### Organising data

Organizing data is a systematic way of bringing data together in a way that makes it easier to read. Tally marks (I) can be used to record data in different categories as shown in the callout. Tallies are grouped into five to make it easier to count them quickly.

**Tally marks**  
 A count of three = III  
 A count of five = IIII  
 A count of seven = IIII II

3. Use the information you have collected on recycling material per gender to construct a tally table.

| Recycling material | Tally for boys | Frequency | Tally for girls | Frequency |
|--------------------|----------------|-----------|-----------------|-----------|
| Plastic            |                |           |                 |           |
| Paper              |                |           |                 |           |
| Tin                |                |           |                 |           |
| Bottle             |                |           |                 |           |
| Box                |                |           |                 |           |

**Learners own collected data and present in class.**

- (a) Portia's collected data is organized in the table below:

| Recycling material<br>Gender | Plastic bags | Paper | Tins | Bottles | Boxes |
|------------------------------|--------------|-------|------|---------|-------|
| Girls                        | 2            | 10    | 12   | 5       | 1     |
| Boys                         | 5            | 3     | 6    | 12      | 4     |

Complete the table using Portia's data.

| Recycling material | Tally (total number of learners) | Frequency (total number of learners) |
|--------------------|----------------------------------|--------------------------------------|
| Plastic bags       |                                  | 7                                    |
| Paper              |                                  | 13                                   |
| Tins               |                                  | 18                                   |
| Bottles            |                                  | 17                                   |
| Boxes              |                                  | 5                                    |

- (b) The table below represents the quantity of recycling material collected Year 4 learners. Construct a frequency table for this set of data values, using grouped data, grouped in five groups with intervals of 10.

|    |    |    |    |    |    |    |    |
|----|----|----|----|----|----|----|----|
| 58 | 68 | 60 | 71 | 53 | 62 | 64 | 72 |
| 63 | 46 | 61 | 52 | 67 | 54 | 63 | 78 |
| 78 | 62 | 68 | 55 | 69 | 81 | 76 | 62 |
| 52 | 64 | 65 | 74 | 59 | 66 | 63 | 76 |
| 55 | 74 | 74 | 59 | 51 | 59 | 67 | 71 |

| Quantity of recycling material | Tally | Frequency |
|--------------------------------|-------|-----------|
| 40 – 50                        |       | 1         |
| 51 – 60                        |       | 12        |
| 61 – 70                        |       | 16        |
| 71 – 80                        |       | 10        |
| 81 – 90                        |       | 1         |

## 7.2 Summarising Data

### Measures of central tendency and measures of spread and dispersion

A measure of central tendency is a single value that attempts to show a central position of a set of data. There are three types of measures of central tendency i.e. mode, median and mean.

#### Mode

The mode is the data value that appears most often in a set of data. No calculation is needed to find the mode. You just find the value that appears most frequently. If no number appears more than the other numbers, then there is no mode. It is important to arrange data in ascending or descending order in order to easily identify the mode.

#### Median

The median is the middle number in the data set. To determine the median, you have to write all the numbers in the data set from the smallest to the highest and the number in the middle will be your median. If there is more than one number in the middle (i.e. if the data contains an even number of data values) add the two numbers in the middle and divide the answer by 2.

#### Mean

The mean or the average is the most common measure of central tendency that is used, but it can be easily influenced by high or low numbers in the data set. It is calculated by adding all the values together and dividing by the number of values in the data set.

#### Range

The range of a data set is the difference between the highest value and the lowest value. For example, the range of the data 18; 12; 10, 10, 11 is:  $18 - 10 = 8$ .

The bigger the range, the more the data is spread out. The smaller the range, the more the data is clustered around similar values



## Example

Use the combined data (for both boys and girls) from Portia's findings to determine the mode, median, mean and the range.

| Recycling material<br>Gender | Plastic bags | Paper | Tins | Bottles | Boxes |
|------------------------------|--------------|-------|------|---------|-------|
| Girls                        | 2            | 10    | 12   | 5       | 1     |
| Boys                         | 5            | 3     | 6    | 12      | 4     |

Data in ascending order: 1; 2; 3; 4; 5; 5; 6; 10; 12; 12

|        |                                                                  |              |
|--------|------------------------------------------------------------------|--------------|
| Mode   | 1; 2; 3; 4; <b>5; 5</b> ; 6; 10; <b>12; 12</b>                   | <b>5, 12</b> |
| Median | 1; 2; 3; 4; <b>5; 5</b> ; 6; 10; 12; 12                          | <b>5</b>     |
| Mean   | $\frac{1; 2; 3; 4; 5; 5; 6; 10; 12; 12}{10} = \frac{60}{10} = 6$ | <b>6</b>     |
| Range  | $12 - 1 = 11$                                                    | <b>11</b>    |

1. Use the data you collected (about types of recycling material) for both girls and boys to determine the mode, median, mean and range.

|        |  |  |
|--------|--|--|
| Mode   |  |  |
| Median |  |  |
| Mean   |  |  |
| Range  |  |  |

Responses will depend on each learner's collected data.

2. Use the following data set to calculate the range, mode, median and the mean.

The table represents times in seconds taken by the different learners in Year 3 in a certain school to wash their hands:

|    |    |    |    |    |    |    |    |    |    |    |    |    |
|----|----|----|----|----|----|----|----|----|----|----|----|----|
| 20 | 30 | 36 | 14 | 20 | 14 | 29 | 39 | 15 | 37 | 35 | 24 | 29 |
| 29 | 18 | 16 | 38 | 13 | 24 | 27 | 22 | 38 | 29 | 11 | 38 |    |

3. The table represents times in seconds taken by the different learners in Year 4 in the same school to wash their hands:

|    |    |    |    |    |    |    |    |    |    |    |    |    |
|----|----|----|----|----|----|----|----|----|----|----|----|----|
| 20 | 12 | 39 | 22 | 16 | 37 | 36 | 15 | 14 | 13 | 16 | 10 | 14 |
| 13 | 16 | 37 | 36 | 15 | 14 | 13 | 16 | 10 | 14 | 26 | 11 |    |

(a) Calculate the range for each set of data

Year 3:  $39 - 11 = 28$

Year 4:  $39 - 10 = 29$

(b) What can you say about the two groups by looking at the range?

Range is big for both of them therefore the data is spread out.

(c) Calculate the mean (average) for each group

| Class  | Total | Number | Mean |
|--------|-------|--------|------|
| Year 3 | 645   | 25     | 25,8 |
| Year 4 | 485   | 25     | 19,4 |

(d) Compare the two set of data using the mean

On average the year 3 learners take more time on washing their hands than year 4 learners.

(e) Find the median for each class

| Class  | Scores from highest to lowest                                                 | Middle position | Median |
|--------|-------------------------------------------------------------------------------|-----------------|--------|
| Year 3 | 11 13 14 14 15 16 18 20 20 22 24 24<br>27 29 29 29 29 30 35 36 37 38 38 39 39 | 27              | 27     |
| Year 4 | 10 10 11 12 13 13 13 14 14 14 14 15 15<br>16 16 16 16 20 22 26 36 36 37 37 39 | 15              | 15     |

(f) Compare the two sets of data using the median

Year 3 learners' median is higher than year 4 learners which shows that year 3 learners spend more time washing their hands.

(g) Find the mode for each class

| Class  | Highest frequency | Mode      |
|--------|-------------------|-----------|
| Year 3 | 3                 | 29 and 38 |
| Year 4 | 4                 | 14 and 16 |

(h) Compare the two sets of data using the mode

Year 3 learners' mode is 29 and 38 (3 learners). Year 4 learners' mode is 14 and 16 (4 learners). This shows that more year 4 learners spend more time washing their hands than in year 3.

## 7.3 Represent data

Data can be represented using different types of graphs. They help with displaying and reporting data and make it easier to report patterns and relationships. We will be working with three graphs i.e. bar graphs, double bar graphs, histograms, pie charts and broken line graph.

### Bar graph

A bar graph usually shows categories (or classes) of data along the horizontal axis, and the frequency of each category along the vertical axis.

A bar graph is a visual display used to compare the frequency of occurrence of different characteristics of data.

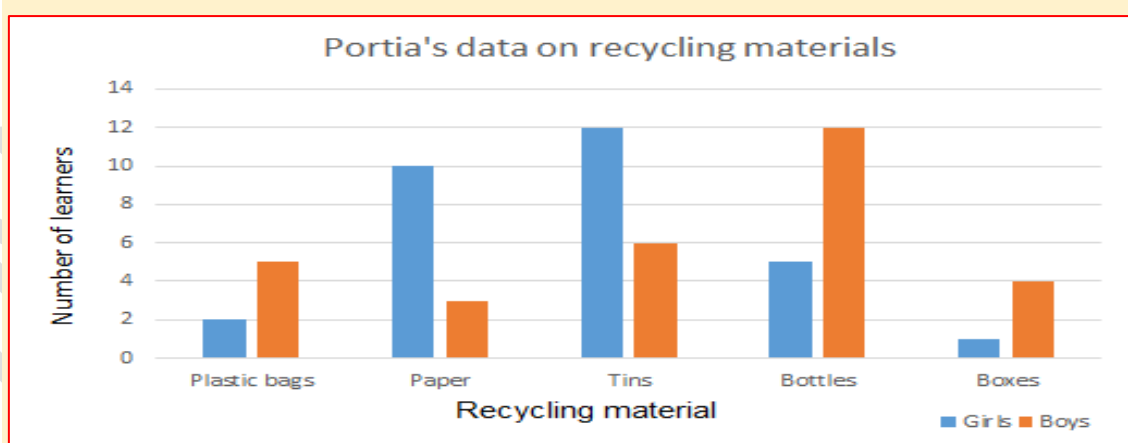
### Double bar graph

A double bar graph shows two sets of data for each category (or class).

#### Example

Use Portia's frequency table to represent data in a double bar graph.

| Recycling material \ Gender | Plastic bags | Paper | Tins | Bottles | Boxes |
|-----------------------------|--------------|-------|------|---------|-------|
| Girls                       | 2            | 10    | 12   | 5       | 1     |
| Boys                        | 5            | 3     | 6    | 12      | 4     |



1. Use your frequency table (on recycling materials per gender) to represent data in a bar graph (only girls)

Learner's own collected data.

## Histogram

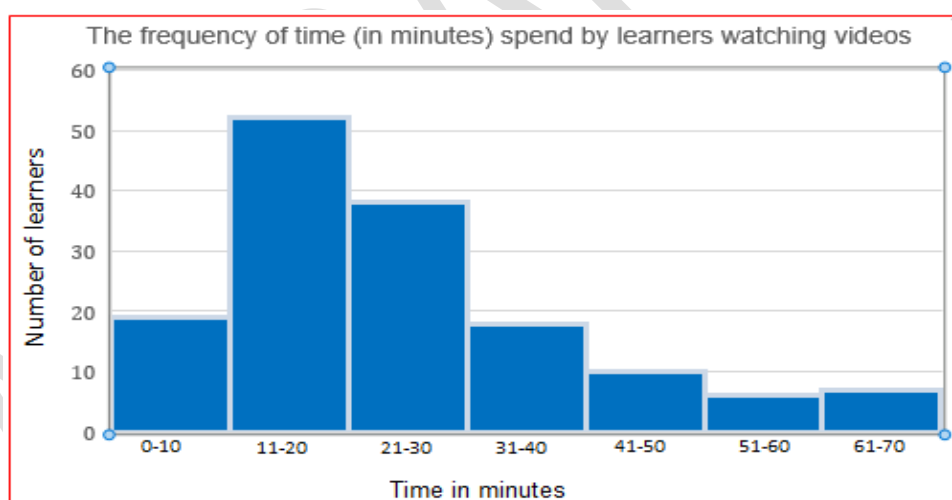
2. The learners from Year 1 to 4, at school B were asked how much time (in minutes) did they spend watching videos in YouTube. The following table shows the results from a sample of 150 learners:

| Time in minutes | Frequency |
|-----------------|-----------|
| 0 – 10          | 19        |
| 11 – 20         | 52        |
| 21 – 30         | 38        |
| 31 – 40         | 18        |
| 41 – 50         | 10        |
| 51 – 60         | 6         |
| 61 – 70         | 7         |

- (a) In which interval are most learners on you tube?

11 to 20 minutes

- (b) Draw the histogram to illustrate the data



- (c) Describe in your own words what the histogram shows.

It shows a pattern of most learners spending 0 to 30 minutes watching videos on YouTube (109 out of 150); and the number of those who spend more than 50 minutes are less (13 out of 150)

- (d) What would you expect this data to look like for a school in a farming area?

Few learners who will be able to connect, can spend between 0 to 30 minutes; because cost of data and connectivity, and lack of access to smart phones and other gadgets.

## Pie chart

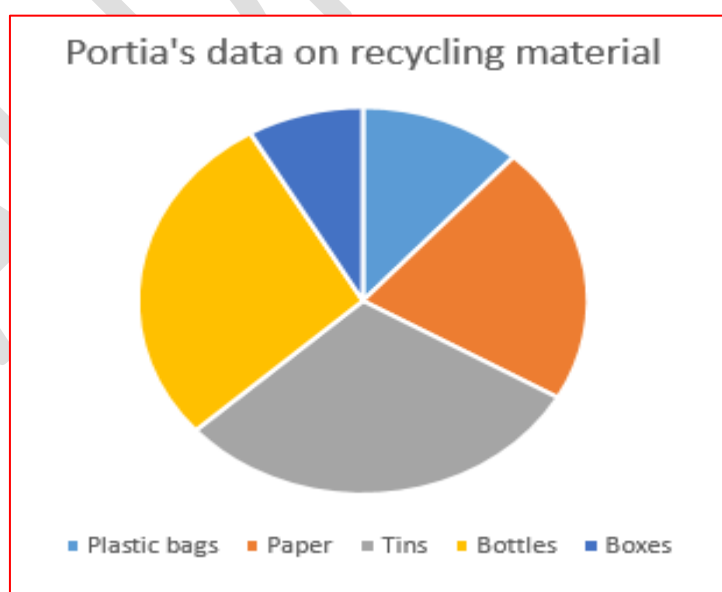
A pie chart is a circular graph in which the circle is divided into sectors. Each sector visually represents an item in a data set to match the amount of the item as a percentage or fraction of the total data set. (the whole circle) Pie charts are useful to compare different parts of a whole amount. They are often used in media to show clear and important differences, but they cannot show shape and spread of data

3. Use your frequency table (on recycling material per gender) to represent your data in a pie chart (only boys)

Learner's own collected data.

4. Use Portia's frequency table to represent the data in a pie chart (total number of learners in each gender)

| Recycling material<br>Gender | Plastic bags | Paper | Tins | Bottles | Boxes |
|------------------------------|--------------|-------|------|---------|-------|
| Girls                        | 2            | 10    | 12   | 5       | 1     |
| Boys                         | 5            | 3     | 6    | 12      | 4     |



## Broken line graph

A broken line graph is a line that joins consecutive data points plotted on a set of axes. It is useful to show how something has changed or stayed the same over time; and where the data values vary considerably e.g. where the rainfall goes up and down. Meteorologists use line graphs to show monthly rainfall. Businesses often use line graphs to show information about profits. With some line graphs it might be possible to continue the line to show what might happen in the future.

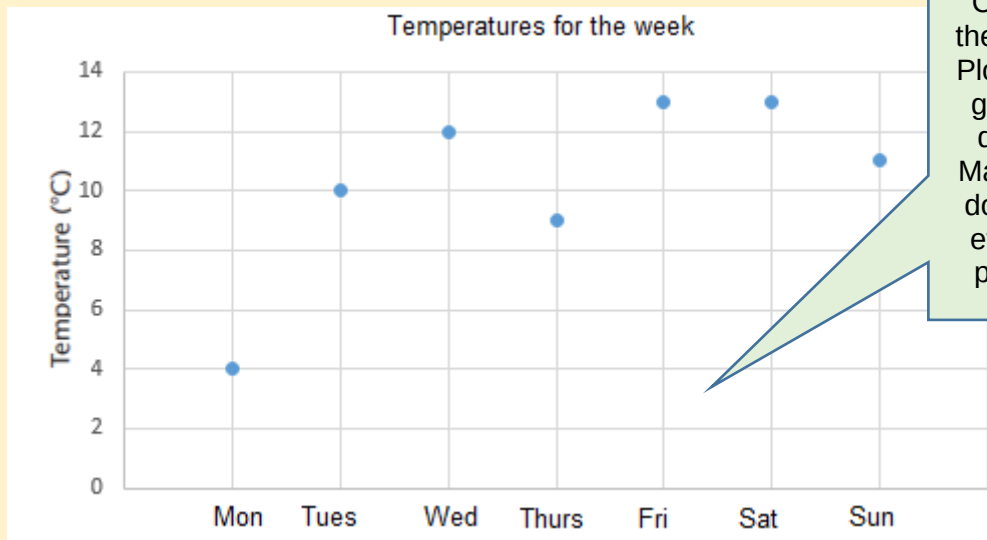
### Steps to draw the broken line:

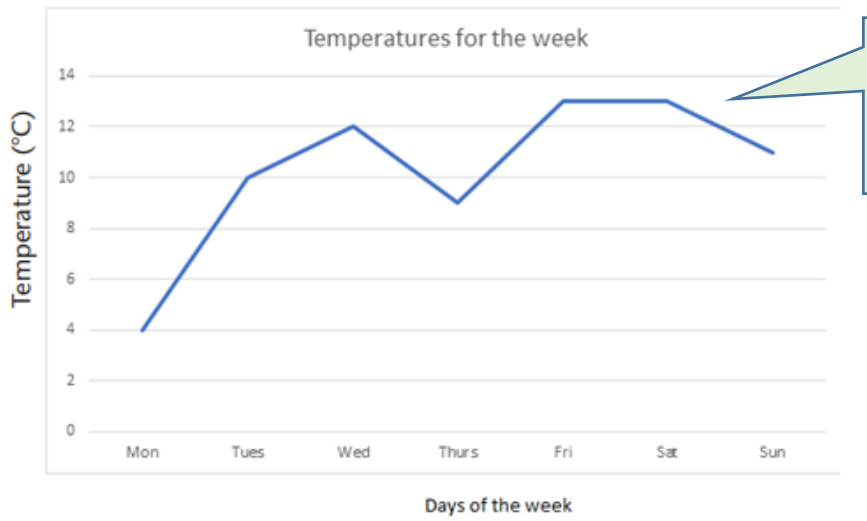
- Copy the axes.
- Plot the given data on the set of axes.
- Make a dot for every point that you plot
- Use a ruler to join the dots in order. You now have drawn a broken-line graph

### Example:

The following table shows the average temperature in Mahikeng recorded every day for one week:

| Day              | Mon | Tues | Wed | Thurs | Fri | Sat | Sun |
|------------------|-----|------|-----|-------|-----|-----|-----|
| Temperature (°C) | 4   | 10   | 12  | 9     | 13  | 13  | 11  |





Use a ruler to join the dots

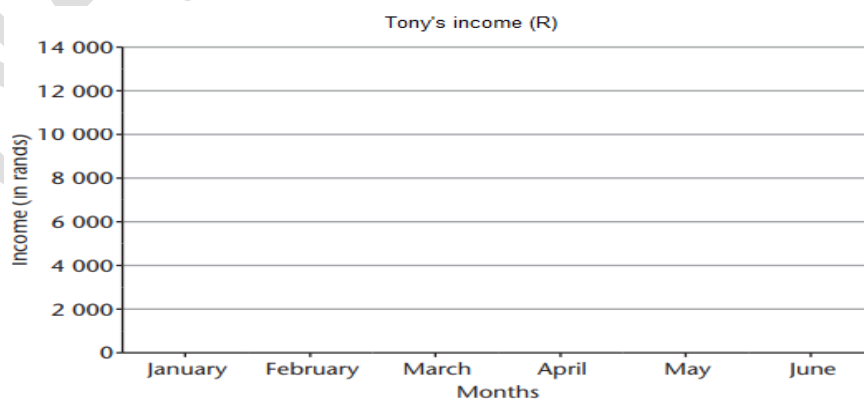
5. Search for a broken line graph in a newspaper or the internet. Redraw it and then describe it.

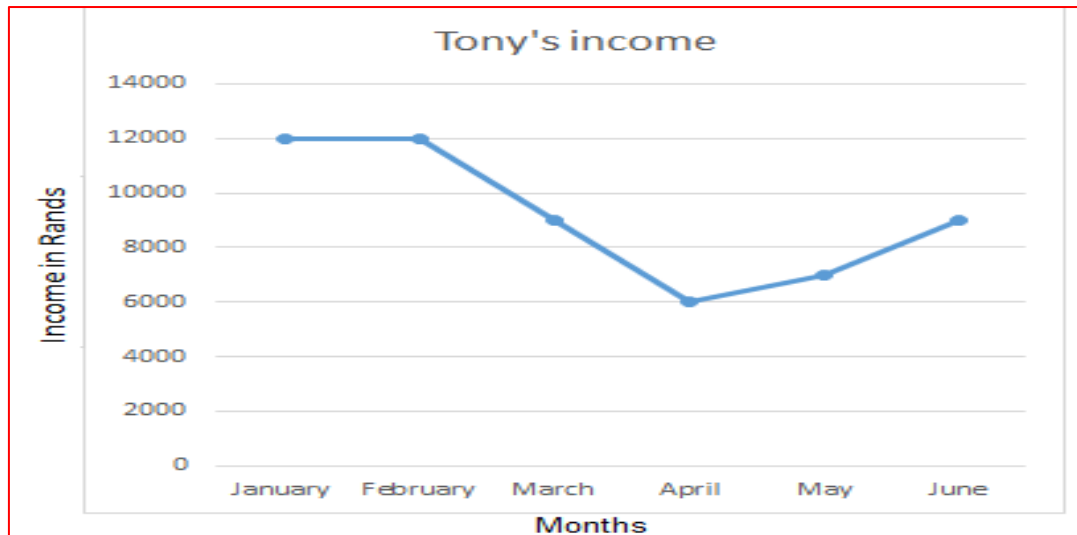
**Learner's own broken line and description**

6. The following table shows the income of Tony's small business and Tonya's small business over six months:

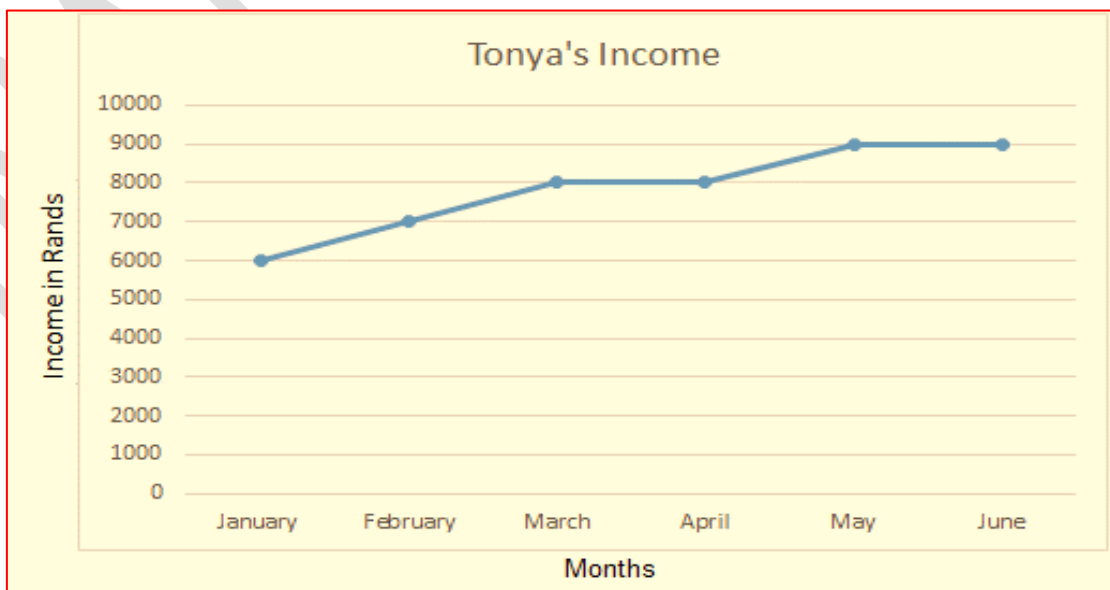
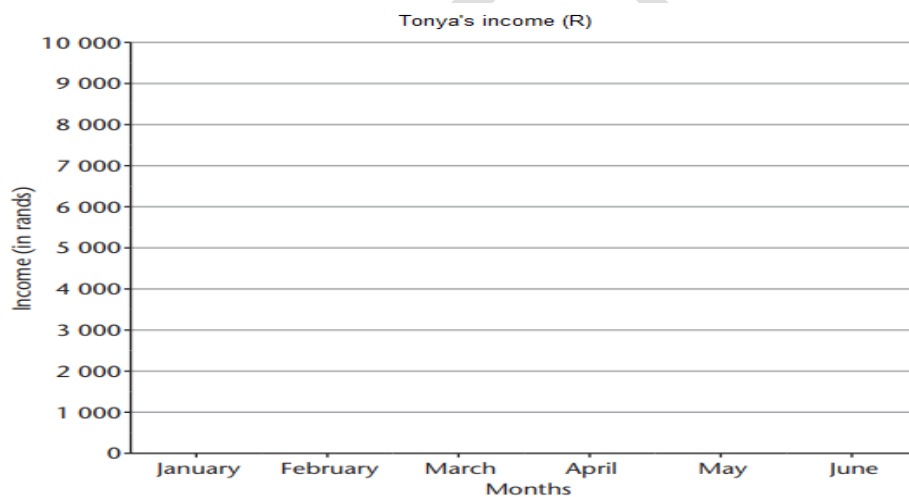
| Month              | January | February | March | April | May   | June  |
|--------------------|---------|----------|-------|-------|-------|-------|
| Tony's income (R)  | 12 000  | 12 000   | 9 000 | 6 000 | 7 000 | 9 000 |
| Tonya's income (R) | 6 000   | 7 000    | 8 000 | 8 000 | 9 000 | 9 000 |

- (a) Copy the axes below and draw a broken-line graph showing Tony's income





(b) Copy the axes below and draw a broken-line graph showing Tonya's income





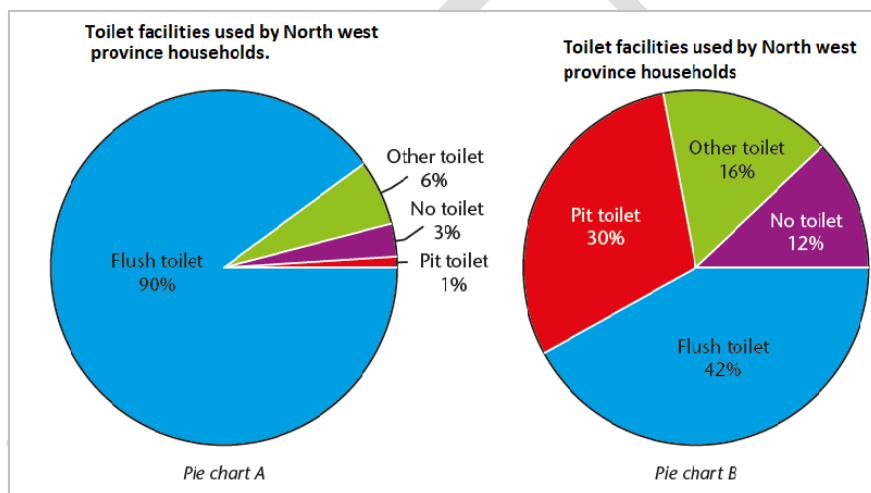
## 7.4 Interpret and reporting

Data **analysis** and **interpretation** is a process of assigning meaning to the collected information and determining the conclusion, significance and implications of the findings. There are four steps to data interpretation i.e. collect data required; develop findings; develop conclusions and develop recommendations.

There are four steps to data interpretation.

- Collect the information you need.
- Develop findings
- Develop conclusions
- Develop recommendations.

1. The pie chart below represent data collected about toilet facilities in different years in households in North West province.



- (a) According to pie chart A, what type of toilet facility do most people have and what percentage of households is this?

Flush toilets, 90% of households.

- (b) How will your answer for (a) be different if you use pie chart B to answer the question?

According to pie chart B, 42% of households have flush toilets.

- (c) Write a short report in one paragraph about the data in the pie charts.

Pie chart A shows that 90% of households use flush toilets and pie chart B shows that less than half (42%) use flush toilets. 1% use pit toilets in pie chart A while in chart B they are 30%. In pie chart A 6% of households use other toilet facilities, while in pie chart B they use 16%. In pie chart

A 3% of households have no toilets and pie chart B 12% do not have toilets.

There is a big difference between the two sets of data

(d) What can you conclude from the data in pie chart A?

Almost all households have flush toilets.

(e) What can you conclude from the data in pie chart B?

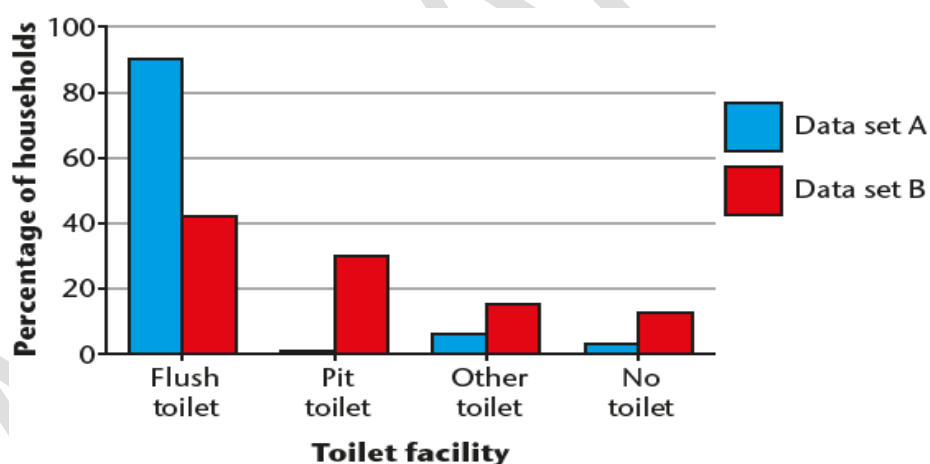
Less than half (42% of all households have flush toilets. Almost a third (30%) use pit toilets.

(f) How could the year of data collection account for the differences in the data?

Possible answer: The data in pie chart A was probably collected many years after the data in pie chart B. By then more flush toilets had replaced other types of toilets.

2. The graph below shows the same data as the two pie charts above.

**Toilet facilities used by North west province households**



(a) Which graph allows us to compare the two sets of data more easily.

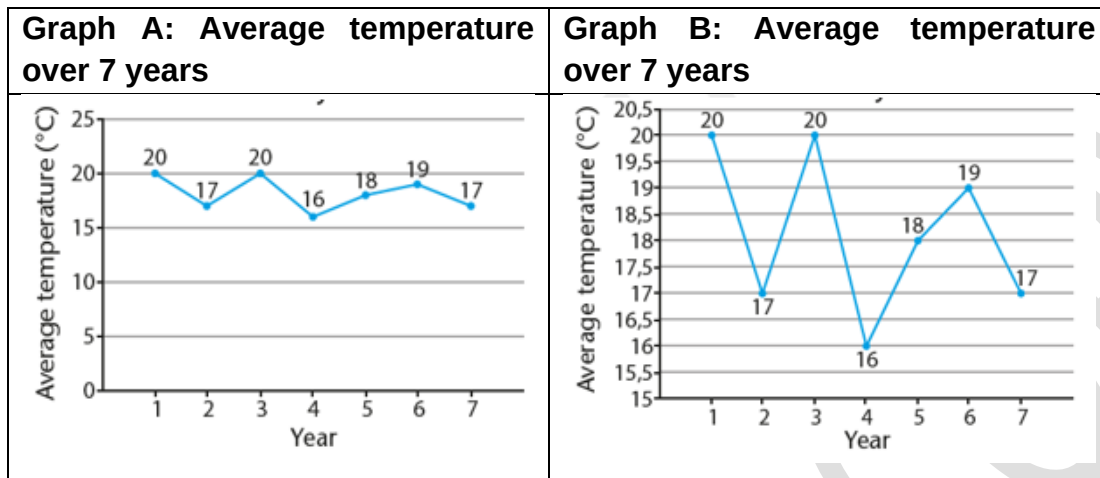
Motivate your answer

The double bar graph, because you have both data in one graph so can easily see the difference.

(b) Do the pie chart or the double bar graph show the percentage of types of toilet facilities used in North West province better?

The pie charts (in the pie chart the percentages are indicated as they are while in the double bar graph they are indicated in a scale.

3. The graphs represent the average temperatures recorded in the same place, at the same time over a period of seven years.



(a) Do both graphs show exactly the same data? **Yes, they do.**

(b) Why do the graphs look so different?

Graph A uses a vertical scale that increases in units of 5. Graph B does not start at zero and it increases in units of 0,5 °C. Therefore, the vertical distances between two consecutive data points are much smaller in Graph A (or much bigger in Graph B).

(c) Which of the graphs would people use to emphasise that there are big differences in the temperatures over the years? Explain your answer

They would use Graph B, because the steep up and down slopes of the broken-line graph give the impression that there are big variations (changes) between the temperatures.

(d) Suggest a way to change the vertical scale of Graph A to emphasise even more that there are no big differences between the temperatures over the years.

Change the scale to units bigger than 5, °C or example units of 10°C

(e) Write a short report on Graph A. Also include a prediction of the temperatures in years 8 and 9.

The graph shows that the temperatures stay more or less constant, between 16°C and 20°C over the seven years. Therefore, the temperatures for Years 8 and 9 would most likely also stay in this temperature range.

## 7.5 Reporting Data

To report data one needs to consider the following:

**Plan:** What data do you need? Who will you get it from? How will you collect it? How will you record it? How will you make sure the data is reliable? Give reasons for the choices you made.

**Analysis:** This is where you do the calculations and draw charts.

- Compare groups with the mean and median.
- The range is a measure of how spread out the group is.
- Graphs are good for representing visual data.

**Conclusions:** Do your results agree with the hypothesis? How confident are you? What went wrong? How did you deal with it? What would you do differently if you did the research again?

1. Read the following paragraph and answer the questions that follows:

A recent study revealed that 35% of Year 4 learners smoke cigarettes, 49% drink alcohol and 8% abuse drugs. This is an indication of the general poor health and social problems of the teenage population in our country.

(a) Do you agree that the figures are high enough to conclude that the habits of these teenagers are unhealthy?

Yes, the percentages are a concern.

(b) Can we tell the following from the data above?

- |                                      |    |
|--------------------------------------|----|
| ▪ What was the sample of this study? | No |
| ▪ Where was the data collected?      | No |
| ▪ When was the data collected?       | No |

(c) If the sample consisted of ten teenagers all located in an area known for drug and alcohol abuse, would the data be a good reflection of all teenagers in the country?

No

(d) Describe what you think would be a better sample.

Possible answer: A sample that consisted of 1 000 teenagers chosen from various areas all over South Africa

(e) Why is it important to know the date when this data set was collected?

Situations and facts tend to change over time. Therefore, data also changes over time. For example, even if the data above were true 10 years ago, the data is likely to have changed since then. Data should be collected and reported on at the time when it is still applicable. It is useful to compare new data to older data.

Hypothesis: Girls prefer Accounting and Maths above Physical Sciences and Social Sciences.

2. Use the data set provided to write a report on your findings. Include your frequency table, graphs and conclusions. Also compare the favourite subjects of girls to those of boys.

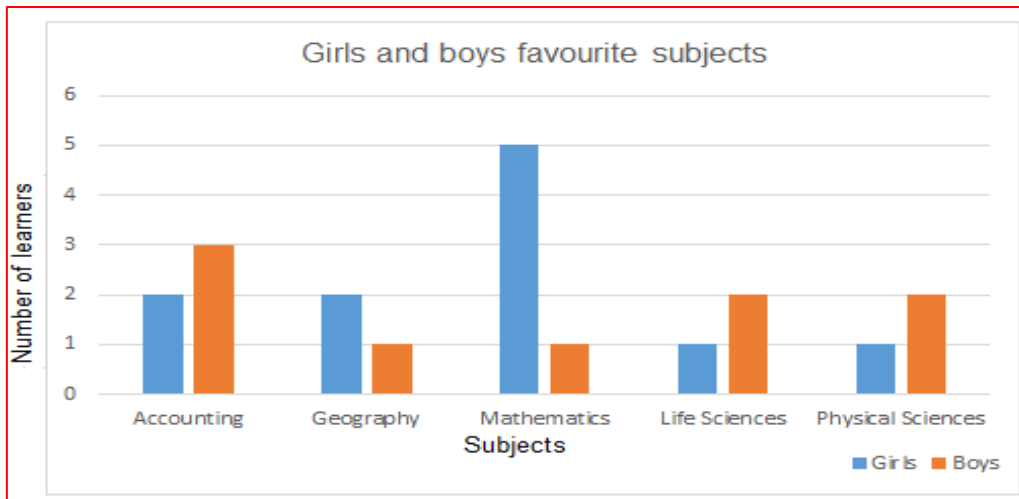


| Name      | Favourite subjects | Name      | Favourite subjects |
|-----------|--------------------|-----------|--------------------|
| Thabo     | Accounting         | Victoria  | Mathematics        |
| George    | Life sciences      | Lungile   | Mathematics        |
| Rorisang  | Mathematics        | David     | Accounting         |
| Andile    | Accounting         | Kate      | Geography          |
| Tshepo    | Mathematics        | Nelly     | Mathematics        |
| Wonderboy | Geography          | Elizabeth | Physical Sciences  |
| Modise    | Physical Sciences  | Peter     | Life Sciences      |
| Rebaona   | Accounting         | Themba    | Accounting         |
| Lerato    | Life Sciences      | Kabelo    | Physical Sciences  |
| Faith     | Mathematics        | Rejoice   | Geography          |

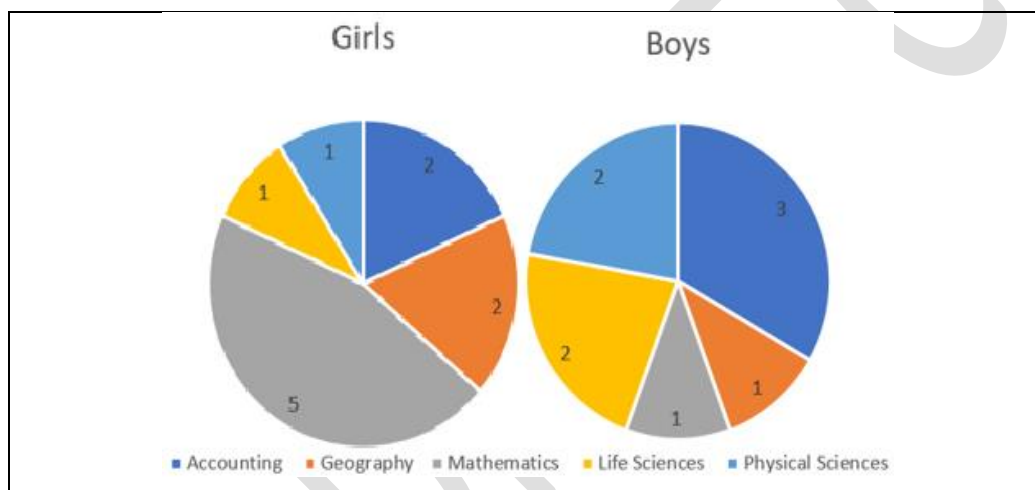
Frequency table

| Learners favourite subjects | Tallies for Girls | Frequency | Tallied for boys | Frequency |
|-----------------------------|-------------------|-----------|------------------|-----------|
| Accounting                  | //                | 2         | ///              | 3         |
| Geography                   | //                | 2         | /                | 1         |
| Mathematics                 | ///               | 5         | /                | 1         |
| Life Sciences               | /                 | 1         | //               | 2         |
| Physical Sciences           | /                 | 1         | //               | 2         |

Double bar graph



### Pie chart



### Conclusion

The gathered data shows that most girls favourite subject is Mathematics and Geography, while most boys favourite subjects is Physical Sciences, Life Sciences and Accounting.

However, this data cannot be used to generalise, as except for Mathematics; the favourite subject numbers for boys and girls is close together. If the sample of learners can be increased the results can go either way

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# TERM 4

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## UNIT 1

### WHOLE NUMBERS

#### 1.1 Addition and Subtraction

1. Calculate the following using any method and verify your answers using the calculator:

(a)  $584\,364 + 417\,120 = 1\,001\,484$

(b)  $734\,249 + 46\,342 = 780\,591$

(c)  $563\,482 - 278\,356 = 285\,126$

(d)  $735\,593 - 149\,256 = 586\,337$

#### 1.2 Multiplication and Division

1. Write the first 5 multiples of:

(a) 7 (M<sub>7</sub>: 7; 14; 21; 28; 35)

(b) 9 (M<sub>9</sub>: 9; 18; 27; 36; 45)

(c) 15 (M<sub>15</sub>: 15; 30; 45; 60; 75)

2. Write down the factors of:

(a) 28 (F<sub>28</sub>: 1; 2; 4; 7; 14; 28)

(b) 26 (F<sub>26</sub>: 1; 2; 13; 26)

(c) 60 (F<sub>60</sub>: 1; 2; 3; 4; 5; 6; 10; 12; 15; 20; 30; 60)

**Multiplication: Use a range of techniques to perform written calculations with whole numbers**

3. Multiply the following numbers and check your answers using a calculator.

(a)  $4\,625 \times 213 = 985\,125$

(b)  $2\,417 \times 124 = 299\,708$

(c)  $7\,128 \times 252 = 1\,796\,256$

**Division: Use a range of techniques to perform written calculations with whole numbers**

4. Calculate the following using any method and check your answers using a calculator:

(a)  $3\,300 \div 25 = 132$

(b)  $4\,599 \div 32 = 143 \text{ remainder } 23$

(c)  $7\,028 \div 251 = 28$

### Perform multiple operations on whole numbers with or without brackets

1. Calculate.

(a)  $48 \div 6 + (4 \times 6) \times 2 + 6 = 62$

(b)  $3\,024 + (60 \times 2 \div 3) = 3\,064$

(c)  $(5\,340 + 160) + 2 - (240 \times 2 - 230) = 5\,252$

(d)  $5\,978 \div 427 + 51 \div 17 \times 2 = 20$

### 1.3 Ratio and Rate

Solve the following problems

1. A florist has 7 red roses and 3 white roses in her garden.

(a) What is the ratio of white roses to red roses and of red roses to white roses?

3:7 and 7:3

(b) Write your answer as a common fraction and a percentage

Common fraction  $\frac{3}{10}$  and  $\frac{7}{10}$

Percentage 30% and 70%

2. Karabo drove a total of 230 km and used 23 litres of petrol. What is the rate of his petrol consumption in kilometres per litre?

10 kilometers per litre.

3. A tennis player hits 60 aces in 12 matches. Determine the rate of his aces per match.

5 aces per match.

4. Khaya cycles 21 kilometers per hour. What distance does he cycle in 12 hours?

He cycles 252 kilometers in 12 hours.

## UNIT 2

### FRACTIONS

#### 2.1 Common Fractions

**Addition and Subtraction of common fractions with the same denominators.**

1. Calculate each of the following:

$$(a) \frac{3}{12} + \frac{5}{12} = \frac{8}{12} \text{ or } \frac{2}{3}$$

$$(b) \frac{7}{8} - \frac{3}{8} = \frac{4}{8} \text{ or } \frac{1}{2}$$

**Addition and Subtraction of mixed numbers with the same denominators.**

2. Add:

$$(a) 9\frac{7}{10} + 5\frac{3}{10} = 14\frac{10}{10} = 14 + 1 = 15$$

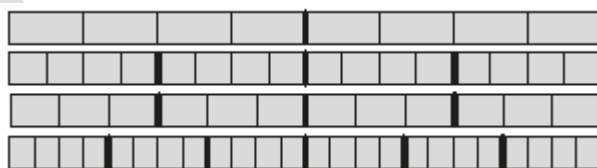
$$(b) 5\frac{2}{9} + 3\frac{5}{9} = 8\frac{7}{9}$$

**Fractions of whole numbers**

3. Calculate how much money is  $\frac{1}{5}$  of R200 R40

**Recognise, describe and use the equivalence of fractions**

4. Study the following diagram and consider the fraction quarters. It can be written as  $\frac{3}{4}$ .



(a) Multiply both the numerator and the denominator by 2 to form a new fraction.

$$\frac{3}{4} \times \frac{2}{2} = \frac{6}{8}$$

(b) Is the new fraction equivalent to  $\frac{3}{4}$ ? You may check on the diagram.

$\frac{6}{8}$  yes, it is equivalent to  $\frac{3}{4}$

(c) Multiply both the numerator and the denominator by 3 to form a new fraction. Is the new fraction equivalent to  $\frac{3}{4}$ .  $\frac{9}{12}$ , yes

(d) Multiply both the numerator and the denominator by 4 to form a new fraction. Is the new fraction equivalent to  $\frac{3}{4}$ ?  $\frac{12}{16}$ , yes

(e) Multiply both the numerator and the denominator by 6 to form a new fraction. Is the new fraction equivalent to  $\frac{3}{4}$ ?  $\frac{18}{24}$ , yes

### Percentages of whole numbers

5. How much is 80% of each of the following?

- |            |        |
|------------|--------|
| (a) R500   | R400   |
| (b) R480   | R384   |
| (c) R850   | R680   |
| (d) R2 400 | R1 920 |

### Problem solving in context involving common fractions including grouping and equal sharing.

6. Calculate:

(a) Jamie has 16 bottles of juice and needs 3 quarters ( $\frac{3}{4}$ ) of a bottle to fill one glass.

(i) How many quarters of a bottle of juice are there in 16 full bottles?

$16 \times 4 = 64$  quarters

(ii) How many glasses can he fill with these quarters?

$64 \div 3 = 21$  glasses and there is one quarter of a bottle left (one third of a glass)

(b) Jamie has 25 bottles of juice and needs  $\frac{3}{5}$  of a bottle to fill one glass. How many glasses can he fill?

25 bottles have  $25 \times 5 = 125$  fifths. He can fill  $125 \div 3 = 41$  glasses and two fifths of a bottle is left over.

With two fifths of a bottle, he can fill two thirds of a glass.

He can fill 41 glasses and one more glass will be only two thirds full.

### Problem solving in context involving percentages

7. Solve

- (a) Jannie eats a quarter of a watermelon. What percentage of the watermelon is this?

$\frac{1}{4}$  of 1 watermelon = 25% of the watermelon.

- (b) Sibü drinks 75% of the milk in a bottle. What fraction of the milk in the bottle has he drunk?

He has drunk  $\frac{75}{100}$  of the milk or  $\frac{3}{4}$  of the milk.

- (c) Building costs of houses increased by 20%. What is the new building cost for a house that previously cost R110 000 to build?

$$20\% \times R110\ 000 = R22\ 000$$

$$R110\ 000 + R22\ 000 = R132\ 000$$

**Recognize and use equivalent forms of common fractions (fractions in which one denominator is a multiple of another).**

8. Calculate:

(a)  $5\frac{1}{9} - 4\frac{2}{3}$

$$1\frac{1}{9} - \frac{2}{3} = \frac{10}{9} - \frac{6}{9} = \frac{4}{9}$$

(b)  $\frac{13}{20} - \frac{2}{5}$

$$\frac{13}{20} - \frac{8}{20} = \frac{5}{20} \text{ or } \frac{1}{4}$$

**Recognize equivalence between common fraction and percentage forms of the same number.**

9. Find the percentage of the following fractions

(a)  $\frac{12}{100}$

$$= 12\%$$

(b)  $\frac{41}{50}$

$$= 82\%$$

10. Write the following percentages as fractions

(a) 70%

$$= \frac{7}{10}$$

(b) 66%

$$= \frac{33}{50}$$

## 2.2 Decimal Fractions

### Compare and order decimal fractions to at least two decimal places

1. Order the following numbers from biggest to smallest. Explain your method.

(a) 0,8; 0,05; 0,5; 0,15; 0,465; 0,55; 0,75; 0,4; 0,62

0,8; 0,75; 0,62; 0,55; 0,5; 0,465; 0,4; 0,15; 0,05

2. Below are the results of some of the 2012 London Olympic events. In each case, order them from first to last place. Use the column provided.

Women: Long jump – Final

| NAME                | COUNTRY | DISTANCE | PLACE           |
|---------------------|---------|----------|-----------------|
| Anna Nazarova       | RUS     | 6,77m    | 5 <sup>th</sup> |
| Brittney Reese      | USA     | 7,12m    | 1 <sup>st</sup> |
| Elena Sokolova      | RUS     | 7,07m    | 2 <sup>nd</sup> |
| Ineta Radevica      | LAT     | 6,88m    | 4 <sup>th</sup> |
| Janay De Loach      | USA     | 6,89m    | 3 <sup>rd</sup> |
| Lyudmila Kolchanova | RUS     | 6,76m    | 6 <sup>th</sup> |

### Place value of digits to at least two decimal places

3. What is the place value of the underlined digits?

(a) 900,009

Tens and Tenths

(b) 4,672

Thousandths

### Rounding off decimal fractions to at least 2 decimal places

4. Round off the following decimal fractions to at least two decimal places.

(a) 365,8760

365,88

(b) 564,234

564,23

### Addition and subtraction of decimal fractions with at least two decimal places.

5. Calculate the value of the following:

(a)  $8,26 + 5,79 - 2,42$

=11,63

(b)  $21,01 - 5,64 - 0,39$

=14,98

(c)  $0,78 + 17,64 + 6,65$

=25,07

### Multiply decimal fractions by 10 and 100

6. Calculate the value of the following:

|     | Number | Multiply by 10 | Multiply by 100 | Multiply by 1000 |
|-----|--------|----------------|-----------------|------------------|
| (a) | 82,679 | 826,79         | 8 267,9         | 82 679           |
| (b) | 640,39 | 6403,9         | 64 039          | 640 390          |
| (c) | 0,785  | 7,85           | 78,5            | 785              |

### Solve problems in context involving decimal fractions

7. Alison buys a cool drink for R5,95; a chocolate for R3,25 and a packet of chips for R4,60. She pays with a R20 note.

(a) How much did she spend?

$$R5,95 + R3,25 + R4,60 = R13,80$$

(b) How much change did she get?

$$R20 - R13,80 = R6,20$$

8. A tractor uses 11,25 l of fuel in 0,75 hours. How many litres does it use in one hour?

$$11,25 \times 3 \div 4 \text{ (0,75 is the same as three quarters)} = 8,43 \text{ litres per hour}$$

### Recognize equivalence between common fraction, decimal fraction and percentage forms of the same number

9. Write the following as common fraction, decimal fraction and percentage forms of the same number

| No. | Percentage | Common Fraction   | Decimal Fraction |
|-----|------------|-------------------|------------------|
| (a) | 1,25%      | $\frac{1}{80}$    | 0,0125           |
| (b) | 24%        | $\frac{12}{50}$   | 0,24             |
| (c) | 1,8%       | $\frac{18}{1000}$ | 0,018            |

## UNIT 3

### ALGEBRAIC LANGUAGE

1. Complete the table below by writing each statement as an algebraic equation.

| No. | Statement                                    | Algebraic expression |
|-----|----------------------------------------------|----------------------|
| (a) | The sum of 6 and a number.                   | $6 + x$              |
| (b) | A number multiplied by 3 and increased by 4. | $3x + 4$             |
| (c) | 4 more than a number.                        | $x + 4$              |
| (d) | The difference between a number and 10       | $x - 10$             |
| (e) | The difference between 10 and a number       | $10 - x$             |
| (f) | 14 less than a number                        | $x - 14$             |
| (g) | 8 less than the product of 5 and a number    | $5x - 8$             |
| (h) | The sum of a number and 3 multiplied by 7    | $(x + 3) \times 7$   |
| (i) | The square root of a number                  | $\sqrt{x}$           |
| (j) | The square of a number                       | $x^2$                |

2. Solve equations by inspection

(a) Determine the value of  $x$  if  $x$  plus 4 is equals to 9

$$x = 5, \text{ therefore, } 5 + 4 = 9$$

(b) Solve for  $x$  if  $2x + 4 = 16$

$$x = 6, \text{ therefore, } 12 + 4 = 16$$

(c) Solve for  $x$  if  $x^3 = 27$

$$x = 3, \text{ therefore, } 3^3 = 27$$

(d) Solve for  $m$  if  $m^2 = 25$

$$m = 5, \text{ therefore, } 5^2 = 25$$

(e) Solve for  $p$  if  $\frac{p}{3} + 5 = 7$

$$p = 6, \text{ therefore } \frac{6}{3} + 5 = 7$$

3. Solve the following equations using trial and improvement

(a)  $b - 14 = 6$

| $b$ | $b - 14$ | Comment        |
|-----|----------|----------------|
| 10  | -4       | Smaller than 6 |



|    |    |                |
|----|----|----------------|
| 12 | -2 | Smaller than 6 |
| 16 | 2  | Smaller than 6 |
| 18 | 4  | Closer to 6    |
| 20 | 6  | Correct answer |

(b)  $3x + 5 = 20$

| $x$ | $3x + 5$ | Comment         |
|-----|----------|-----------------|
| 2   | 11       | Smaller than 20 |
| 3   | 14       | Smaller than 20 |
| 4   | 17       | Closer to 6     |
| 5   | 20       | Correct answer  |

(c)  $\frac{x}{5} + 9 = 12$

| $x$ | $\frac{x}{5} + 9$ | Comment         |
|-----|-------------------|-----------------|
| 5   | 10                | Smaller than 12 |
| 10  | 11                | Closer to 12    |
| 15  | 12                | Correct answer  |

4. Solve the equations by substitution

- (a)  $y + 8 = 24$        $16 + 8 = 24$        $y = 16$
- (b)  $z - 9 = 9$        $18 - 9 = 9$        $z = 18$
- (c)  $6x + 35 = 65$        $18 - 9 = 9$        $x = 5$
- (d)  $\frac{x}{4} + 12 = 17$        $\frac{20}{4} + 12 = 17$        $x = 20$
- (e)  $x^2 = 36$        $6^2 = 36$        $x = 6$
- (f)  $b^3 = 64$        $4^3 = 64$        $b = 4$

5. Identify variables and constants in the table.

| No. | Expression     | Variable(s) | Constant |
|-----|----------------|-------------|----------|
| (a) | $(2x + y) = 6$ | $x$ and $y$ | 6        |
| (b) | $z^3 - 10$     | $z$         | -10      |

|     |          |     |    |
|-----|----------|-----|----|
| (c) | $4p^7$   | $p$ | 0  |
| (d) | $k + 17$ | $k$ | 17 |

6. Interpret and write the following rules in words

(a)  $\frac{1}{4}w - 4$

Quarter of a number minus 4

(b)  $(5k + 11) \times 5$

Multiply the sum of a number and 11 by 5

(c)  $10n - 5$

Multiply a number by 10 and subtract 5

(d)  $(z \times 6) + 14$

The product of a number and 6 is increased by 14

(e)  $x^3 + \sqrt{q}$

The sum of a number cubed and the square root of another number

7. Substitute to determine the numerical value of the following expressions.

| No. | $x$                | -1  | 0   | 2  |
|-----|--------------------|-----|-----|----|
| (a) | $x - 4$            | -5  | -4  | -2 |
| (b) | $5x + 4 + 3x - 2$  | -6  | 2   | 18 |
| (c) | $(4x + 7)(3x - 2)$ | -15 | -14 | 60 |

## UNIT 4

### NUMERIC AND GEOMETRIC PATTERNS

Investigate, extend and describe numeric and geometric patterns

1. What are the next three numbers in each of these sequences?

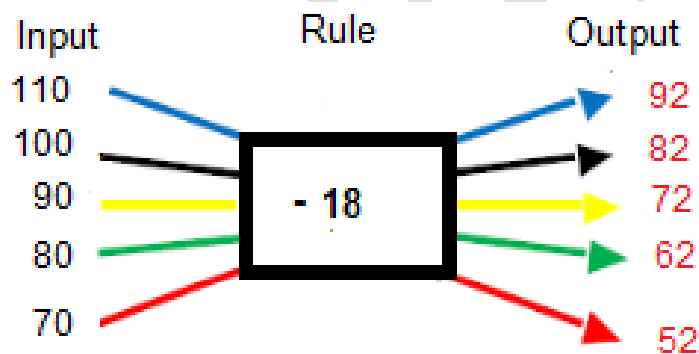
- |                        |               |
|------------------------|---------------|
| a) 4; 7; 10; 13; 16;   | 19; 22; 25    |
| b) 22; 18; 14; 10; 6;  | 2; -2; -6     |
| c) 37; 42; 47; 52; 57; | 62; 67; 72    |
| d) 7; 14; 28; 56; 112; | 224; 448; 896 |

2. Identify whether the sequence has constant difference or constant ratio

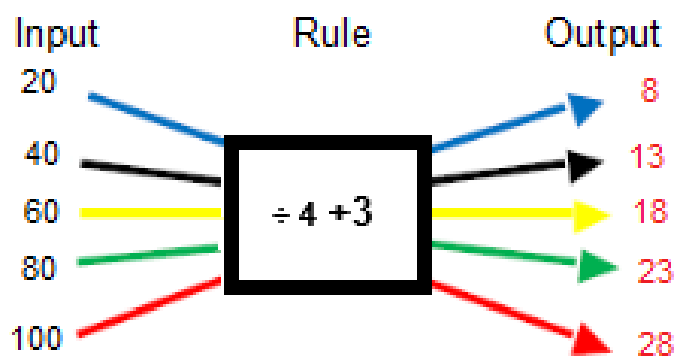
- |                        |                     |
|------------------------|---------------------|
| (a) 60; 65; 70; 75     | Constant difference |
| (b) 200; 175; 150; 125 | Constant difference |
| (c) 625; 125; 25; 5    | Constant ratio      |
| (d) 6; 18; 54; 162     | Constant ratio      |

3. Complete the flow diagrams

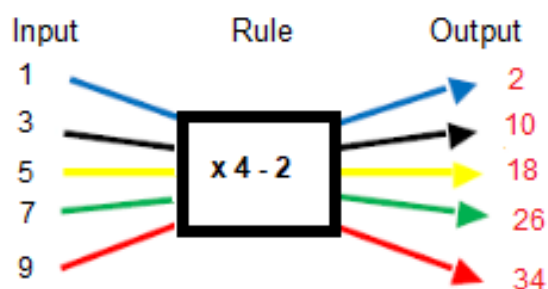
(a)



(b)



(c)



4. Complete the following tables using the given rules:

(a) The rule for calculating area of a square for a garden is "Multiply the length of a side of the square by itself"

|                    |    |    |    |     |     |
|--------------------|----|----|----|-----|-----|
| Length of the side | 4  | 6  | 8  | 10  | 12  |
| Area of square     | 16 | 36 | 64 | 100 | 144 |

(b) Add 37 to the previous term to get the next one.

|              |   |    |    |     |     |     |
|--------------|---|----|----|-----|-----|-----|
| Position (n) | 1 | 2  | 3  | 4   | 5   | 6   |
| Term         | 3 | 40 | 77 | 114 | 151 | 188 |

(c) Subtract 3 from the previous term to get the next one.

|             |    |    |    |    |    |    |
|-------------|----|----|----|----|----|----|
| Position(n) | 1  | 2  | 3  | 4  | 5  | 6  |
| Term        | 70 | 67 | 64 | 61 | 58 | 55 |

(d) Multiply by 4

|             |   |   |    |    |    |    |
|-------------|---|---|----|----|----|----|
| Position(n) | 1 | 2 | 3  | 4  | 5  | 6  |
| Term        | 4 | 8 | 12 | 16 | 20 | 24 |

5. Create your own numeric pattern, represent it in words, in a flow diagram and in a table.

Accept any reasonable correct answer.

6. Study the pattern in the table below

|             |   |   |    |    |    |    |
|-------------|---|---|----|----|----|----|
| Position(n) | 1 | 2 | 3  | 4  | 5  | 6  |
| Term        | 2 | 5 | 10 | 17 | 26 | 37 |

a) What is the value of the 7<sup>th</sup> term?

$$T_7 = 50$$

b) Describe the rule in words to determine value of any term ( $n^{\text{th}}$  term) in the sequence.

The value of any term in the sequence is equal to the position of the term squared plus one

c) Write down the formula to determine the value of any term in the sequence.

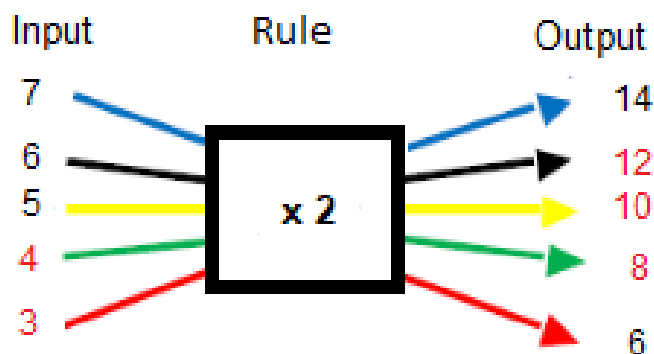
$$T_n = n^2 + 1$$

d) Use your formula to determine the 10<sup>th</sup> term of the pattern

$$\begin{aligned} T_{10} &= 10^2 + 1 \\ &= 100 + 1 \\ &= 101 \end{aligned}$$

7. Complete the flow diagram and use it to determine the equivalence

(a) Flow diagram



(b) Represent the relationship in the flow diagram verbally

Multiply the input by two to get the output number

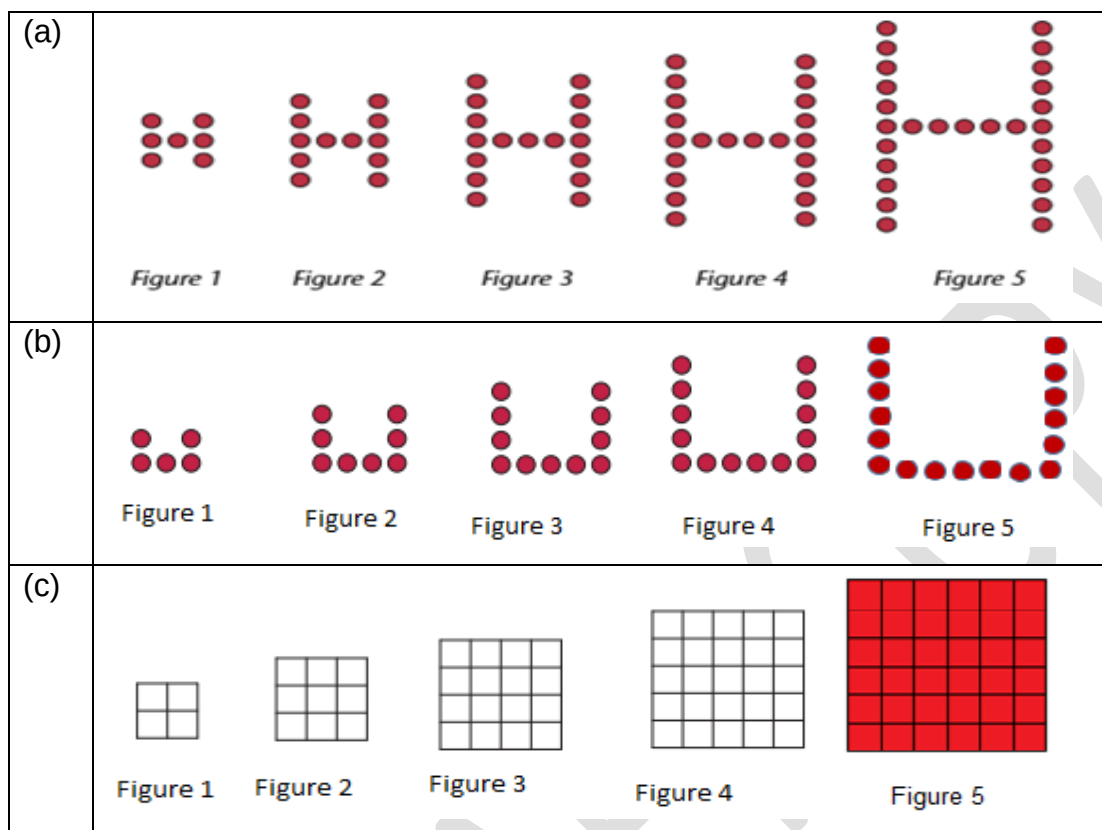
(c) Represent in a table

|          |    |    |    |   |   |
|----------|----|----|----|---|---|
| Input(n) | 7  | 6  | 5  | 4 | 3 |
| Output   | 14 | 12 | 10 | 8 | 6 |

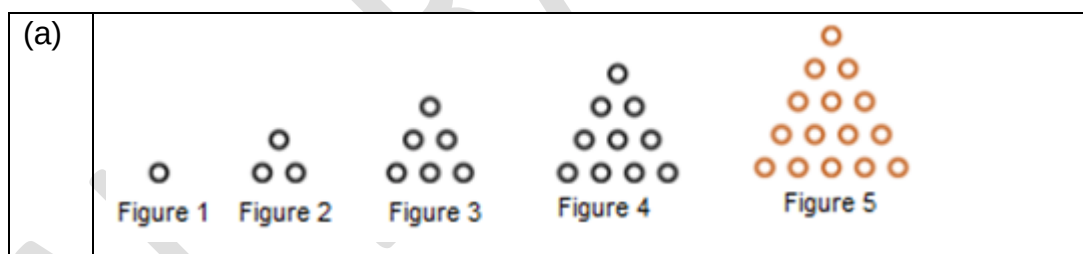
(d) Write the general rule or formula

$$T_n = 2n$$

8. Study the following geometric patterns and draw figure 5.



9. Mpho builds a pattern using circles, study her pattern and answer the questions that follows:



- (a) How many circles are added in each figure?

To get figure 2 we add 2 circles to figure 1, to get figure 3 we add 3 circles to figure 2 (to get the next figure you add the figure number to the previous figure).

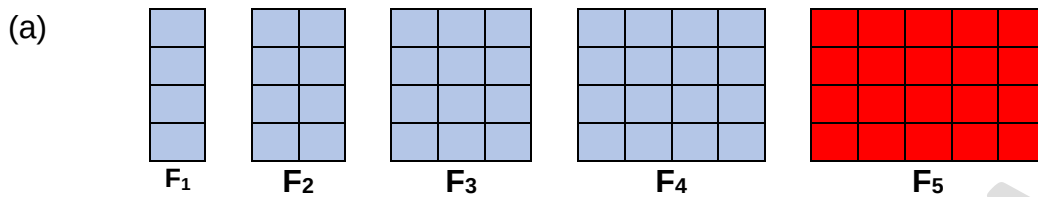
- (b) Write down the formula to determine the number of matchsticks in each figure.

$$T_n = (n^2 + n) \div 2$$

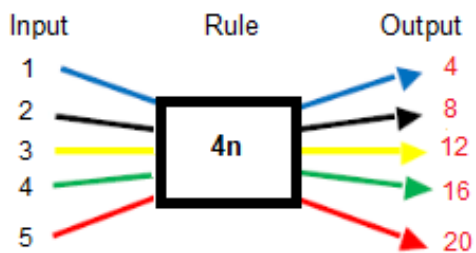
- (c) Use your formula to determine the number of matchsticks in the 12<sup>th</sup> figure.

$$\begin{aligned}
 T_{12} &= (12^2 + 12) \div 2 \\
 &= 78
 \end{aligned}$$

10. Study the diagram below. Represent the equivalence.



(b) The flow diagram



(c) The table

| Figure         | 1 | 2 | 3  | 4  | 5  |
|----------------|---|---|----|----|----|
| No. of squares | 4 | 8 | 12 | 16 | 20 |

(d) The constant difference?

4

(e) The pattern in own words

Multiply the input number by 4 to get the output

## UNIT 5

### MEASUREMENT

#### 5.1 Length

**Solve problems in contexts involving length, mass and capacity/volume**

1. Joseph is a designer and has 10 rolls of material each roll is 6 metres each.

(a) How much material does he have altogether?

**He has 60 metres of the material.**

(b) If he uses 4,5 metres to design one dress, how many metres will he need to design 13 dresses?

**He will need 58,5 metres to design 13 dresses.**

(c) How much material will be left?

**For thirteen dresses he will use 58,5 metres and he will be left with one and a half metres material.**

2. In a marathon, Buli ran 12 300 metres and Lethu ran 12,3 kilometres. Who ran the shortest distance, explain your answer?

**They both ran an equal distance because 12 300 metres is equal to 12,3 kilometres.**

#### 5.2 Mass

Solve the following problems.

1. A farmer sells oranges for R24,99 a bag. Each bag weighs 7kg and he supplies twelve retail stores with 36 bags each. How many kilograms will be received by each retail store and how much will he be paid altogether?

**Each retail store will receive 252 kilograms. He will be paid R10 795,68.**

2. A retail store has a sale for cornflakes. They sell 4 times 1,2 kilograms for R150. The original price of the same cornflakes is R24,99 for a 750 grams box. Which is the best buy and why?

**Sale:  $4 \times 1,2 \text{ kg} = 4,8 \text{ kg}$   $\therefore 4,8 \text{ kg}$  costs R150**

**Normal Price: 750 g sells for R24,99**

**$750 \text{ g} \times 5 = 4,5 \text{ kg}$  costs R124,95**



$$R150 \div 4,8 \text{ kg} = R31,25/\text{kg}$$

$$R124,95 \div 4,5 \text{ kg} = R27,77/\text{kg}$$

Therefore 750 grams for R24,99 is the best buy.

3. Sendi and Lerato are twins, Sendi's birth weight was 2,15 kg and Lerato's was 728. Who had more weight? Explain your answer.

Sendi had more weight because 728 grams is equal to 0,728 kilograms which is less than 2,15 kilograms.

4. My mother bought a 10 kg bag of rice. She gave her sister one quarters of the rice. How much rice did she give her sister, write your answer in grams?

She gave her sister 2,5 kilograms of rice which equal to 2 500 grams.

### 5.3 Capacity and Volume

Solve the following problems.

1. A cube – shaped water reservoir has a side length of 12 m.

(a) What is the capacity of the reservoir? Write your answer in litres.

$$\text{Volume} = s^3$$

$$= 12\text{m} \times 12\text{m} \times 12\text{m}$$

$$= 1\,728 \text{ m}^3 =$$

$$1 \text{ m}^3 \text{ volume} = 1\text{k}\ell \text{ capacity therefore capacity is } 1\,728\text{k}\ell = 1\,728\,000 \text{ litres}$$

(b) If the reservoir is quarter full of water, what does it hold in kilolitres?

$$\frac{1}{4} \times 1\,728\,000 = 432\,000 \ell = 432 \text{ k}\ell$$

2. In a bucket there are 30 litres of juice. How many learners can get 375 millilitres of juice?

80 learners can get 375 millilitres juice.

3. The monthly water use for a certain family is as follows:

76 000 millilitres to wash the dishes; 150 litres to bath; 26,5 litres for cooking;

$12\frac{1}{2}$  litres for gardening and 12 000 millilitres for drinking. How much water

does the family use monthly? Write your answer in litres.

The family use 277 litres monthly.

4. For my sister's 16<sup>th</sup> birthday party, my aunt made 5 000 millilitres of cocktail juice. She used 375 millilitres of mango juice and 4,5 litres of strawberry juice. How many millilitres of granadilla juice has she added to get the 5 000 millilitres of cocktail?

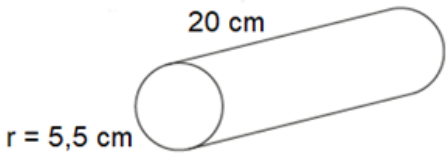
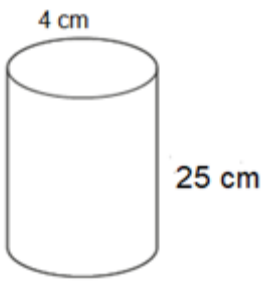
She has added 125 millilitres of granadilla.

5. A motor car uses 15,50 ℓ of fuel in 0,25 hours. How many litres does it use in one hour?

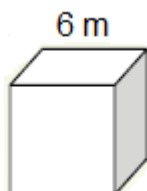
62,00 litres or 62 litres

6. Calculate the volume and capacity of the following cylinders. Use  $\pi = 3,14$  and round off all the answers to two decimal places.

Volume of a cylinder = Area of base  $\times h = \pi r^2 \times h$

|     |                                                                                     |                                                                                                           |
|-----|-------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| (a) |   | $V = \pi r^2 \times h$<br>$1\,899,7\text{ cm}^3$                                                          |
| (b) |  | $V = \pi r^2 \times h$<br>$1\,256\text{ cm}^3$<br>$V = \ell \times w \times h$<br>$= 72\,000\text{ mm}^3$ |

7. Use a formula to calculate the volume of water that will fill each cube. Convert your answer into millimetres, litres and kilolitres.

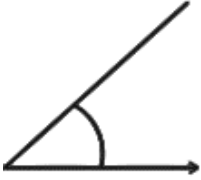
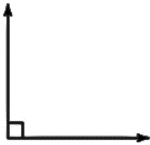
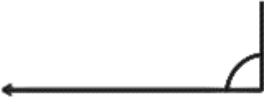
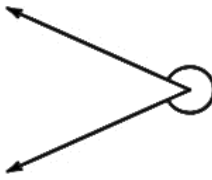


$$216\text{ m}^3 = 216\,000\,000\text{ m}\ell = 216\,000\text{ litres} = 216\text{ k}\ell$$

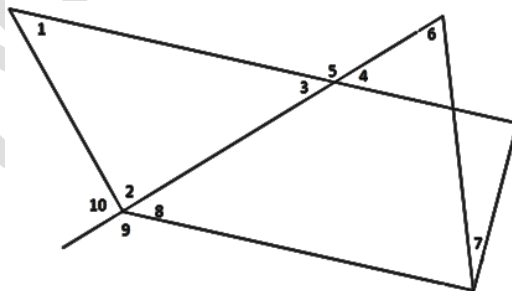
## UNIT 6

### CONSTRUCTION OF GEOMETRIC FIGURES

1. Measure the following angles using your protractor.

| No. | Angle                                                                                | Size |
|-----|--------------------------------------------------------------------------------------|------|
| (a) |     | 43°  |
| (b) |     | 90°  |
| (c) |     | 90°  |
| (d) |   | 120° |
| (e) |  | 180° |
| (f) |   | 328° |

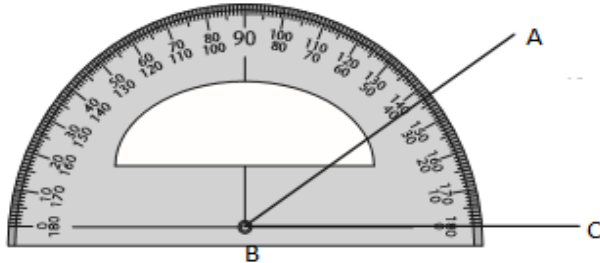
2. Measure all the numbered angles in the following figure.



| Angle | 1   | 2   | 3   | 4   | 5    | 6   | 7   | 8   | 9    | 10  |
|-------|-----|-----|-----|-----|------|-----|-----|-----|------|-----|
| Size  | 47° | 88° | 45° | 45° | 135° | 65° | 21° | 44° | 136° | 92° |

3. Construct the following angles.

(a)  $\hat{ABC} = 37^\circ$  using a protractor



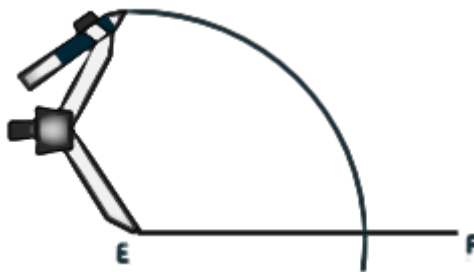
(b)  $\hat{XYZ} = 110^\circ$  using a protractor

(c)  $\hat{UWV} = 240^\circ$  using a protractor

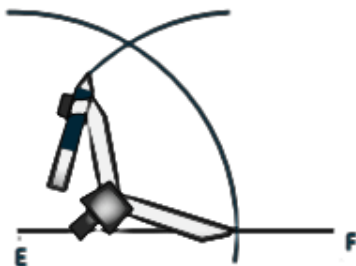
Angles  $\hat{XYZ}$  and angle  $\hat{UWV}$  can be constructed using the skill shown in (a) above

(d)  $\hat{DEF} = 60^\circ$  using a compass

Step 1 Draw a line segment (EF). With the compass on point E, draw an arc across EF and up over above point E.

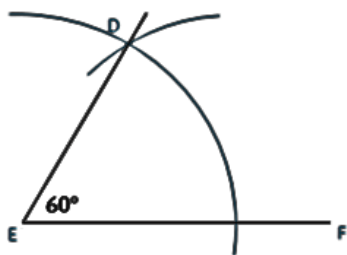


Step 2: Without changing the compass width, move the compass to the point where the arc crosses EF, and draw an arc that crosses the first one.



Step 3: Join point E to the point where the two arcs meet (point D).

$\hat{DEF} = 60^\circ$



(e) Angle  $\widehat{KLM}$  can be constructed by using the skill in (d) above

4. Construct a circle with a :

(a) Radius of 4 cm

Measure a radius of 4 cm on a ruler.

Mark a centre point on a page.

Place the compass on the centre point and draw a circle.

(b) Diameter of 7 cm

Measure a radius of 3,5 cm on a ruler.

Mark a centre point on a page.

Place the compass on the centre point and draw a circle.

5. Construct perpendicular lines.

(a) From point P to the line segment

P

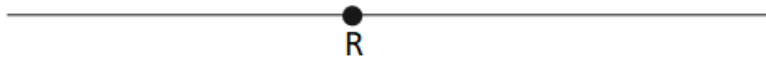
**Step 1:** Place the sharp point of the compass on the given point. Open the compass to a radius of your choice such that it can make arcs on the line segment.

**Step 2:** Without adjusting the radius, draw arcs on either side of the line segment.

**Step 3:** With the same radius, place your compass on each point of intersection between the line segment and the arc. Draw arcs that intersect on the opposite side of the given point P.

**Step 4:** Use a ruler to join point P to the point of intersection of the arcs.

(b) Point R in the line segment below.



Open the compass to a radius of your choice such that it can make arcs on the line segment. Place your compass on point R.

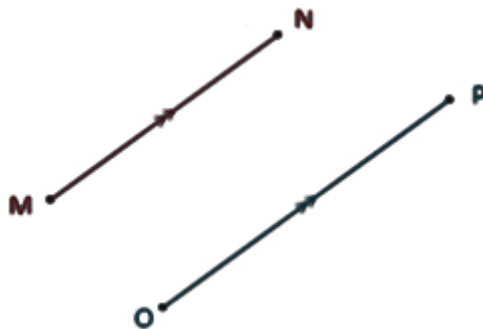
Without adjusting the radius, draw arcs intersecting the line segment on each side of point R.

Open the radius so that it is wider than the distance from one of the arcs to point R.

Place your compass on each point of intersection between the line segment and the arc. Draw arcs that intersect either above or below point R.

Use your ruler to join the point of intersection of the arcs and point R.

6. Construct parallel lines with line MN parallel to line OP.



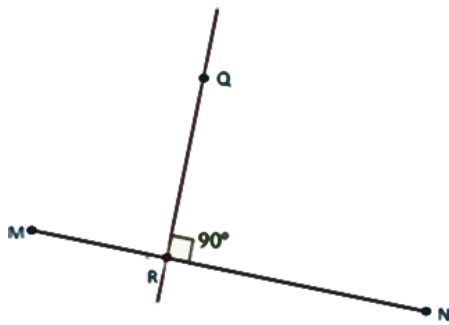
7. Draw the perpendicular line QR which is parallel to lines MN and OP using a protractor and a ruler. Given line MN and point Q draw a parallel line OP to MN that passes through point Q.



Step1: Draw a perpendicular line between Q and MN.

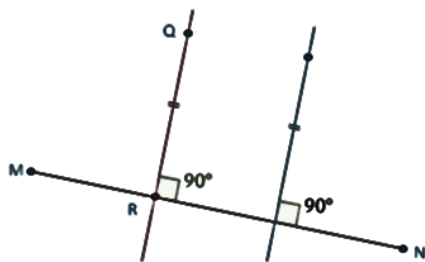
Use your protractor to draw a line that goes through Q and is at  $90^\circ$  to MN.

Label the point R where your new line touches MN



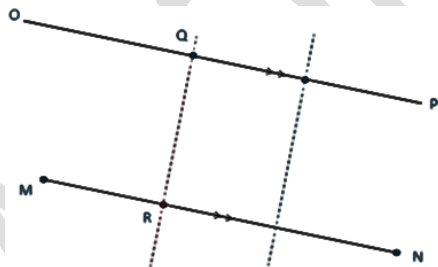
Step2: Measure the perpendicular distance between the point and the line MN

Draw another line that is perpendicular to line MN. Mark off the same length as QR on that line.



Step 3: Draw the parallel line

Join Q with the new point that is an equal distance away from MN. You now have a parallel line.



8. Use a protractor to construct the following.

(a) Equilateral triangle

Step 1: Use the line provided as your starting point. Use the end point of your line to measure the  $60^\circ$  angle.

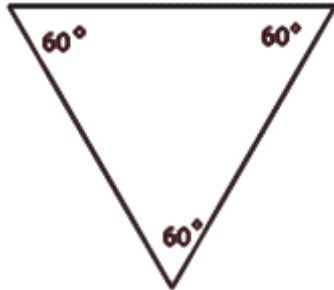
Step 2: Place the protractor with its base on the line and its origin exactly on top of the mark of the first end point.

Step 3: Make a small, clear mark at the edge of the protractor above the  $60^\circ$  mark.

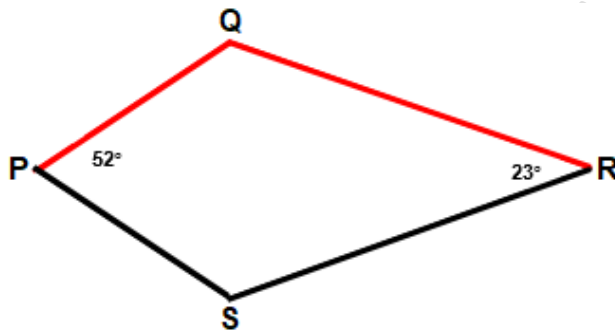
Step 4: Repeat step 2 and 3 on the second end point.

Step 5: Use a ruler to line up the two marks and draw a straight line that passes exactly through them.

Step 6: The two lines drawn should meet at one point



(b) Kite, the angle P must be  $52^\circ$  and the one at R must be  $23^\circ$



Step 1: Use the lines provided as your starting point. Use the end point P to measure angle  $52^\circ$  and at point R to measure angle  $23^\circ$ .

Step 2: Place the protractor with its base line on the line and its origin exactly on top of the end point P and measure angle  $52^\circ$  and do the same to measure angle  $23^\circ$  at point R.

Step 3: Make a small, clear mark at the edge of the protractor above the angles  $52^\circ$  and  $23^\circ$

Step 4: Use a ruler to join point P and the mark, and point R and the mark.

Step 6: The two lines drawn should meet at one-point Q



## UNIT 7

### GEOMETRY OF STRAIGHT LINES

1. Define the following terms

(a) Line segment

A set of points with a definite starting-point and an end-point.

(b) Ray

A set of points with definite starting-point and no definite end-point

(c) Straight line

A set of points with no definite starting-point and end-point.

(d) Parallel lines

When two lines on the same plane are a constant / same distance apart

(e) Perpendicular lines

When two lines intersect at right angle

(Accept any reasonable correct definition for (a) to (e))

2. Highlight the parallel lines in this picture



Picture A



Picture B



Picture C



Picture D

The learners should correctly highlight parallel lines in each picture

3. Identify the parallel and the perpendicular lines in the pictures.



Picture A



Picture B



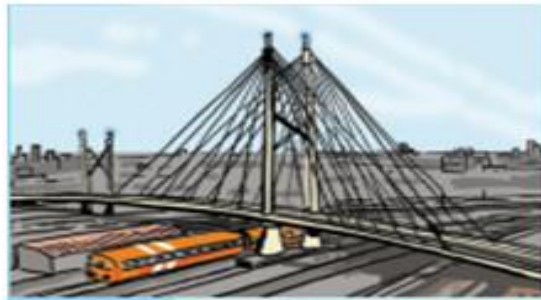
Picture C

Correct identification of the parallel and perpendicular lines in the picture

4. Circle the intersecting lines which forms vertically opposite angles in the pictures.



Picture A

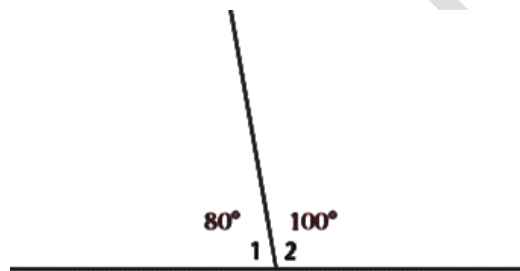


Picture B

Correct identification of intersecting lines which forms vertically opposite angles

5. Use a protractor to measure the sizes of all the angles in each figure. Write your answers in each figure.

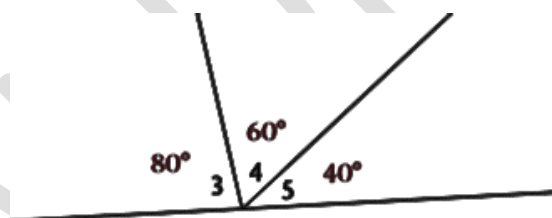
(a)



$$\hat{1} + \hat{2} = 180^\circ$$

supplementary

(b)



$$\hat{3} + \hat{4} + \hat{5} = 180^\circ$$

supplementary

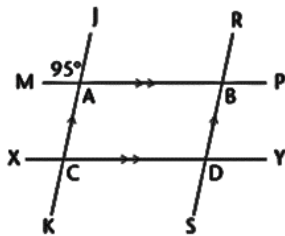
- (a) Use your answers in (a) and (b) above to say if the angles are supplementary or complementary. See the previous table.

6. Study the diagram of girders below, identify and label the complementary and supplementary angles



Correct identification of all complementary and supplementary angles

7. Find the sizes of all the angles. (Can you see two transversals and two sets of parallel lines?)



$$\hat{KAM} = 85^\circ [\angle \text{ on a str. line}]$$

$$\hat{JCX} = 95^\circ [\text{co-int. } \angle \text{ with } \hat{KAM}; MP \parallel XY]$$

$$\hat{KCK} = 85^\circ [\text{corr. } \angle \text{ with } \hat{KAM}; MP \parallel XY]$$

$$\hat{JAP} = 85^\circ [\text{vert. opp. } \angle \text{ s}]$$

$$\hat{PAK} = 95^\circ [\text{vert. opp. } \angle \text{ s}]$$

$$\hat{JCY} = 85^\circ [\angle \text{ on a str. line}]$$

$$\hat{YCK} = 95^\circ [\angle \text{ on a str. line}]$$

$$\hat{RBM} = 95^\circ [\text{corr. } \angle \text{ with } \hat{JAP}; JK \parallel RS]$$

$$\hat{MBS} = 85^\circ [\text{corr. } \angle \text{ with } \hat{MAK}; JK \parallel RS]$$

$$\hat{RDY} = 95^\circ [\text{corr. } \angle \text{ with } \hat{JCY}; JK \parallel RS]$$

$$\hat{XDS} = 85^\circ [\text{corr. } \angle \text{ with } \hat{XCK}; JK \parallel RS]$$

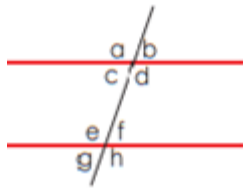
$$\hat{RBP} = 85^\circ [\text{corr. } \angle \text{ with } \hat{JAP}; JK \parallel RS]$$

$$\hat{PBS} = 95^\circ [\text{corr. } \angle \text{ with } \hat{PAK}; JK \parallel RS]$$

$$\hat{RDY} = 85^\circ [\text{corr. } \angle \text{ with } \hat{JCY}; JK \parallel RS]$$

$$\hat{YDS} = 95^\circ [\text{corr. } \angle \text{ with } \hat{PBS}; JK \parallel RS]$$

8. Identify pairs of equal angles and write their special name.



Vertically opposite angles:  $a = d$ ;  $b = c$ ;  $e = h$ ;  $f = g$

Corresponding angles:  $a = e$ ;  $b = f$ ;  $c = g$ ;  $d = h$

Alternate angles:  $c = f$ ;  $d = e$ ;

Co-interior angles:  $c$  &  $e$ ;  $d$  &  $f$

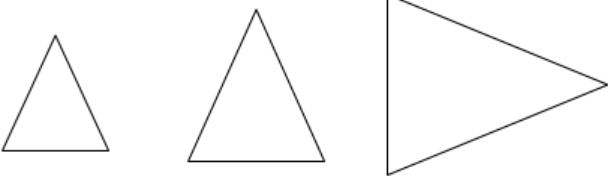
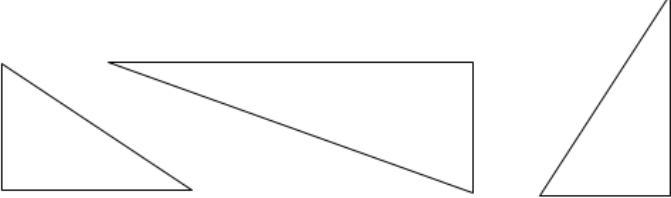
## UNIT 8

### GEOMETRY OF 2-D SHAPES

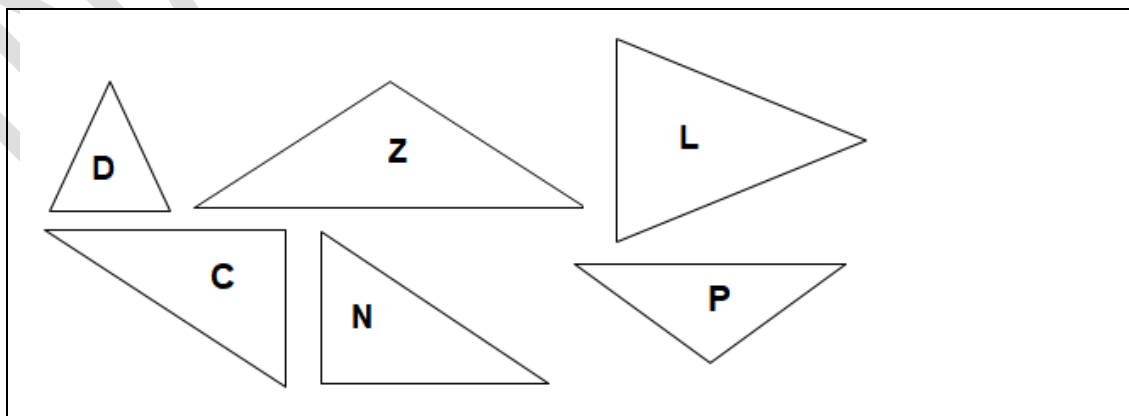
1. Identify triangles in terms of their sides and angles

- (a) equilateral triangles
- (b) isosceles triangles
- (c) right-angled triangles

Measure the **sides** of each triangle below. Classify the triangles according to the measurements you got. Motivate why you classified them the way you did.

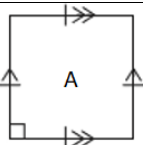
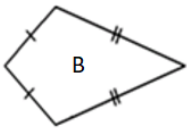
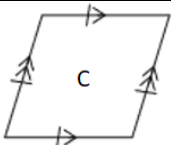
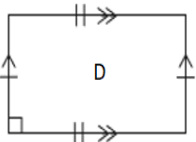
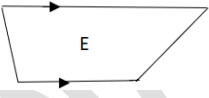
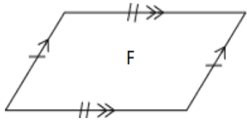
|                                                                                     |                                                                                                      |
|-------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
|    | <p>All the sides of each triangle above are equal. They are called <b>equilateral triangles</b>.</p> |
|   | <p>Two sides of each triangle above are equal. They are called <b>isosceles triangles</b>.</p>       |
|  | <p>All the sides not equal. They are called <b>scalene triangles</b>.</p>                            |

2. Measure the **angles** of each triangle below. Classify the triangles according to the measurements you got. Motivate why you classified them the way you did.



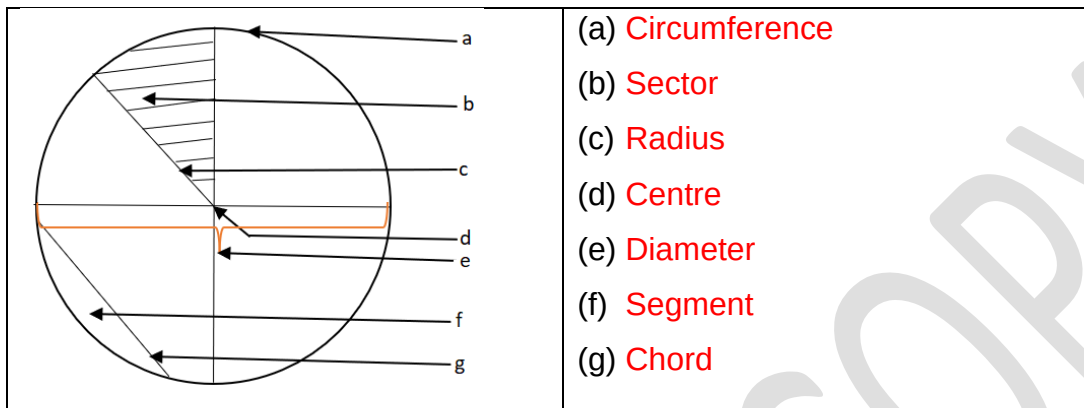
- All the angles of triangle D and L above are acute, therefore they are acute-angled triangles
- One of the angles of triangles P and Z above is an obtuse angle, therefore they are obtuse-angled triangles
- One of the angles of triangles C and N above is a right angle, therefore they are right-angled triangles.

3. Study the figures below and give their properties in terms of the sides and angles.

| Figure                                                                              | Name          | Properties in terms of:                                                  |                                    |
|-------------------------------------------------------------------------------------|---------------|--------------------------------------------------------------------------|------------------------------------|
|                                                                                     |               | Sides                                                                    | Angles                             |
|    | Square        | All the sides are equal.<br>The opposite sides are parallel              | All the angles are right angles.   |
|   | Kite          | Each pair of adjacent sides equal.                                       | One pair of opposite angles equal. |
|  | Rhombus       | All the sides are equal in length. The opposite sides are parallel.      | The opposite angles are equal.     |
|  | Rectangle     | The opposite sides are equal in length. The opposite sides are parallel. | All the angles are right angles.   |
|  | Trapezium     | One pair of opposite sides parallel.                                     |                                    |
|  | Parallelogram | The opposite sides are equal in length. The opposite sides are parallel. | The opposite angles are equal.     |

### Describing and naming parts of a circle.

4. Label the parts of the circle marked (a-g) below:

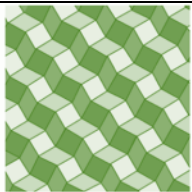


## UNIT 9

### TRANSFORMATIONS

Use transformations to make composite shapes and tessellations

- Which shapes have been used in the patterns below? Are these patterns tessellations? In which way have the shapes been tessellated? (translation, reflection or rotation)

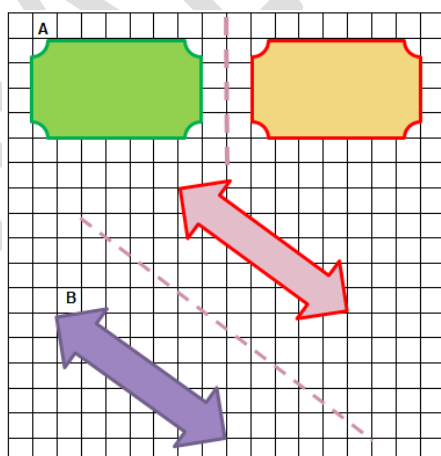
|                                                                                       |                                                              |                                                                                         |                                                                                 |
|---------------------------------------------------------------------------------------|--------------------------------------------------------------|-----------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| (a)  | Rhombus and Square<br>The patterns tessellate<br>Translation | (b)   | Ovals<br>The patterns do not tessellate                                         |
| (c)  | Circles<br>Patterns do tessellate                            | (d)  | Square, hexagon and dodecagon<br>Patterns tessellate<br>Rotation and reflection |

### Reflection, Translation and Rotation

- Perform the following transformations on a grid.

(c) Reflect shape A along the vertical line.

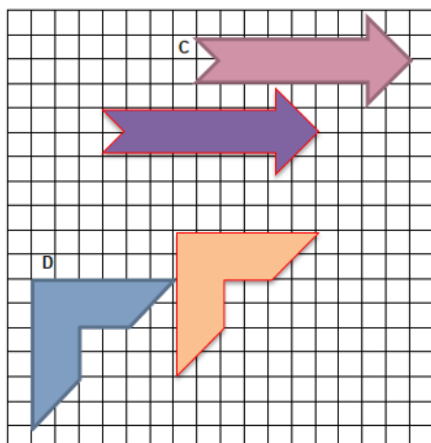
Reflect shape B along the diagonal line.





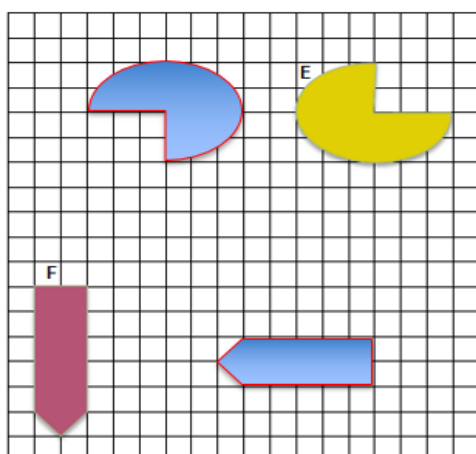
(d) Translate shape C, 4 units to the left and 3 units down.

Translate shape D, 2 units up and 6 units to the right.



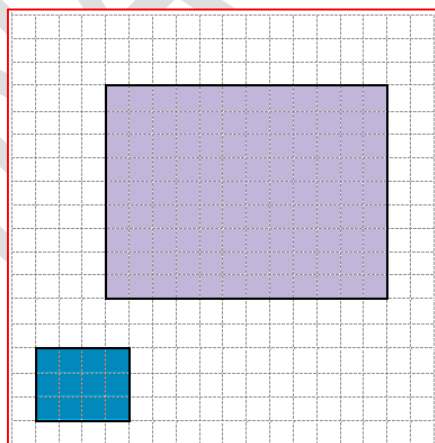
(e) Rotate shape E  $180^\circ$  anticlockwise.

Rotate shape F  $90^\circ$  clockwise.

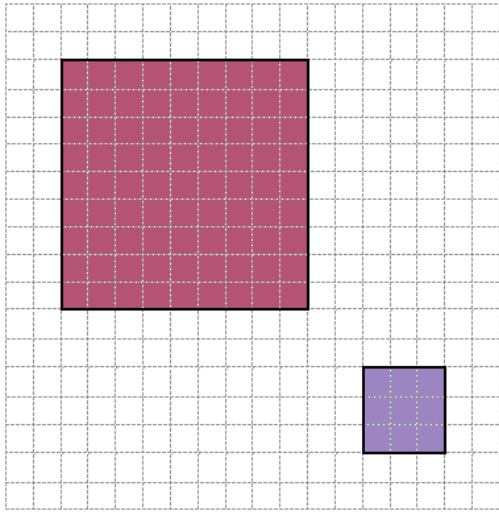


### Enlargement and reduction

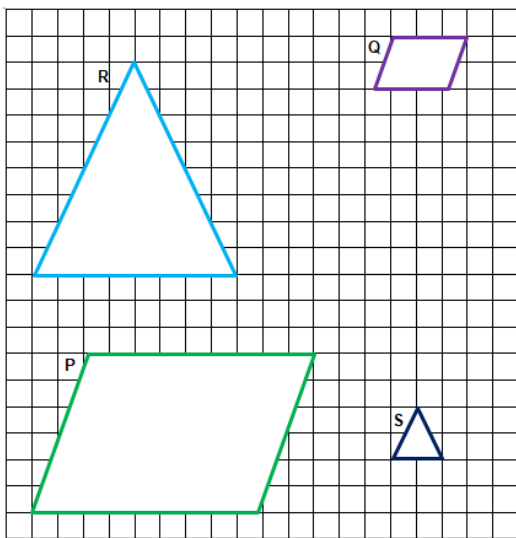
1. On grid paper, enlarge the rectangle below with a scale factor of 3



2. On grid paper, enlarge the square below with a scale factor of  $\frac{1}{3}$



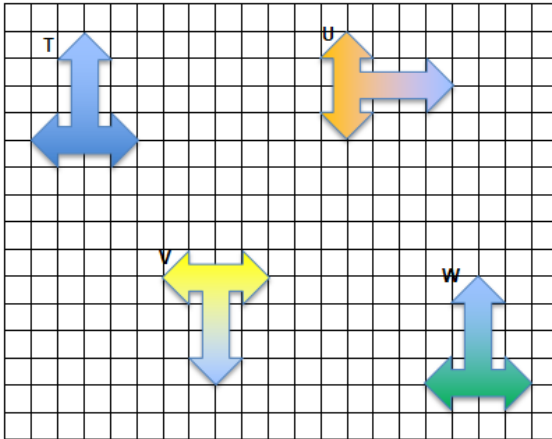
3. Describe the transformation from R to S and from Q to P



R to S, R the original shape has been enlarged by the scale factor of  $\frac{1}{4}$

Q to P, Q the original shape has been enlarged by the scale factor of 3

4. Shape T is the original shape. Describe the type of transformation from T to U, T to V and T to W



T to U – Rotation ( 90° clockwise)

T to V – Reflection (horizontal)

Rotation (180°)

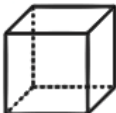
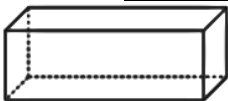
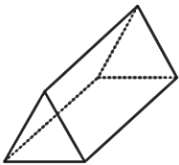
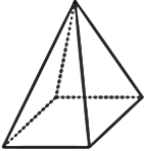
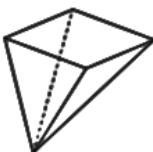
T to W – Translation

## UNIT 10

### GEOMETRY OF 3-D OBJECTS

Describe, sort and compare polyhedra

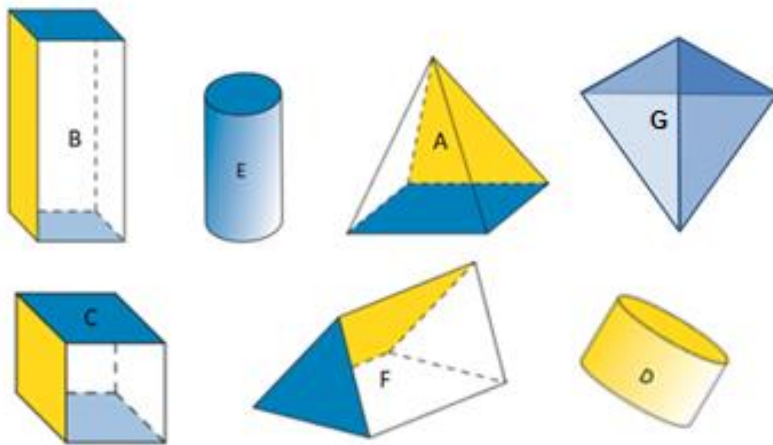
1. Complete the following table:

| Figure                                                                              | Name                      | No. of faces | No. of Edges | No. of vertices |
|-------------------------------------------------------------------------------------|---------------------------|--------------|--------------|-----------------|
|    | Cube                      | 6            | 12           | 8               |
|    | Cuboid                    | 6            | 12           | 8               |
|    | Triangular based prism    | 5            | 9            | 6               |
|   | Rectangular based Pyramid | 5            | 8            | 5               |
|  | Square-based pyramid      | 5            | 8            | 5               |
|  | Triangular based pyramid  | 3            | 6            | 4               |

2. Differentiate between **prisms** and **pyramids**.

|                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                   |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>Prisms</b></p> <p>Prisms are objects with two faces that are congruent and parallel. These two faces are called bases and are connected by lateral faces that are parallelograms. Prisms are named using their base. The two opposite congruent faces are the base.</p> | <p><b>Pyramids</b></p> <p>Pyramids have one base. The lateral faces of a pyramid are always triangles. These triangles meet at the same vertex called the <b>apex</b>. Pyramids are named using their base. The base is not always the face it is sitting on.</p> |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

3. Sort the objects below into prisms and pyramids.



**Prisms: B, C, D, E, F**

**Pyramids: A, G**

4. Write the name of the object that would be formed by each net below:

| Net | Name of the object |
|-----|--------------------|
|     | Rectangular prism  |
|     | Hexagonal prism    |
|     | Triangular prism   |
|     | Hexagonal pyramid  |
|     | Cube               |

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